# Brain-age research summary: cohorts, experiments, the best model and what mattered

📄 **PDF version of this notebook** (figures, tables and formulas, no code; regenerate with `scripts/notebook_to_pdf.py notebooks/brain_age_research_summary.ipynb`).

One notebook for the whole EEG brain-age study on TUH (2026-10-05 → 10-16), in three phases:

- **Phase 1, TUAB only.** About 50 fine-tunes of LaBraM-base on the TUH Abnormal corpus (TUAB), aimed at memorization: schedule, regularizers, mixup/EMA, loss shape, codebook regularization, heads, training from scratch (`docs/age_overfitting_experiments.md`).
- **Phase 2, full TUH.** The same recipe on cohorts drawn from the whole TUH EEG corpus (TUEG, about 60k recordings), plus data, input and head changes (`docs/data/tuh_cohort_eval_v1.csv`).
- **Phase 3, cross-validation.** 5-fold patient-grouped CV of the three strongest phase-2 configurations (`notebooks/tuh_cv_summary.ipynb`, `docs/data/cv5_eval_v2.csv`).

It answers four questions:

1. Which datasets and cohorts were used, and how they differ (Section 1).
2. Which experiments ran and which came out near the best (Section 2).
3. What exactly the best experiment does, step by step (Section 3).
4. What changed relative to LaBraM as published (Section 4), and which of those changes measurably moved the error (Section 5).

**Contents.** 0 Metrics and evaluation sets · 1 Data · 2 Experiments · 3 The best experiment · 4 Changes versus the LaBraM recipe · 5 Which changes mattered · 6 Conclusions.

**Inputs.**

- In git: the registry, the two CSVs, `docs/data/line_removal_ab.json`, the configs, `docs/data/brain_age_clearml_runs.csv` (each run's ClearML experiment) and `scripts/submit_tuh_cohorts.sh`. Every phase-2 run's configuration is rebuilt from `labram/eval/research_summary.py::PHASE2` with the same config loader the trainer uses. `tests/test_research_summary.py` checks that catalog against the submit script.
- Optional, on the TUH data volume (`LABRAM_TUH_ROOT`, default `/data/datasets/EEG-public/TUH_EEG`):
  - the cohort manifests behind Section 1's counts;

- the TUH recording index and NEDC's clinical sheet behind the TUEG specification (Tables 1.6–1.13);
- the per-recording CV predictions behind the bootstraps ( `LABRAM_CV_EVAL` ).

They stay out of git under the TUH data-use agreement. Without them those cells print a note and skip, and the outputs show only aggregate counts.

# 0. Metrics and evaluation sets

Every error is computed **per recording**. A recording's predicted age is the mean of its window predictions (Eq. (3.7)), in years. For a set $S$ of recordings with true age $y_i$ and predicted age $\hat{y}_i$ :

$$\mathrm{MAE}(S) = \frac{1}{|S|} \sum_{i \in S} |\hat{y}_i - y_i| \tag{0.1}$$

$$R^2(S) = 1 - \frac{\sum_{i \in S} (\hat{y}_i - y_i)^2}{\sum_{i \in S} (y_i - \bar{y}_S)^2} \tag{0.2}$$

$$\mathrm{bias}(S) = \frac{1}{|S|} \sum_{i \in S} (\hat{y}_i - y_i) \tag{0.3}$$

where $\bar{y}_S$ is the mean true age of $S$ ( `labram/eval/cv_compare.py::metrics` ). The effect of a change from a baseline run $b$ to a variant run $v$ is

$$\Delta\mathrm{MAE} = \mathrm{MAE}_v(S) - \mathrm{MAE}_b(S), \tag{0.4}$$

so **negative is better**. Section 5 defines when a $\Delta$ is larger than run-to-run noise.

**All errors are before bias correction.** Every MAE in this notebook scores the raw prediction $\hat{y}_i$. The exceptions are Table 3.5, which compares raw and bias-corrected errors, and the `cal.` columns of Tables 2.5–2.7 (a leak-free correction, Eq. (3.11)).

The **bias slope** $\beta$ measures regression to the mean, the typical failure of an age model: old patients are predicted too young and young ones too old. It is the least-squares slope of the error $e_i = \hat{y}_i - y_i$ against the true age, with population moments over $S$ ( `labram/utils/regression_metrics.py` , `age_bias_slope` ):

$$\beta(S) = \frac{\mathrm{Cov}_S(y, e)}{\mathrm{Var}_S(y)} \tag{0.5}$$

- $\beta = 0$: the error does not depend on age.
- $\beta = -1$: the prediction ignores age entirely; it is a constant.
- $\beta = -0.4$: a patient 10 years older than average is predicted about 4 years too young.

**Evaluation sets.**

- **TUAB test:** TUAB's official evaluation set, 274 recordings of 251 patients (149 normal, 125 abnormal). It is the same 274 recordings in every phase (checked in Section 1) and no training split contains any of its patients, so it is the one yardstick every run shares.
- **Val:** each run's own validation split. It differs between cohorts, so it compares runs only within a cohort.
- **CV test / TUEG out-of-fold:** each recording of the CV pool scored by the one fold model that never saw its patient.

**No test label in the target normalization.** The age is z-scored with the mean and sd of the training windows only (Eq. (0.6), Table 0.1).

**Caveat.** Phases 1 and 2 compared many configurations on the TUAB test set, which makes their best single-split numbers optimistically selected. The CV numbers (Phase 3) and the TUEG out-of-fold error were not used to choose anything.

```
registry: 45 phase-1 runs; cohort CSV: 21 phase-2 runs; CV CSV: 4 experiments x 5 folds
TUH data volume: found; per-recording CV predictions: found
```

## Table 0.1. The age normalization uses training ages only

Training regresses a z-scored age, and predictions are mapped back to years before any metric ( `labram/data/window_selection.py::_train_target_stats` , `losses/regression.py` ):

$$z = \frac{y - \mu}{\sigma}, \qquad \hat{y} = \mu + \sigma\,\hat{z}, \qquad \mu = \frac{1}{N}\sum_{w \in \text{train}} y_w, \quad \sigma^2 = \frac{1}{N-1}\sum_{w \in \text{train}} (y_w - \mu)^2$$

where $w$ runs over the $N$ **selected training windows** of the run (after trimming), $y_w$ is the age of the window's recording, and $\hat{z}$ is the model output. If $\mu$ or $\sigma$ included a test recording, the test ages would leak into training and into every test prediction.

The check ( `scripts/check_target_norm_leak.py` , `docs/data/target_norm_leak_check.csv` ) rebuilds each training setup exactly as the trainer does (cohort, CV fold, window selection) twice. The first build is as recorded; the second has every **test** recording's age shifted by +40 years. Columns:

- `train` / `test recordings` and `train ∩ test recordings` (must be 0);
- `mean` / `std` : $\mu$, $\sigma$ of Eq. (0.6) as the run uses them, in years;
- `mean` / `std, test ages +shift` : the same after the test ages were shifted. They must not change;
- `= train windows' own stats` : $\mu$, $\sigma$ equal the training windows' own mean and sample sd;
- `test targets moved by` : the median change of the test targets, 40, confirming the shift took effect;
- `normalization uses test ages` : "no" when the two (mean, std) pairs are identical.

Phase 1 is checked on the local TUAB v3.0.1 copy (the phase-1 runs used v3.0.0, with the same split and code). The CV rows check fold 0 in both CV schemes. The other folds use the same code on other recordings.

| setup | train recordings | test recordings | train ∩ test recordings | mean | std | mean, test ages +shift | std, test ages +shift | = train windows' own stats | test targets moved by | normalization uses test ages |
|---|---|---|---|---|---|---|---|---|---|---|
| **phase 1: TUAB (D recipe)** | 2145 | 274 | 0 | 48.88 | 17.82 | 48.88 | 17.82 | True | 40.00 | no |
| **phase 2: cohort all_both_v3 (Exp4*)** | 25613 | 274 | 0 | 52.34 | 17.59 | 52.34 | 17.59 | True | 40.00 | no |
| **phase 3: CV fold 0, test = fold (cvbase/cvctrl/cvaug)** | 17607 | 5663 | 0 | 52.49 | 17.28 | 52.49 | 17.28 | True | 40.00 | no |
| **phase 3: CV fold 0, test = TUAB held out (cvaugval)** | 23372 | 274 | 0 | 52.74 | 17.45 | 52.74 | 17.45 | True | 40.00 | no |

# 1. Data: every dataset and cohort

All data comes from the TUH EEG corpus. TUAB (TUH Abnormal, about 3k recordings with a normal/abnormal label) is a subset of TUEG (the full TUH EEG corpus, about 60k recordings). The age label is parsed from each EDF header ( `labram/data/tuh_metadata.py` ). Every dataset uses the same signal format:

- 23 referential channels at 200 Hz in µV, 0.1–75 Hz band-pass;
- 10 s windows;
- the first and last 60 s of each recording dropped;
- evaluation on the first 5 minutes after the trim ( `labram/data/window_selection.py` ).

The datasets differ in which recordings they contain, how mains interference is removed, and how patients are split.

**Reading order.**

- Table 1.1: the two **raw corpora**, TUAB and TUEG, as downloaded.
- Table 1.2: why each removed recording was removed.
- Table 1.3: how each training dataset is built from them, one inclusion rule at a time, with the recordings and patients each rule removes.
- Tables 1.4–1.5: the resulting datasets and their splits; Figure 1.1: the final training set by source (TUAB / TUEG / total) and status.
- Tables 1.6–1.13 and Figure 1.2: a closer look at TUEG.

**Lineage.**

- `TUAB v3.0.0` (phase 1) → `TUAB v3.0.1 interp` (phase-2 control): same recordings and split, mains removed by 50 + 60 Hz spectrum interpolation instead of a 50 Hz notch.
- `normal_TUEG_v1` , `normal_both_v1` : normal recordings only (TUEG alone; TUAB + TUEG).
- `all_both_v1` → `v2` → `v3` : every recording (normal, abnormal, unlabelled), with TUAB patients kept in their TUAB split.

    - `v2` gives TUEG one patient split shared by every cohort, so validation is comparable across cohorts.
    - `v3` adds the TUEG recordings that lack only the T1/T2 electrodes.
- `v3_compact` / `v3_cvcompact`: byte-for-byte subsets of `v3` that store only the windows a run reads (`dataset_maker/make_compact_cohort.py`). `cvcompact` keeps both training and evaluation windows of every recording, so any CV fold can use any recording.

Tables 1.6–1.13 and Figure 1.2 at the end of this section specify TUEG itself: size, subsets, labels and age.

All cohorts are built by `dataset_maker/make_cohort_dataset.py`, which drops every TUEG recording of a TUAB-test patient and asserts that no patient is in two splits.

## Table 1.1. Raw data: TUAB and TUEG, and what the final cohort selects

One row per part of the raw data, split by status (`rs.raw_summary`). TOTAL counts every recording once:

- **TUAB train**, **TUAB eval (test)**: the two official splits of TUAB v3.0.1 (2,993 recordings), with TUAB's own normal / abnormal label.
- **TUEG (not in TUAB)**: the rest of TUEG v2.0.2. Every TUAB recording is also in TUEG, so these rows hold only what TUEG adds. Its status comes from NEDC's clinical report sheet; recordings without a sheet entry have **no label**.
- **TOTAL**: all of TUEG, 70,830 recordings, each counted once.

Columns, left to right:

- `patients`, `recordings`: the raw counts. A recording is one EDF file, a "case". A patient with recordings of different status counts in each status row, so the status rows' patients add up to more than the subtotal.
- `readable EDF, ≥ 60 s`: the recordings converted to the npy format (Section "step process" below).
- `age known`: of those, the ones with a valid age in the EDF header (TUH redacts 90+).
- `T1/T2 missing (selected)`: selected recordings that lack only the T1/T2 electrodes. They are kept, and the model leaves the two channels out per sample.
- `selected recordings`, `selected patients`: what the final cohort `all_both_v3` keeps. That is the dataset of the best models: train 25,613 + val 3,422 + test 274 = 29,309 recordings.
- `removed`: `recordings` − `selected recordings`. Table 1.2 gives the reason for each removed recording, and Table 1.3 follows the rules step by step for every dataset.

Phase 1 trains on TUAB only and selects differently; Table 1.3 gives that flow too.

| part | status | patients | recordings | readable EDF, ≥ 60 s | age known | T1/T2 missing (selected) | selected recordings | selected patients | removed |
|---|---|---|---|---|---|---|---|---|---|
| TUAB train | normal | 1237 | 1371 | 1370 | 1365 | 0 | 1258 | 1167 | 113 |
| | abnormal | 893 | 1346 | 1345 | 1337 | 0 | 1092 | 812 | 254 |
| | all | 2076 | 2717 | 2715 | 2702 | 0 | 2350 | 1944 | 367 |
| TUAB eval (test) | normal | 148 | 150 | 150 | 149 | 0 | 149 | 147 | 1 |
| | abnormal | 105 | 126 | 126 | 125 | 0 | 125 | 104 | 1 |
| | all | 253 | 276 | 276 | 274 | 0 | 274 | 251 | 2 |
| TUEG (not in TUAB) | normal | 6403 | 25340 | 20680 | 20289 | 2055 | 10197 | 5506 | 15143 |
| | abnormal | 8132 | 32373 | 27503 | 27045 | 3474 | 14298 | 7407 | 18075 |
| | no label | 808 | 10124 | 9212 | 9122 | 302 | 2190 | 662 | 7934 |
| | all | 13859 | 67837 | 57395 | 56456 | 5831 | 26685 | 12457 | 41152 |
| TOTAL | | 14987 | 70830 | 60386 | 59432 | 5831 | 29309 | 13801 | 41521 |

## Table 1.2. Why each raw recording was removed

Table 1.1's `removed` column, broken down by reason for the final cohort `all_both_v3` (`rs.removal_reasons`). Every raw recording counts once, under the **first** rule it fails, in the order the builder applies them (`dataset_maker/make_cohort_dataset.py`). The rows sum to the `raw` row, and `selected` matches Table 1.1. The rules:

- **EDF shorter than 60 s** / **EDF unreadable**: never converted to npy. One unreadable TUEG EDF has none of the 23 channels.
- **age unknown**: no valid age in the EDF header. TUH redacts ages of 90 and over.
- **age outside the cohort's range**: the TUEG cohorts use 18–89 years, so children and teenagers are out, including TUAB's.
- **recording shorter than the minimum length**: under 5 minutes. Too little EEG is left after the 60 s trims for the 5-minute evaluation.
- **channels other than the allowed ones missing**: the recording lacks a channel other than T1/T2, e.g. A1/A2 in the `*_a` montages. Recordings lacking only T1/T2 are kept.
- **patient is in the TUAB test set**: all other recordings of the 251 test patients are kept out of training and validation.
- **over the per-patient cap**: at most 10 train/val recordings per patient, drawn with a fixed seed.
  - It removes recordings, not patients: every patient with a usable recording keeps up to 10.
  - Some patients were recorded hundreds of times.
  - The cap counts a patient's TUAB and TUEG recordings together. TUAB patients who also have many TUEG recordings can therefore lose TUAB recordings to the draw.

`patients` counts the patients with at least one recording removed for that reason. A patient can appear under several reasons.

| | TUAB train | | TUAB eval (test) | | TUEG (not in TUAB) | | TOTAL | |
|---|---|---|---|---|---|---|---|---|
| reason | recordings | patients | recordings | patients | recordings | patients | recordings | patients |
| **EDF shorter than 60 s (not converted)** | 0 | 0 | 0 | 0 | 10441 | 2471 | 10441 | 2471 |
| **EDF unreadable (not converted)** | 2 | 2 | 0 | 0 | 1 | 1 | 3 | 3 |
| **age unknown (missing, or 90+ redacted)** | 13 | 13 | 2 | 2 | 939 | 322 | 954 | 329 |
| **age outside the cohort's range** | 30 | 30 | 0 | 0 | 1543 | 760 | 1573 | 779 |
| **recording shorter than the minimum length** | 0 | 0 | 0 | 0 | 6662 | 2638 | 6662 | 2638 |
| **channels other than the allowed ones missing** | 0 | 0 | 0 | 0 | 5114 | 716 | 5114 | 716 |
| **patient is in the TUAB test set** | 0 | 0 | 0 | 0 | 417 | 74 | 417 | 74 |
| **over the per-patient cap** | 322 | 164 | 0 | 0 | 16035 | 756 | 16357 | 756 |
| **selected** | 2350 | 1944 | 274 | 251 | 26685 | 12457 | 29309 | 13801 |
| **raw** | 2717 | 2076 | 276 | 253 | 67837 | 13859 | 70830 | 14987 |

## How the datasets are built: the step process and the data split

**Steps, in order.** Table 1.3 counts each one.

1. **Index.** Both corpora are scanned into one recording index, one row per EDF (`dataset_maker/tuh_index.py`). Age, sex and the TUAB label come from the EDF headers and folders; the report finding comes from NEDC's clinical sheet (`dataset_maker/tuh_clinical.py`).

2. **Convert** (`dataset_maker/make_TUH_npy.py`). Every readable EDF of at least 60 s becomes one npy file:

   - the 23 model channels in a fixed order; a missing channel is zero-filled and flagged in a channel mask;
   - 0.1–75 Hz band-pass and mains removal (50 + 60 Hz spectrum interpolation);
   - resampled to 200 Hz, in µV.

3. **Select per dataset** (`dataset_maker/make_cohort_dataset.py`):

   - age 18–89 years;
   - recording at least 5 minutes long;
   - channel rule: all 23 channels, or in v3 only T1/T2 may be missing;
   - for the `normal_*` cohorts only: report-normal, no neurological diagnosis, no sedating medication.

4. **Protect the test set.** Every recording of a patient in TUAB's eval split is removed from training and validation. TUAB's eval split itself (274 recordings with a known age) is the test set of every dataset.

5. **Split by patient**, never by recording:

   - TUAB patients keep the split of phase 1: TUAB's train split divided into train and a seeded 20 % val, patient-disjoint;
   - TUEG patients: `v1` draws a seeded 10 % patient val per cohort. `v2` / `v3` use one shared patient split for all cohorts (`dataset_maker/tueg_patient_split.py`), so a model can be validated on another cohort's val without patient leakage.

   The builder asserts that no patient is in two splits.

6. **Cap per patient.** At most 10 train/val recordings per patient, drawn with a fixed seed, so that patients recorded hundreds of times do not dominate.

7. **Windows.** Each training recording contributes at most 20 minutes of 10 s windows. Val and test score the first 5 minutes after 60 s trims at both ends (Section 0).

8. **Phase 3, cross-validation.** The train + val of `all_both_v3` (29,035 recordings) is split into 5 patient-grouped folds. Fold $k$ tests, fold $k+1$ validates, and the other three train. The TUAB test set stays outside every fold.

## Table 1.3. From the raw corpora to the training datasets: inclusion flow

Each row is the set of recordings left after one more inclusion rule. Columns as in Table 1.1, plus `removed`, the recordings that rule took out. The rows are replayed from each dataset's recorded build parameters with the builder's own functions:

- `rs.tuab_flow` for TUAB (`data/age_splits.py`, `make_TUAB.py`);
- `rs.cohort_flow` for the TUEG cohorts (`dataset_maker/make_cohort_dataset.py`).

Each flow's final counts are checked against the dataset's split files, so the table is exact.

The datasets:

- **TUAB (phase 1):** TUAB only. Ages 1–89, all 23 channels. TUAB's train split is divided patient-disjointly into train and val; its eval split is the test set.
- **TUEG cohorts (phase 2–3):** built from the TUEG npy export.
  - Shared rules (first block): readable EDF of at least 60 s, age 18–89, recording at least 5 minutes.
  - Per-cohort rules (later blocks): channel completeness and, for the `normal_*` cohorts, the normal filter (report-normal, no neurological diagnosis, no sedating medication).
  - Then every recording of a TUAB-test patient is removed, so no test patient is ever trained on.
  - At most 10 train/val recordings are kept per patient.
  - The test set is always TUAB's eval split.
  - `all_both_v1` and `v2` contain the same recordings and differ only in how TUEG patients are split into train and val. `all_both_v3` is the cohort of the best models, and its train + val (29,035 recordings) is the CV pool of phase 3.

| dataset | cohort | step | recordings | patients | normal | abnormal | no label | all 23 channels | only T1/T2 missing | other channels missing | removed |
|---|---|---|---|---|---|---|---|---|---|---|---|
| TUAB (phase 1) | | TUAB recordings (TUH index) | 2993 | 2329 | 1521 | 1472 | 0 | 2990 | 3 | 0 | – |
| | | age known (1–89 y; 90+ redacted) | 2978 | 2314 | 1515 | 1463 | 0 | 2975 | 3 | 0 | 15 |
| | | all 23 channels | 2975 | 2313 | 1515 | 1460 | 0 | 2975 | 0 | 0 | 3 |
| | | TUAB train split = train + val | 2701 | 2062 | 1366 | 1335 | 0 | 2701 | 0 | 0 | – |
| | | of which train | 2145 | 1650 | 1075 | 1070 | 0 | 2145 | 0 | 0 | – |
| | | of which val (patient-disjoint, seeded) | 556 | 412 | 291 | 265 | 0 | 556 | 0 | 0 | – |
| | | test: TUAB eval split | 274 | 251 | 149 | 125 | 0 | 274 | 0 | 0 | – |
| TUEG cohorts | shared rules | TUEG recordings (TUH index) | 70830 | 14987 | 26861 | 33845 | 10124 | 55182 | 8479 | 7169 | – |
| | | converted to npy (readable EDF, ≥ 60 s) | 60386 | 14979 | 22200 | 28974 | 9212 | 46247 | 7576 | 6563 | 10444 |
| | | age known, 18–89 y | 57857 | 14019 | 21045 | 27909 | 8903 | 44645 | 6918 | 6294 | 2529 |
| | | recording ≥ 5 min | 51195 | 14001 | 18203 | 25062 | 7930 | 40065 | 6016 | 5114 | 6662 |
| | | all 23 channels | 40065 | 10553 | 14886 | 19126 | 6053 | 40065 | 0 | 0 | 11130 |
| | normal_TUEG_v1 | normal only: report-normal, no neurological diagnosis, not sedated | 8798 | 3585 | 8798 | 0 | 0 | 8798 | 0 | 0 | 31267 |
| | | without TUAB patients (TUEG only) | 5485 | 2389 | 5485 | 0 | 0 | 5485 | 0 | 0 | 3313 |
| | | ≤ 10 recordings per patient = train + val | 4158 | 2389 | 4158 | 0 | 0 | 4158 | 0 | 0 | 1327 |
| | | test: TUAB eval recordings with a known age | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | – |
| | normal_both_v1 | all 23 channels | 40065 | 10553 | 14886 | 19126 | 6053 | 40065 | 0 | 0 | 11130 |
| | | normal only: report-normal, no neurological diagnosis, not sedated | 8798 | 3585 | 8798 | 0 | 0 | 8798 | 0 | 0 | 31267 |
| | | without recordings of TUAB-test patients | 8592 | 3453 | 8592 | 0 | 0 | 8592 | 0 | 0 | 206 |
| | | ≤ 10 recordings per patient = train + val | 6293 | 3453 | 6293 | 0 | 0 | 6293 | 0 | 0 | 2299 |
| | | test: TUAB eval recordings with a known age | 274 | 251 | 149 | 125 | 0 | 274 | 0 | 0 | – |
| | all_both_v1 = v2 (same recordings) | all 23 channels | 40065 | 10553 | 14886 | 19126 | 6053 | 40065 | 0 | 0 | 11130 |
| | | without recordings of TUAB-test patients | 39420 | 10304 | 14591 | 18895 | 5934 | 39420 | 0 | 0 | 645 |
| | | ≤ 10 recordings per patient = train + val | 23273 | 10304 | 9458 | 11920 | 1895 | 23273 | 0 | 0 | 16147 |
| | | test: TUAB eval recordings with a known age | 274 | 251 | 149 | 125 | 0 | 274 | 0 | 0 | – |
| | all_both_v3 | all 23 channels, or only T1/T2 missing | 46081 | 13799 | 16998 | 22690 | 6393 | 40065 | 6016 | 0 | 5114 |
| | | without recordings of TUAB-test patients | 45392 | 13550 | 16679 | 22448 | 6265 | 39420 | 5972 | 0 | 689 |
| | | ≤ 10 recordings per patient = train + val | 29035 | 13550 | 11455 | 15390 | 2190 | 23204 | 5831 | 0 | 16357 |
| | | test: TUAB eval recordings with a known age | 274 | 251 | 149 | 125 | 0 | 274 | 0 | 0 | – |

# Table 1.4. The training datasets (cohorts): contents, rules and splits

One row per dataset. Names show the corpora in capitals (TUAB, TUEG); the directories on disk under `$LABRAM_TUH_ROOT/cohorts/` use the same names in lower case (e.g. `normal_tueg_v1`).

| column | meaning |
|---|---|
| `phase` | the phase that used the dataset |
| `contents` | which recordings it holds |
| `line removal` | how mains interference is removed (from the dataset's `manifest.json`) |
| `rules` | inclusion rules: `age` range in years; `≥ min` minimum recording length in minutes; `≤ k/patient` at most *k* train/val recordings per patient; `T1/T2 may be missing` |
| `size (train / val / test)` | recordings per split, from the dataset's `age_split.json` (`rs.split_sizes`); the CV runs split train + val into 5 folds |
| `TUEG split` | how TUEG patients are split; TUAB patients always keep their TUAB split |
| `used by` | the runs of Tables 2.2–2.7 trained on it |

| dataset | phase | contents | line removal | rules | size (train / val / test) | TUEG split | used by |
|---|---|---|---|---|---|---|---|
| **TUAB v3.0.0** | 1 | TUAB, all recordings (normal + abnormal) | notch:50 | age 1–89 (TUAB) | 2,145 / 556 / 274 | TUAB: seeded 20 % val, official test | phase-1 runs (Table 2.2) |
| **TUAB v3.0.1 interp** | 2 | TUAB, all recordings | interp:50,60 | age 1–89 (TUAB) | 2,145 / 556 / 274 | TUAB: seeded 20 % val, official test | Control |
| **normal_TUEG_v1** | 2 | TUEG normal, patients outside TUAB; no test split | interp:50,60 | age 18–89, ≥ 5 min, ≤ 10/patient | 3,763 / 395 / 0 | seeded 10 % val | Exp2 |
| **normal_both_v1** | 2 | TUAB normal + TUEG normal | interp:50,60 | age 18–89, ≥ 5 min, ≤ 10/patient | 5,450 / 843 / 274 | seeded 10 % val | Exp1, Exp1b, Exp1t, Exp1f |
| **all_both_v1** | 2 | TUAB + TUEG, every recording (normal / abnormal / no label) | interp:50,60 | age 18–89, ≥ 5 min, ≤ 10/patient | 20,368 / 2,905 / 274 | seeded 10 % val | Exp3, Exp3b, Exp3q, Exp3c |
| **all_both_v2** | 2 | as v1, TUEG patients on the shared split | interp:50,60 | age 18–89, ≥ 5 min, ≤ 10/patient | 20,470 / 2,803 / 274 | shared patient split v2 | Exp3v2, Exp3cond, Exp3clscond |
| **all_both_v3** | 2 | v2 + TUEG recordings missing only T1/T2 | interp:50,60 | age 18–89, ≥ 5 min, ≤ 10/patient, T1/T2 may be missing | 25,613 / 3,422 / 274 | shared patient split v2 | Exp4drop, Exp4base, Exp4ctrl, Exp4aug, Exp4aug15 |
| **all_both_v3_compact** | 2 | v3, only the windows Exp4base reads | interp:50,60 | age 18–89, ≥ 5 min, ≤ 10/patient, T1/T2 may be missing | 25,613 / 3,422 / 274 | shared patient split v2 | Exp4basec |
| **all_both_v3_cvcompact** | 3 | v3, training + evaluation windows of every recording (CV pool = train + val) | interp:50,60 | age 18–89, ≥ 5 min, ≤ 10/patient, T1/T2 may be missing | 25,613 / 3,422 / 274 (CV pool 29,035) | shared patient split v2 | cvbase, cvctrl, cvaug |

# Table 1.5. Size and composition per split

For each dataset and split:

- `recordings` and `patients`;
- `from TUAB`: recordings that are also in TUAB;
- `normal` / `abnormal` / `no label`: the recording's report status (TUEG recordings without a normal/abnormal report have no label);
- `age mean` and `age sd`: the true age's mean and sample standard deviation (years).

Computed from each dataset's `cohort.parquet` / `manifest.json`, or from `age_split.json` + `age_metadata.json` for the TUAB-only datasets (`labram/eval/research_summary.py::cohort_summary`, `tuab_split_summary`).

| dataset | split | recordings | patients | from TUAB | normal | abnormal | no label | age mean | age sd |
|---|---|---|---|---|---|---|---|---|---|
| TUAB v3.0.0 | train | 2145 | 1650 | 2145 | 1075 | 1070 | 0 | 49.2 | 17.7 |
| | val | 556 | 412 | 556 | 291 | 265 | 0 | 48.4 | 15.9 |
| | test | 274 | 251 | 274 | 149 | 125 | 0 | 50.4 | 18.1 |
| TUAB v3.0.1 interp | train | 2145 | 1650 | 2145 | 1075 | 1070 | 0 | 49.2 | 17.7 |
| | val | 556 | 412 | 556 | 291 | 265 | 0 | 48.4 | 15.9 |
| | test | 274 | 251 | 274 | 149 | 125 | 0 | 50.4 | 18.1 |
| normal_TUEG_v1 | train | 3763 | 2150 | 0 | 3763 | 0 | 0 | 48.6 | 17.6 |
| | val | 395 | 239 | 0 | 395 | 0 | 0 | 49.9 | 16.9 |
| normal_both_v1 | train | 5450 | 2995 | 775 | 5450 | 0 | 0 | 47.2 | 17.2 |
| | val | 843 | 458 | 196 | 843 | 0 | 0 | 46.6 | 15.7 |
| | test | 274 | 251 | 274 | 149 | 125 | 0 | 50.4 | 18.1 |
| all_both_v1 | train | 20368 | 9069 | 1879 | 8204 | 10610 | 1554 | 52.0 | 17.5 |
| | val | 2905 | 1235 | 478 | 1254 | 1310 | 341 | 51.3 | 16.3 |
| | test | 274 | 251 | 274 | 149 | 125 | 0 | 50.4 | 18.1 |
| all_both_v2 | train | 20470 | 9087 | 1879 | 8277 | 10617 | 1576 | 51.9 | 17.5 |
| | val | 2803 | 1217 | 478 | 1181 | 1303 | 319 | 51.6 | 16.5 |
| | test | 274 | 251 | 274 | 149 | 125 | 0 | 50.4 | 18.1 |
| all_both_v3 | train | 25613 | 12017 | 1868 | 10067 | 13714 | 1832 | 52.4 | 17.5 |
| | val | 3422 | 1533 | 482 | 1388 | 1676 | 358 | 52.1 | 16.8 |
| | test | 274 | 251 | 274 | 149 | 125 | 0 | 50.4 | 18.1 |
| all_both_v3_compact | train | 25613 | 12017 | 1868 | 10067 | 13714 | 1832 | 52.4 | 17.5 |
| | val | 3422 | 1533 | 482 | 1388 | 1676 | 358 | 52.1 | 16.8 |
| | test | 274 | 251 | 274 | 149 | 125 | 0 | 50.4 | 18.1 |
| all_both_v3_cvcompact | train | 25613 | 12017 | 1868 | 10067 | 13714 | 1832 | 52.4 | 17.5 |
| | val | 3422 | 1533 | 482 | 1388 | 1676 | 358 | 52.1 | 16.8 |
| | test | 274 | 251 | 274 | 149 | 125 | 0 | 50.4 | 18.1 |

```
TUAB test: 274 recordings; identical in every dataset with a test split: True
{'TUAB v3.0.0': True, 'TUAB v3.0.1 interp': True, 'normal_both_v1': True, 'all_both_v
1': True, 'all_both_v2': True, 'all_both_v3': True, 'all_both_v3_compact': True, 'all_b
oth_v3_cvcompact': True}
```

**Reading Table 1.5.**

- **Scale:** training grows from 2,145 TUAB recordings (phase 1) to 25,613 recordings of about 12k patients ( `all_both_v3` ).
- **Composition:** in the `all_both` cohorts about half the training recordings are abnormal and about 7 % have no label.
- **Normal-only cohorts:** `normal_*` hold only normal recordings, but their test split is still the full TUAB test set, abnormal recordings included.
- **Test set:** the 274 TUAB test recordings are identical across datasets (the check above), so every TUAB test number in Sections 2–5 is on the same recordings.

## Figure 1.1. Composition and age of the training set

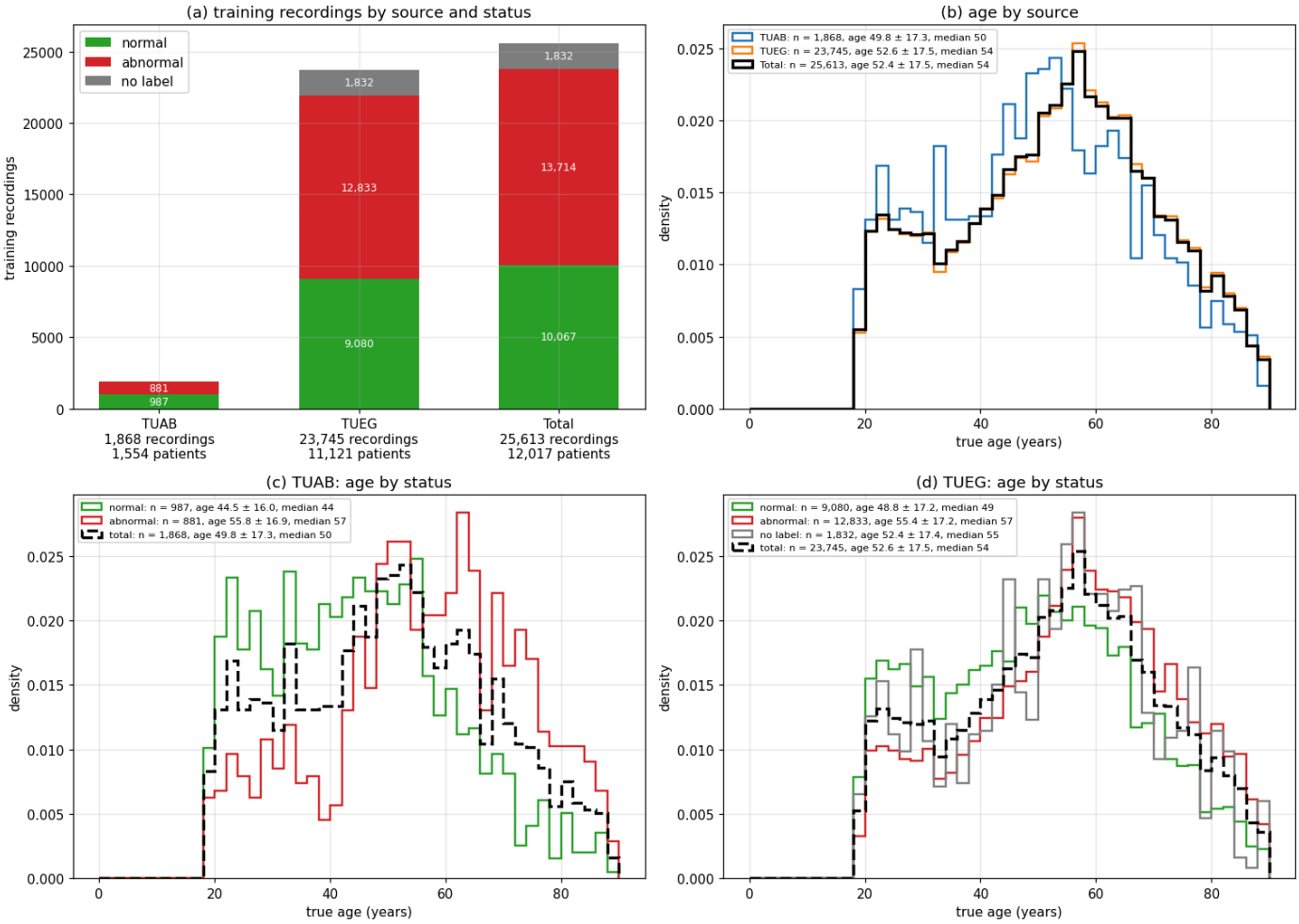
The training set of the final cohort `all_both_v3` , the one the best models use: its train split, 25,613 recordings. It is split by source:

- **TUAB:** recordings that are in TUAB;
- **TUEG:** the TUEG recordings that are not in TUAB;
- **Total:** both.

Status is the recording's normal / abnormal / no-label status (Eq. (1.2)). The other datasets of Table 1.4 are subsets or reshuffles of the same recordings, so they are not drawn again. Table 1.5 gives each dataset's counts per split.

- (a) Training recordings per source, stacked by status. Each segment shows its count, and the bar's label gives the total and the number of patients.
- (b) Distribution of true age (years) per source, as a density (step histogram, 2-year bins).
- (c) TUAB by status, as in (b).
- (d) TUEG by status, as in (b).

Each legend gives $n$ (recordings), the mean ± sample standard deviation of age and its median.




**TUEG, the source corpus.** Every cohort above draws from TUEG v2.0.2, the full TUH EEG corpus ( `$LABRAM_TUH_ROOT/tueg/v2.0.2` ). Tables 1.6–1.13 and Figure 1.2 specify it from two files:

- the TUH recording index, one row per EDF over all TUH corpora ( `index/tuh_index.parquet` , `dataset_maker/tuh_index.py` );
- NEDC's clinical sheet, one row per session, joined on patient + session ( `index/nedc/metadata_v12.xlsm.xlsx` , `dataset_maker/tuh_clinical.py` ).

`labram/eval/research_summary.py::tueg_recordings` builds the joined table. For a set $S$ of the $N = |\mathrm{TUEG}|$ recordings, with recording $i$ lasting $d_i$ seconds:

$$H(S) = \frac{1}{3600} \sum_{i \in S} d_i, \qquad \%(S) = 100 \, \frac{|S|}{N} \tag{1.1}$$

A recording's **status** is the TUAB folder label where the recording is in TUAB. Otherwise it is the report finding of the clinical sheet: abnormal when the sheet's "abnormal due to" field is filled. Recordings whose session is not in the sheet have no label (`dataset_maker/make_cohort_dataset.py::label_of`, the label the `all_both` cohorts use):

$$\text{status}_i = \begin{cases} \text{TUAB label} & i \in \text{TUAB} \\ \text{abnormal if the report gives a cause, else normal} & i \notin \text{TUAB}, \text{ session} \\ \text{no label} & \text{otherwise} \end{cases}$$

Ages are in years, from the EDF header. TUH writes 999 for patients aged 90 or older (a HIPAA redaction), and those recordings count as age-unknown. Age statistics use the recordings with a known age $a_i$: mean $\bar{a}$, sample standard deviation and quartiles $Q_1$, median, $Q_3$:

$$\bar{a} = \frac{1}{n} \sum_{i=1}^{n} a_i, \qquad s_a = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (a_i - \bar{a})^2} \tag{1.3}$$

Data volume required.

## Table 1.6. TUEG at a glance: all recordings and the selected ones

One row per quantity (`rs.tueg_overview`), in two columns:

- `all TUEG`: every TUEG recording, 70,830, TUAB included;
- `selected (all_both_v3)`: the recordings the final cohort keeps, train + val + test (Tables 1.1–1.2).

Rows:

- `size`: recordings, patients, sessions and hours (Eq. (1.1)).
  - A **session** is one visit of a patient (`<patient>_sNNN`). It can hold several recordings (EDF files `t000`, `t001`, ...), so there are more recordings than sessions.
  - The last size row splits the selected sessions into train / val / test.
- `recording`: length, year, sampling rate and montage. The year range omits a few EDF headers with impossible dates.
- `channels`: whether the recording has all 23 channels the models read.
- `patient`: sex and age (Eq. (1.3)), counted over recordings, so a patient with several recordings counts several times.

| group | quantity | all TUEG | selected (all_both_v3) |
|---|---|---|---|
| size | recordings (EDF files) | 70,830 | 29,309 |
| | patients | 14,987 | 13,801 |
| | sessions | 26,845 | 21,783 |
| | sessions in train / val / test | – | 18,923 / 2,586 / 274 |
| | hours of EEG | 27,021 | 15,940 |
| | recordings per patient, median (max) | 1 (713) | 1 (10) |
| recording | length (min), median [IQR] | 10.0 [5.0, 20.3] | 20.1 [10.0, 22.6] |
| | year, median [5 %, 95 %] | 2014 [2003, 2016] | 2012 [2003, 2016] |
| | sampling rate (Hz) | 256: 52%, 250: 37%, 1000: 4% | 250: 60%, 256: 30%, 1000: 7% |
| | montage | 01_tcp_ar: 73%, 02_tcp_le: 18%, 03_tcp_ar_a: 9% | 01_tcp_ar: 70%, 02_tcp_le: 30%, 03_tcp_ar_a: 0% |
| channels | all 23 channels | 55,182 (77.9 %) | 23,478 (80.1 %) |
| | missing only T1/T2 | 8,479 (12.0 %) | 5,831 (19.9 %) |
| | missing other channels | 7,169 (10.1 %) | 0 (0.0 %) |
| patient | sex F / M / unknown (recordings) | 36,519 / 33,785 / 526 | 15,072 / 14,183 / 54 |
| | age known (1–89 y) | 69,747 (98.5 %) | 29,309 (100.0 %) |
| | age 90+ (redacted as 999) | 714 (1.0 %) | 0 (0.0 %) |
| | age missing | 369 (0.5 %) | 0 (0.0 %) |
| | age (y), mean ± sd; median [IQR] | 51.0 ± 18.1; 53 [38, 64] | 52.4 ± 17.4; 53 [39, 65] |

## Table 1.7. Subsets of TUEG

The other TUH corpora are subsets of TUEG: the same EDF files, each corpus adding its own label. One recording can be in several of them, so the rows overlap ( `rs.tueg_subsets` ). The last four rows are the subsets the brain-age cohorts select from:

- `TUEG only` : in no other corpus;
- `in the NEDC clinical sheet` : has a report status (Eq. (1.2));
- `adult age (18–89 y)` : the age range of the cohorts;
- `normative` : the pool of the `normal_*` cohorts ( `dataset_maker/tuh_clinical.py::normative_mask` ).

Columns:

- `label counts (recordings)` : the subset's label per category, where it is per-recording and categorical:
  - TUAB: normal / abnormal;
  - TUEP: epilepsy / no epilepsy, a patient-level diagnosis;
  - clinical sheet: the report finding;
  - `normative` : the status (Eq. (1.2)). TUSZ, TUAR and TUSL annotate *events* inside a recording (seizures, artifacts, slowing) rather than one category per recording; their annotation files are not on the data volume, hence "–". Age is numeric.
- `recordings` , `% of TUEG` and `hours` (Eq. (1.1)), and `patients` .

The `normative` subset holds 75 recordings that are report-normal but in TUAB's *abnormal* folder. The `normal_*` cohorts drop these too ( `make_cohort_dataset.py::select` ). TUEV is not listed: none of its recordings is in TUEG v2.0.2.

| subset | label it carries | label counts (recordings) | record-ings | % of TUEG | pa-tients | hours |
|---|---|---|---|---|---|---|
| **in TUAB** | normal / abnormal (TUAB folder) | normal 1,521 / abnormal 1,472 | 2993 | 4.2 | 2329 | 1141.0 |
| **in TUSZ** | seizure events | – | 8140 | 11.5 | 675 | 1480.7 |
| **in TUEP** | epilepsy / no epilepsy (patient) | epilepsy 2,244 / no epilepsy 564 | 2808 | 4.0 | 200 | 635.5 |
| **in TUAR** | artifact events | – | 310 | 0.4 | 213 | 95.2 |
| **in TUSL** | slowing events | – | 112 | 0.2 | 38 | 26.8 |
| **TUEG only** | – | – | 57836 | 81.7 | 13448 | 24024.5 |
| **in the NEDC clinical sheet** | report finding, 15 diagnosis flags, medications | report normal 26,846 / abnormal 33,426 | 60272 | 85.1 | 14508 | 24956.2 |
| **adult age (18–89 y)** | age | – | 68077 | 96.1 | 14025 | 25795.6 |
| **normative** | report-normal, no neurological diagnosis, not se-dated, 18–89 y | normal 14,268 / abnormal 75 | 14343 | 20.2 | 4773 | 4569.7 |

# Table 1.8. TUEG status labels and age per label

One row per status (Eq. (1.2)) and label source ( `rs.tueg_status` ):

- `TUAB folder` : TUAB's normal/abnormal label;
- `NEDC report` : the clinical sheet's finding;
- `all` : both sources;
- `total` : all of TUEG.

Columns:

- `recordings` , `% of TUEG` (Eq. (1.1));
- `patients` : a patient whose recordings differ in status counts under each status, so the rows sum to more than the total;
- `age n` : recordings with a known age;
- `age mean` , `age sd` (Eq. (1.3)), and `age Q1` , `age median` , `age Q3` in years.

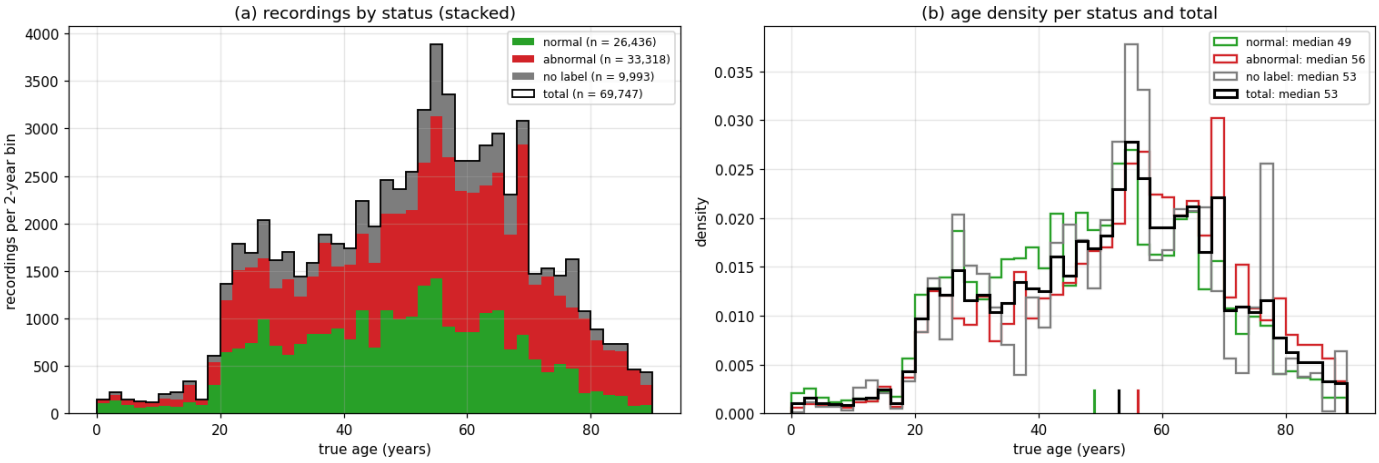
Where TUAB and the report both label a session, they agree on about 95 % of its recordings.

| status | source | recordings | % of TUEG | patients | age n | age mean | age sd | age Q1 | age median | age Q3 |
|---|---|---|---|---|---|---|---|---|---|---|
| | **NEDC report** | 25340 | 35.8 | 6403 | 24921 | 48.2 | 17.9 | 34.0 | 50.0 | 62.0 |
| **normal** | **TUAB folder** | 1521 | 2.1 | 1385 | 1515 | 44.0 | 16.1 | 31.0 | 44.0 | 55.0 |
| | **all** | 26861 | 37.9 | 7348 | 26436 | 47.9 | 17.8 | 34.0 | 49.0 | 62.0 |
| | **NEDC report** | 32373 | 45.7 | 8132 | 31855 | 53.3 | 18.1 | 41.0 | 56.0 | 67.0 |
| **abnormal** | **TUAB folder** | 1472 | 2.1 | 998 | 1463 | 54.4 | 17.2 | 44.0 | 55.0 | 67.0 |
| | **all** | 33845 | 47.8 | 8664 | 33318 | 53.4 | 18.0 | 41.0 | 56.0 | 67.0 |
| **no label** | **–** | 10124 | 14.3 | 808 | 9993 | 51.1 | 17.7 | 39.0 | 53.0 | 63.0 |
| **total** | **all** | 70830 | 100.0 | 14987 | 69747 | 51.0 | 18.1 | 38.0 | 53.0 | 64.0 |

# Figure 1.2. TUEG age distribution per status and in total

True age (years) of the TUEG recordings with a known age ($n$ = `age n` of Table 1.8), in 2-year bins, per status (Eq. (1.2)) and for all of TUEG:

- (a) Recordings per bin, stacked by status. The stack's outline is the total.
- (b) The same distributions as densities (each integrates to 1), so their shapes compare despite the different group sizes. The vertical ticks mark each group's median.




## Table 1.9. TUEG diagnosis flags

The clinical sheet's 15 diagnosis flags and two derived flags over the TUEG recordings in the sheet, most frequent first ( `rs.tueg_diagnoses` ):

- `any neurological diagnosis` : seizure (clinical or electrographic), epilepsy, stroke, Parkinson's, Alzheimer's, TBI, dementia or MS;
- `sedating medication` : an anaesthetic, benzodiazepine, barbiturate or opioid on the medication list.

**How the flags relate to the status label.** Table 1.10 gives where each field comes from, and Table 1.11 explains, per kind of flag, why a flagged recording is not necessarily abnormal. The flags are **not** the abnormal label, and they are **not one-hot**:

- **Independent of the status.** Each flag is the patient's diagnosis as recorded in the clinical sheet. The normal / abnormal status (Eq. (1.2)) is the neurologist's reading of *this* EEG. A flagged recording can be normal: an epilepsy patient often has a normal EEG between seizures. The `normal` / `abnormal` columns show the split for every flag, and many abnormal recordings carry no flag at all (Table 1.12).
- **Multi-hot.** Every flag is its own yes/no column, so a recording can carry several at once, or none (Table 1.12). The condition vector of the `path_label` runs (Eq. (3.6)) encodes them the same way.

Columns:

- `recordings` and `patients` with the flag;
- `normal` / `abnormal` : those recordings by status (Eq. (1.2));
- `% of sheet` : Eq. (1.1) relative to the sheet's recordings instead of $N$;
- `age mean` : Eq. (1.3), in years;
- `% abnormal` : the share with status abnormal (Eq. (1.2)).

The last row, **Total**, counts every TUEG recording in the sheet, flagged or not. The flags are multi-hot and many recordings carry none, so the flag rows do not add up to it.

The 14 pathology groups of the condition vector (Eq. (3.6)) come from these flags.

| flag | recordings | patients | normal | abnormal | % of sheet | age mean | % abnormal |
|---|---|---|---|---|---|---|---|
| **sedating medication** | 23067 | 4727 | 9135 | 13932 | 38.3 | 52.3 | 60.4 |
| **any neurological diagnosis** | 11560 | 1736 | 3901 | 7659 | 19.2 | 51.1 | 66.3 |
| **Clinical seizure** | 6980 | 1092 | 2922 | 4058 | 11.6 | 48.5 | 58.1 |
| **Epilepsy** | 4660 | 306 | 1991 | 2669 | 7.7 | 47.0 | 57.3 |
| **Electrographic seizure** | 3015 | 66 | 328 | 2687 | 5.0 | 54.5 | 89.1 |
| **Stroke** | 2590 | 539 | 405 | 2185 | 4.3 | 61.3 | 84.4 |
| **Substance abuse / dependence** | 2083 | 725 | 592 | 1491 | 3.5 | 45.7 | 71.6 |
| **TBI** | 720 | 202 | 147 | 573 | 1.2 | 48.6 | 79.6 |
| **Bipolar disorder** | 502 | 240 | 243 | 259 | 0.8 | 42.7 | 51.6 |
| **Dementia** | 293 | 50 | 62 | 231 | 0.5 | 72.3 | 78.8 |
| **Depression** | 213 | 77 | 137 | 76 | 0.4 | 44.7 | 35.7 |
| **Schizophrenia** | 191 | 43 | 130 | 61 | 0.3 | 52.5 | 31.9 |
| **Anxiety** | 182 | 72 | 110 | 72 | 0.3 | 40.5 | 39.6 |
| **Alzheimer's** | 166 | 6 | 3 | 163 | 0.3 | 68.3 | 98.2 |
| **Multiple sclerosis** | 108 | 37 | 58 | 50 | 0.2 | 48.0 | 46.3 |
| **ADHD** | 78 | 37 | 47 | 31 | 0.1 | 25.4 | 39.7 |
| **Parkinson's** | 9 | 3 | 7 | 2 | 0.0 | 55.4 | 22.2 |
| **Total (all sheet recordings)** | 60272 | 14508 | 26775 | 33497 | 100.0 | 50.9 | 55.6 |

## Table 1.10. Where the labels come from

Every status and flag in this notebook comes from human-written clinical records. None is computed from the EEG signal by an algorithm; the only signal-derived features in the repo are our own ( `labram/features/` ). The NEDC sheet ( `metadata_v12` , sheet `TUEG` , one row per session) has review columns ( `Reviewed spell check` , `Reviewed medication` , `Reviewed abnormal due` ), so NEDC staff evidently extracted its fields from each session's EEG report and reviewed them. The documentation on disk does not state this explicitly.

| field | where it comes from | what it describes | level |
|---|---|---|---|
| status, TUAB recordings | TUAB's normal / abnormal folders | the neurologist's reading of this EEG: the report's impression, curated into TUAB | recording |
| status, other TUEG recordings | NEDC sheet `abnormal due to` : filled = abnormal, empty = normal; session not in the sheet = no label | the neurologist's reading of this EEG, i.e. why the report calls it abnormal | session (its recordings share it) |
| diagnosis flags (15) | NEDC sheet, one yes/no column per diagnosis | the patient's clinical background (history, reason for the EEG), not the EEG | session |
| medications | NEDC sheet `Major Medications` | drugs the patient was taking | session |
| FIRDA / OIRDA / TIRDA / GPD / LPD | NEDC sheet | EEG patterns the neurologist saw in the signal: signal findings, read by a human | session |

A clinical EEG report usually holds both kinds of information:

- **the clinical history and medications**, from the referral: the source of the diagnosis flags;
- **the neurologist's description and impression of the recording**: the source of the status.

The two can therefore disagree: a patient with epilepsy in the history can have a normal EEG (Table 1.11).

## Table 1.11. Why a diagnosis flag is not always abnormal

A flag describes the patient. The status describes whether *this* 20–30-minute EEG shows a specific abnormality (slowing, epileptiform discharges, asymmetry). `% abnormal` is the share of the flag's recordings with status abnormal (Table 1.9).

| kind of flag | % abnormal | why |
|---|---|---|
| **EEG findings and dementias**: electrographic seizure, Alzheimer's, dementia | 89, 98, 79 | Electrographic seizures are seen on the EEG itself. Dementia causes persistent diffuse slowing. Alzheimer's comes from only 6 patients. |
| **structural brain injury**: stroke, TBI | 84, 80 | A lesion shows on the EEG when it is large or cortical. Small, deep or old lesions can leave a normal EEG. |
| **episodic conditions**: epilepsy, clinical seizure | 57, 58 | Seizures come and go. Between seizures, a single routine EEG of an epilepsy patient is often normal. "Clinical seizure" also covers events that turned out non-epileptic (e.g. syncope or psychogenic attacks). |
| **substance use**: substance abuse / dependence | 72 | Intoxication, withdrawal and the drugs given for them can slow or otherwise change the EEG. |
| **psychiatric / developmental**: depression, schizophrenia, anxiety, ADHD, bipolar | 36, 32, 40, 40, 52 | These conditions usually do not change the resting EEG. When it is abnormal, medication or another condition is a frequent cause. |
| **other**: multiple sclerosis, Parkinson's | 46, 22 | MS lesions lie mostly in white matter, and the EEG is often normal. Parkinson's has only 9 recordings. |

Two further reasons the match is loose:

- **Different levels.** The flags are per session, while the status comes from TUAB's label for TUAB recordings and from the report otherwise.
- **Noisy labels.** Both are noisy clinical records. TUAB's label and the report finding disagree on about 5 % of shared recordings (Table 1.8).

## Table 1.12. Diagnosis flags per recording

How many of the 15 diagnosis flags each TUEG recording in the clinical sheet carries ( `rs.flags_per_recording` ), with those recordings by status (Eq. (1.2)) and as a share of the sheet's recordings. A count above 1 shows that the flags are multi-hot. The count-0 row shows how many normal and abnormal recordings have no recorded diagnosis.

| diagnosis flags per recording | recordings | normal | abnormal | % of sheet |
|---|---|---|---|---|
| **0** | 46407 | 22102 | 24305 | 77.0 |
| **1** | 8307 | 2799 | 5508 | 13.8 |
| **2** | 3582 | 1404 | 2178 | 5.9 |
| **3** | 1624 | 338 | 1286 | 2.7 |
| **4** | 316 | 99 | 217 | 0.5 |
| **5** | 33 | 33 | 0 | 0.1 |
| **6** | 3 | 0 | 3 | 0.0 |

## Table 1.13. Shared diagnosis flags and status: pairwise counts

Over the 60,272 TUEG recordings in the clinical sheet, entry (row, column) is the number of recordings that carry **both** the row's and the column's item ( `rs.flag_matrix` ). The items are the status ( `normal` , `abnormal` , Eq. (1.2)) and the 15 diagnosis flags, most frequent first. The diagonal is each item's total (Table 1.9). The matrix is symmetric. `normal` × `abnormal` is 0, because a recording has one status. Flag names are shortened: `el.` / `cl. seizure` = electrographic / clinical seizure, `substance` = substance abuse or dependence, `MS` = multiple sclerosis.

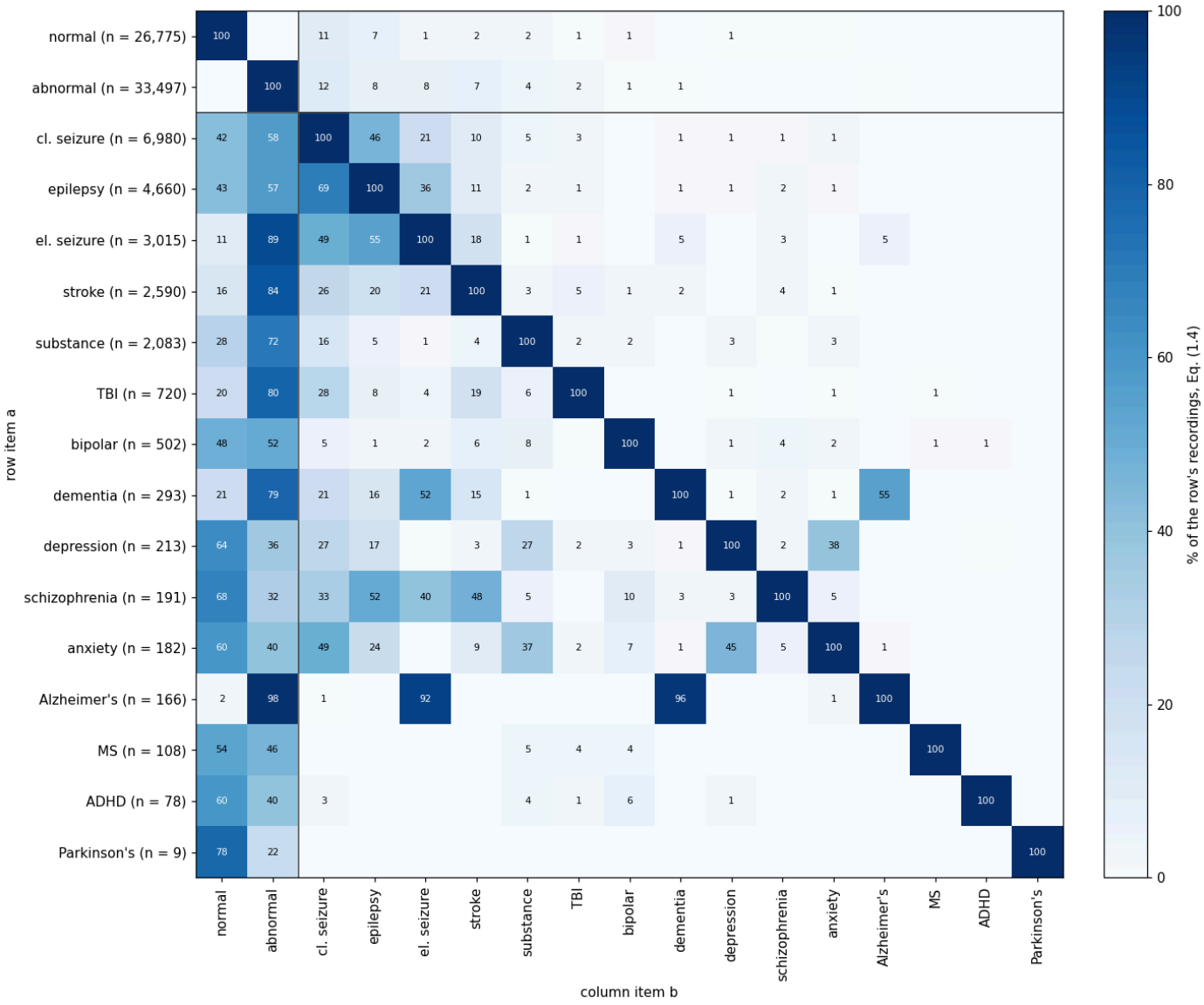
| | normal | abnormal | cl. seizure | epilepsy | el. seizure | stroke | substance | TBI | bipolar | dementia | depression | schizophrenia | anxiety | Alzheimer's | MS | ADHD | Parkinson |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **normal** | 26775 | 0 | 2922 | 1991 | 328 | 405 | 592 | 147 | 243 | 62 | 137 | 130 | 110 | 3 | 58 | 47 | |
| **abnormal** | 0 | 33497 | 4058 | 2669 | 2687 | 2185 | 1491 | 573 | 259 | 231 | 76 | 61 | 72 | 163 | 50 | 31 | |
| **cl. seizure** | 2922 | 4058 | 6980 | 3222 | 1486 | 676 | 324 | 200 | 27 | 61 | 58 | 63 | 90 | 1 | 0 | 2 | |
| **epilepsy** | 1991 | 2669 | 3222 | 4660 | 1660 | 510 | 104 | 60 | 7 | 46 | 37 | 99 | 43 | 0 | 0 | 0 | |
| **el. seizure** | 328 | 2687 | 1486 | 1660 | 3015 | 549 | 18 | 27 | 9 | 153 | 1 | 77 | 0 | 153 | 0 | 0 | |
| **stroke** | 405 | 2185 | 676 | 510 | 549 | 2590 | 81 | 134 | 32 | 43 | 7 | 91 | 17 | 0 | 0 | 0 | |
| **substance** | 592 | 1491 | 324 | 104 | 18 | 81 | 2083 | 41 | 41 | 2 | 57 | 9 | 67 | 0 | 5 | 3 | |
| **TBI** | 147 | 573 | 200 | 60 | 27 | 134 | 41 | 720 | 2 | 1 | 4 | 0 | 4 | 0 | 4 | 1 | |
| **bipolar** | 243 | 259 | 27 | 7 | 9 | 32 | 41 | 2 | 502 | 0 | 7 | 19 | 12 | 0 | 4 | 5 | |
| **dementia** | 62 | 231 | 61 | 46 | 153 | 43 | 2 | 1 | 0 | 293 | 2 | 6 | 2 | 160 | 0 | 0 | |
| **depression** | 137 | 76 | 58 | 37 | 1 | 7 | 57 | 4 | 7 | 2 | 213 | 5 | 82 | 0 | 0 | 1 | |
| **schizophrenia** | 130 | 61 | 63 | 99 | 77 | 91 | 9 | 0 | 19 | 6 | 5 | 191 | 9 | 0 | 0 | 0 | |
| **anxiety** | 110 | 72 | 90 | 43 | 0 | 17 | 67 | 4 | 12 | 2 | 82 | 9 | 182 | 2 | 0 | 0 | |
| **Alzheimer's** | 3 | 163 | 1 | 0 | 153 | 0 | 0 | 0 | 0 | 160 | 0 | 0 | 2 | 166 | 0 | 0 | |
| **MS** | 58 | 50 | 0 | 0 | 0 | 0 | 5 | 4 | 4 | 0 | 0 | 0 | 0 | 0 | 108 | 0 | |
| **ADHD** | 47 | 31 | 2 | 0 | 0 | 0 | 3 | 1 | 5 | 0 | 1 | 0 | 0 | 0 | 0 | 78 | |
| **Parkinson's** | 7 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | |

## Figure 1.3. Shared diagnosis flags and status: share of the row

The counts of Table 1.13 as shares of the row's total:

$$s(a, b) = 100 \, \frac{n(a \wedge b)}{n(a)} \tag{1.4}$$

where $n(a \wedge b)$ is the number of recordings carrying both $a$ and $b$. Read along a row: for example, the `epilepsy` row gives the % of epilepsy recordings that are normal, abnormal, or also carry each other flag. The matrix is not symmetric, because rows are normalized by their own totals. Cells are annotated with the rounded %. The diagonal is 100 %.




## Figures 1.4–1.5. Age per diagnosis flag

True age (years) of the TUEG recordings in the clinical sheet that carry each flag. Each panel is a histogram in 5-year bins, stacked by status: green normal, red abnormal. Its title gives the number of recordings and patients and the median age.

Counts are **recordings**, and the title gives the patients. A spike usually means one or a few patients recorded many times: electrographic seizure has 46 recordings per patient on average. Recordings with an unknown age are left out, so `n` can be slightly below Table 1.9.

- Figure 1.4: the neurological flags, panels (a)–(i), and (j) the recordings with **no** diagnosis flag as a reference.
- Figure 1.5: the psychiatric and other flags, panels (a)–(f).

Figure 1.4. Age per diagnosis flag: neurological flags

As described above; panels (a)–(i) one neurological flag each, (j) recordings with no flag.

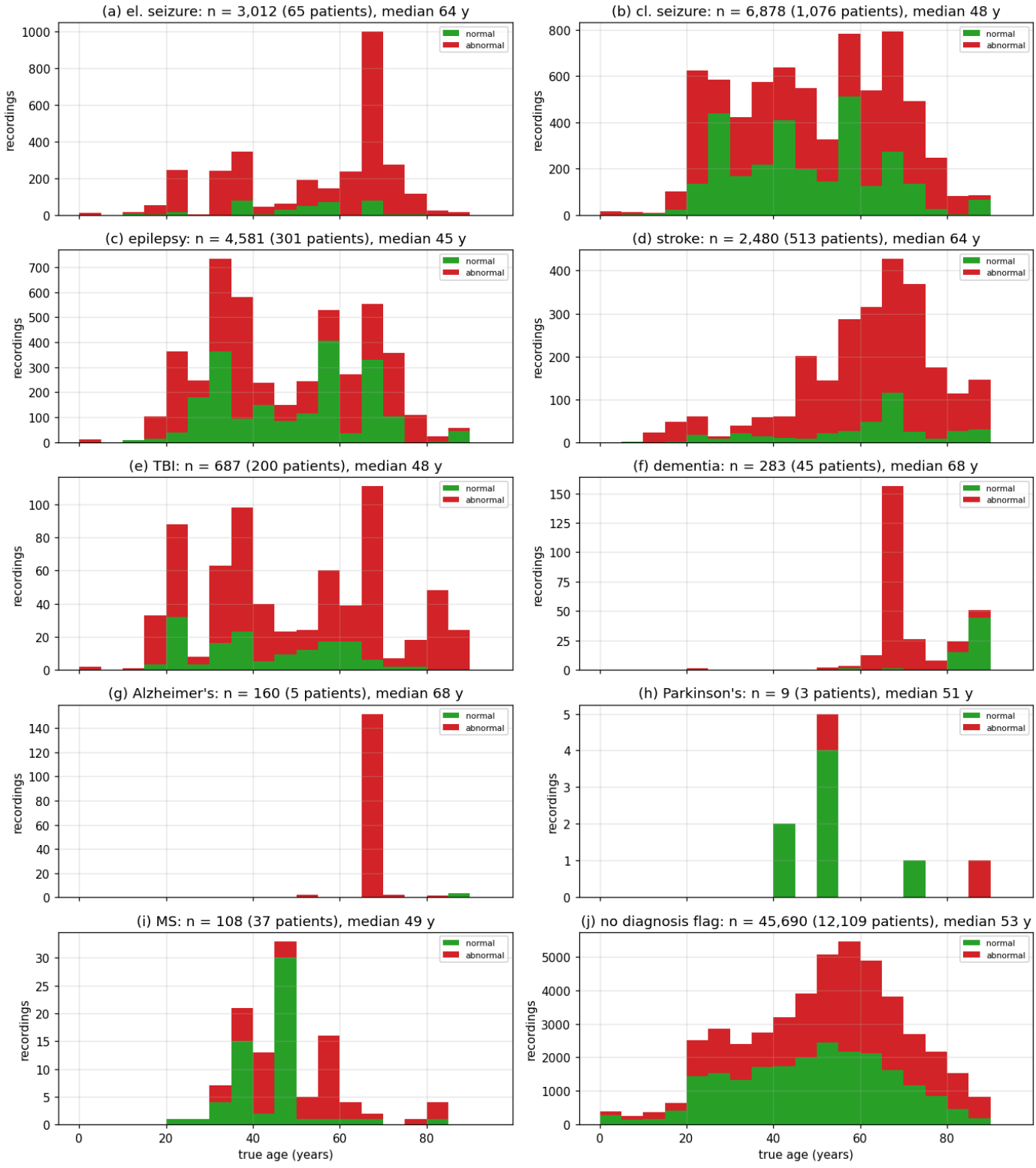
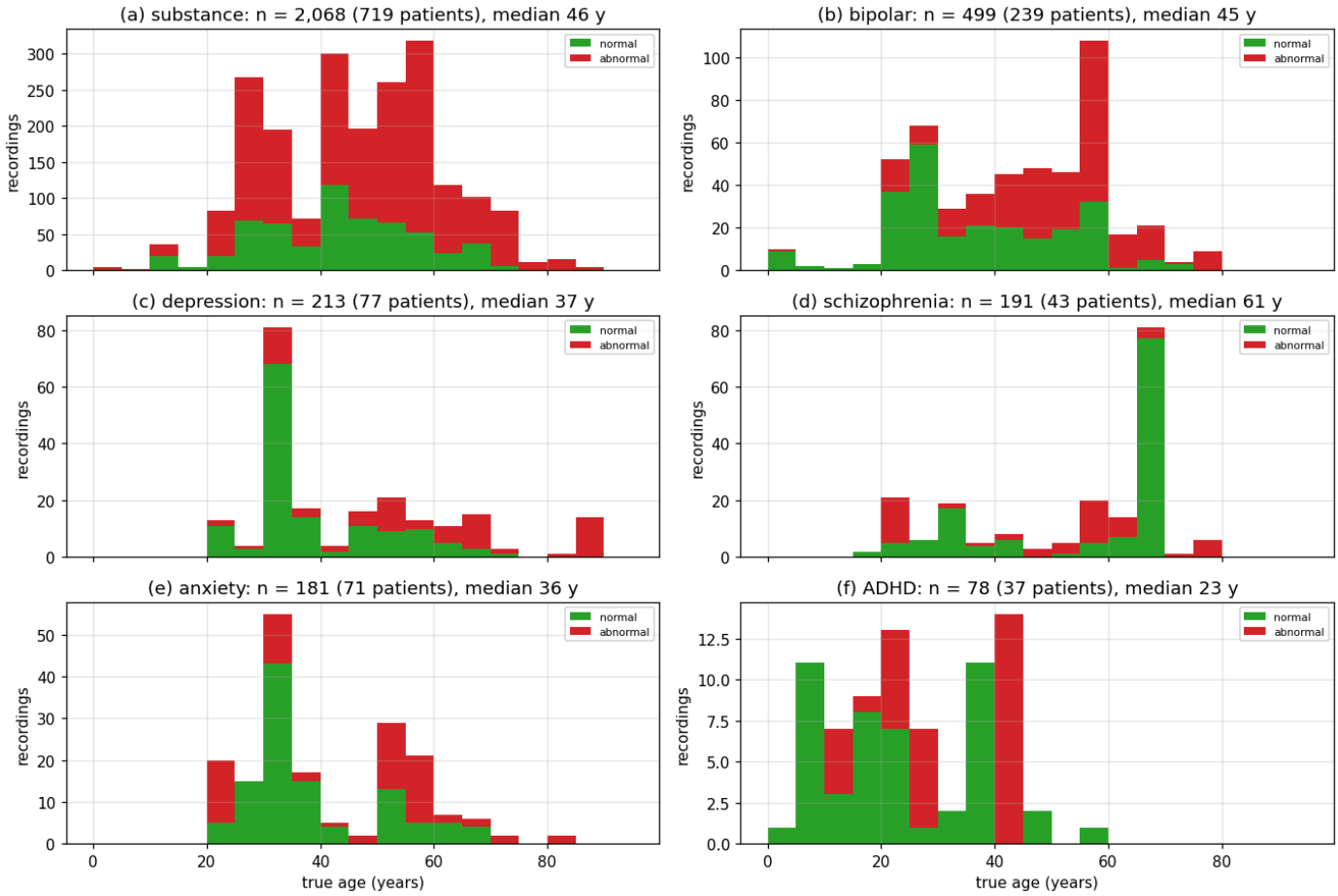



Figure 1.5. Age per diagnosis flag: psychiatric and other flags

As described above; panels (a)–(f) one flag each.




**Reading Table 1.13 and Figures 1.3–1.5.**

- **Seizure flags cluster.** 69 % of epilepsy recordings are also flagged clinical seizure, and 55 % of electrographic-seizure recordings are also flagged epilepsy. The three flags describe one patient group from different angles: the diagnosis, the reported event and the EEG finding.
- **Status follows the kind of flag** (the `normal` / `abnormal` columns of Figure 1.3, as in Table 1.9). Electrographic seizure, Alzheimer's and stroke are mostly abnormal. Psychiatric flags are mostly normal.
- **Small flags come from few patients.** Alzheimer's has 166 recordings from 6 patients. Almost all of them also carry dementia (160) and electrographic seizure (153), so its co-occurrences describe those 6 patients rather than Alzheimer's disease in general (Table 1.9 gives patients per flag).
- **Psychiatric flags overlap each other,** and in this hospital population often co-occur with neurological flags too. 38 % of depression recordings also carry anxiety. Schizophrenia recordings carry epilepsy in 52 % and stroke in 48 % of cases, and anxiety recordings carry clinical seizure in 49 %. Few patients carry each psychiatric flag, so these shares rest on few patients.
- **Age differs strongly by flag** (Figures 1.4–1.5). Dementia and Alzheimer's sit at 65–90 years, and stroke and electrographic seizure peak in the 60s. ADHD sits mostly under 40

(median 23). Epilepsy, clinical seizure, bipolar and substance use are spread over adulthood.
- **Abnormal versus normal age within a flag.** For 10 of the 15 flags the abnormal recordings are older (median), most clearly for depression (55 vs 33), anxiety (50 vs 31) and bipolar (52 vs 32). Epilepsy is the opposite (37 vs 50), and so is schizophrenia (58 vs 66). Stroke shows no difference. Dementia and Alzheimer's have their normal recordings at the oldest ages, from a handful of patients.

**Reading Tables 1.6–1.13 and Figure 1.2.**

- **Size:** TUEG holds about 70.8k recordings (27k hours) of about 15k patients. The cohorts use 25k–29k of them, after the age, length, channel and per-patient rules of Table 1.3.
- **Recordings per patient:** most patients have a single recording, but a few long-monitored patients have hundreds. This is why the cohorts cap train/val recordings per patient and every split is by patient.
- **Labels:** only 4 % of TUEG is in TUAB. The status of nearly all other recordings comes from the NEDC report: about half are abnormal, and 14 % have no label (Table 1.8).
- **Age:** abnormal recordings are older than normal ones by about 5 years in mean and 7 in median (Table 1.8, Figure 1.2(b)). A model trained on mixed statuses can therefore partly read age from pathology, which is one reason Section 5 reports normal and abnormal test recordings separately.

# 2. Experiments

**Phase 1 (TUAB only)** starts from scenario **D**, the default `finetune_tuab_age.json` (Table 2.1 lists how it differs from LaBraM's own fine-tuning recipe):

- LaBraM-base with a **common average reference (CAR)** on the input (Eq. (3.1));
- Huber loss on the z-scored age;
- lr $10^{-4}$, 15 epochs.

**CAR** re-references the EEG. At every time sample it subtracts the mean over all channels from each channel, so every channel is measured against the average of all electrodes instead of the recording's own reference. This removes what all electrodes share: the activity of the reference electrode(s) and noise common to every channel. It also puts the two TUH montages on one reference, average reference ( `tcp_ar` , about 80 % of the recordings) and linked ears ( `tcp_le` ). LaBraM as published feeds the signal on each recording's own reference. CAR is the only part of the LaBraM++ input block D uses: there is no per-patch z-scoring (Table 2.1). In Table 5.2, row P1 tests it.

**E2** shortens it to 4 epochs / 1 warmup epoch, and **M1** = E2 + mixup $\alpha = 0.4$. Every **phase-2** run is M1 plus the change named in its row ( `labram/eval/research_summary.py::M1` ).
**Phase 3** cross-validates the three strongest phase-2 configurations.

**Loss.** Every run trains on a loss from the family of Eqs. (2.1)–(2.11) below, and Tables 2.2, 2.5 and 2.7 name each run's loss in their `loss` column.

**ClearML.** Tables 2.2, 2.5, 2.6, 2.7 and 2.9 link every run to its ClearML experiment; the link text is the experiment's name in ClearML (`docs/data/brain_age_clearml_runs.csv`, written from the ClearML server by `labram/eval/research_summary.py::fetch_clearml_runs`). A CV run links to the ClearML project that holds its five fold experiments.

## Table 2.1. Base configuration D versus the LaBraM fine-tuning recipe

Every configuration field on which LaBraM's TUAB fine-tuning recipe as published and the base configuration D differ:

- `LaBraM paper`: `labram/configs/defaults/finetune_tuab_paper.json`, the normal/abnormal fine-tune of LaBraM-base from the paper and the README;
- `D`: `labram/configs/defaults/finetune_tuab_age.json` as is, without overrides.

Both are rebuilt by `rs.resolve_config` and compared by `labram/eval/research_summary.py::config_diff`. Excluded:

- bookkeeping fields (paths, logging, infrastructure);
- classification-only fields a regression run never reads (`INERT_FOR_REGRESSION`, e.g. label smoothing);
- fields of a section that is off in both.

`group` is the part of the pipeline and `meaning` what the change does. Every other field is the paper's: LaBraM-base from `labram-base.pth`, batch 64, AdamW with weight decay 0.05, layer decay 0.65, drop path 0.1, mean pooling, cosine schedule. Changes that live in the data rather than the configuration (mains-interference removal, splits, labels) are in Table 4.2; how the best configuration differs from the paper's recipe is Table 4.1.

| group | field | LaBraM paper | D | meaning |
|---|---|---|---|---|
| task | data.dataset | TUAB | TUAB_AGE | age regression on TUAB (age from the EDF header) instead of normal/abnormal |
| | model.task | classification | regression | scalar regression head, Huber loss on the z-scored age (Eqs. (3.2), (3.4)) |
| data | data.data_format | pickle | npy | storage only: one float32 npy per recording, windows bit-identical to the pickles |
| input | labram_plus.enabled | False | True | LaBraM++ input block on: common average reference, CAR (Eq. (3.1)) |
| | labram_plus.z_score_patches | True | False | LaBraM++ block with CAR only, no per-patch z-scoring |
| windows | data.eval_minutes | 0 | 5 | score each recording on its first 5 minutes after the trim |
| | data.trim_end_sec | 0 | 60 | drop the last 60 s of every recording |
| | data.trim_start_sec | 0 | 60 | drop the first 60 s of every recording |
| scoring | evaluation.agg_windows | none | mean | one prediction per recording = mean of its windows (Eq. (3.7)) |
| optimizer | optimizer.lr | 0.0005 | 0.0001 | peak lr 1e-4 instead of 5e-4 |
| schedule | optimizer.warmup_epochs | 5 | 2 | 2 warmup epochs instead of 5 |
| | trainer.epochs | 50 | 15 | 15 epochs instead of 50 |

**Loss functions.** The loss is what the optimizer minimizes on every training batch. LaBraM as published fine-tunes TUAB as a binary normal/abnormal classifier. The head emits one logit $s$

per window with label $c \in \{0, 1\}$, and the loss is the binary cross-entropy (`nn.BCEWithLogitsLoss`, `labram/losses/classification.py`):

$$\ell_{\mathrm{BCE}}(s, c) = -\big[c \log \sigma(s) + (1 - c) \log(1 - \sigma(s))\big], \qquad \sigma(s) = \frac{1}{1 + e^{-s}} \tag{2.1}$$

For brain age the head instead emits a scalar $\hat{z}$ per 10 s window. The target is the z-scored age $z = (y - \mu)/\sigma$ (Eq. (3.2), with $\mu$ and $\sigma$ from the training windows). The batch loss averages a per-window loss $\ell$ of the residual over the $B$ windows of the batch (`labram/losses/regression.py::build_regression_criterion`):

$$\mathcal{L} = \frac{1}{B} \sum_{b=1}^{B} \ell(r_b), \qquad r_b = \hat{z}_b - z_b \tag{2.2}$$

**Per-window loss.** The default for D and every later run unless stated otherwise is the **Huber** loss with $\delta = 1$. It is quadratic for small residuals and linear for large ones, so a few recordings with a wrong or extreme age do not dominate the gradient. DX2 uses $\delta = 2$:

$$\ell_\delta(r) = \begin{cases} \frac{1}{2} r^2 & |r| \le \delta \\ \delta\,(|r| - \frac{1}{2}\delta) & \text{otherwise} \end{cases} \tag{2.3}$$

D2 uses the **L1** loss, the absolute error, which directly matches the MAE of Eq. (0.1):

$$\ell_1(r) = |r| \tag{2.4}$$

**Mixup (M1 and every phase-2 run).** Each window $b$ of a batch is blended with a partner window $\pi(b)$ drawn from the same batch, usually from another recording with a different age. $\pi$ permutes the **samples of the batch** (`torch.randperm` over the batch dimension), not the channels. The two windows are averaged point by point, channel with the same channel and time sample with the same time sample, and their targets with the same weight. For example, $\lambda = 0.7$ blends a 30-year-old's window with a 70-year-old's into target $0.7 \cdot 30 + 0.3 \cdot 70 = 42$. One $\lambda \sim \mathrm{Beta}(0.4, 0.4)$ is drawn per batch, usually near 0 or 1, so one window dominates. The loss of Eq. (2.2) is computed on the blended targets (`labram/train/mixup.py::mixup_batch`). A channel missing from either partner (T1/T2 in the `channel_drop` runs) is zeroed rather than blended:

$$\bar{x}_b = \lambda x_b + (1 - \lambda)\, x_{\pi(b)}, \qquad \bar{z}_b = \lambda z_b + (1 - \lambda)\, z_{\pi(b)} \tag{2.5}$$

Plain mixup draws the partner $\pi(b)$ uniformly at random. **C-Mixup** (M1c) prefers partners of similar age, so the blended target stays plausible. With $\sigma_m = 5$ years:

$$P\big(\pi(b) = j\big) \propto \exp\Big(-\frac{(y_b - y_j)^2}{2\sigma_m^2}\Big), \quad j \neq b \tag{2.6}$$

**Soft labels (M4).** The head emits a softmax $p_k$ over age bins with centres $c_k$ = 1, 2, ..., 89 years. The target is a Gaussian $q_k$ around the true age with $\sigma_s = 2.5$ years, the loss is the KL

divergence, and the prediction is the expected age ( `SoftLabelRegressionLoss` ). Centres and $\sigma_s$ are z-scored like $z$:

$$q_k = \frac{\exp\!\big(-(c_k - z)^2 / 2\sigma_s^2\big)}{\sum_{k'} \exp\!\big(-(c_{k'} - z)^2 / 2\sigma_s^2\big)}, \qquad \ell_{\mathrm{KL}} = \sum_k q_k \log \frac{q_k}{p_k}, \qquad \hat{z} = \sum_k p_k \, c_k \tag{2.7}$$

**Age-balanced loss (B1, B2).** Rare ages (children, the very old) are up-weighted. $\rho_k$ is the density of training windows in 1-year age bin $k$, smoothed by a Gaussian with $\sigma = 2$ years. The weight is $\rho^{-1/2}$ (B1) or $\rho^{-1}$ (B2), capped at 10× the smallest weight and normalized to mean 1 over the training windows ( `age_balance_weights` , `BalancedRegressionLoss` ):

$$\mathcal{L} = \frac{1}{B} \sum_b w_{k(y_b)} \, \ell_\delta(r_b), \qquad w_k \propto \rho_k^{-1/2} \text{ or } \rho_k^{-1} \tag{2.8}$$

**Codebook regularization: the spectral reconstruction and quantization terms (DX1, DX4, DX4b only).** LaBraM's tokenizer (VQNSP) was pre-trained to encode each 1 s patch into a codeword and to decode from it the patch's Fourier amplitude and phase. These runs reuse its quantizer and decoder, frozen, on top of the fine-tuned encoder. Extra loss terms then penalize the encoder if its patch tokens drift away from what the frozen decoder can read back ( `model.codebook_reg` , `labram/losses/codebook_regularized.py` ). The age head still reads the mean patch token, as in D.

For each patch token $n$ (one channel × one 1 s patch of 200 samples):

- $\tilde{A}_{n,k}$: the std-normalized FFT amplitude of the patch at frequency bin $k$ (bin $k$ = $k$ Hz);
- $\hat{A}_{n,k}$: the decoder's reconstruction of it;
- $h_n$: the encoder token;
- $e_n$: its nearest codeword; sg is stop-gradient.

The terms are averaged over all tokens (and bins or feature dimensions) of the batch:

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_\delta + w_A \, \mathcal{L}_{\mathrm{amp}} + w_P \, \mathcal{L}_{\mathrm{phase}} + w_Q \, \mathcal{L}_{\mathrm{q}}, \\ \mathcal{L}_{\mathrm{amp}} &= \mathrm{mean}_{n,\, k \le K} \left(\hat{A}_{n,k} - \tilde{A}_{n,k}\right)^2, \\ \mathcal{L}_{\mathrm{q}} &= \beta_q \, \mathrm{mean}_n \left(\mathrm{sg}(e_n) - h_n\right)^2, \qquad \beta_q = 1 \end{aligned} \tag{2.9}$$

$\mathcal{L}_{\mathrm{phase}}$ is the same MSE on the std-normalized FFT phase. $K$ is the number of frequency bins kept: `loss.freq_fraction` × 200. The weights each run used, from its ClearML configuration ( `model.codebook_reg.*_weight` ):

- **DX1:** $w_A = 1$, $w_P = 0$, $w_Q = 0$, so amplitude reconstruction only. `freq_fraction` = 0.5 keeps $K = 100$ bins, 0–99 Hz. That is the whole non-redundant half of the spectrum: for a real signal, the upper 100 bins mirror the lower ones.

- **DX4:** $w_A = 0$, $w_P = 0$, $w_Q = 1$, so the quantization (commitment) term only. No reconstruction.
- **DX4b:** as DX4 with $w_Q = 50$.
- **Phase** ($w_P$) is 0 in all three. The phase reconstruction loss was never applied in any fine-tune of this study.
- **Every other run** has `model.codebook_reg.enabled=false`. The model then has no quantizer or decoder, and all three terms are absent. The `loss.amplitude_weight` / `phase_weight` / `embedding_weight` fields that read 1.0 in every run's ClearML configuration belong to the VQNSP tokenizer's own training and are never read by a fine-tune.

None of the three helped. Compared with D, the test MAE changes by +0.24 (DX1), +0.14 (DX4) and +0.36 (DX4b), all worse but all within seed noise, τ = 0.38 (Eq. (5.2)). Table 5.2 tests DX1 and DX4 in rows P17 and P18.

**Multi-task (MT1).** A second output predicts the normal/abnormal label $c$ with the BCE of Eq. (2.1), weighted 0.5 (`MultiTaskAgeCriterion`). Selection still uses the age MAE:

$$\mathcal{L} = \mathcal{L}_\delta + 0.5\,\frac{1}{B}\sum_b \ell_{\mathrm{BCE}}(s_b, c_b) \tag{2.10}$$

**Channel-dropout consistency (Exp4aug, Exp4aug15, cvaug).** With probability 0.2 per batch, 1–2 of the T1/T2 channels are dropped from every sample. The age loss is computed on that dropped pass. The feature $f$ the head reads should not change: it is pulled towards the feature $f^{\mathrm{full}}$ of a frozen all-channels pass, with $\lambda = 0.1$, over the samples $A$ that lost a channel and the feature dimensions $d = 1..D$ (`labram/train/channel_drop.py`):

$$\mathcal{L} = \mathcal{L}_\delta^{\mathrm{drop}} + \lambda\,\frac{1}{|A|D}\sum_{b\in A}\sum_{d=1}^{D}\left(f_{b,d}^{\mathrm{full}} - f_{b,d}^{\mathrm{drop}}\right)^2 \tag{2.11}$$

Here $\mathcal{L}_\delta$ is Eq. (2.2) with the Huber loss of Eq. (2.3). Changes that do not touch the loss (EMA evaluation, layer decay, dropout, heads, input conditioning) keep Huber $\delta = 1$, with mixup wherever the run's `loss` column says so.

## Table 2.2. Phase 1: TUAB-only runs

The experiment registry's master table (`docs/age_overfitting_experiments.md`, Table 12.1).

Columns:

- `ClearML`: the run's ClearML experiment, linked by its name.
- `family` and `change`: the question the run belongs to, and what it changes relative to D unless it names another run. Abbreviations:
  - CAR common average reference; ep epochs; lr learning rate;

    - ld layer decay; dp drop path; drop dropout; wd weight decay; clip gradient clipping at norm 1;
    - EMA evaluation with an exponential moving average of the weights;
    - LoRA r16 low-rank adapters of rank 16 instead of full fine-tuning;
    - CLS the class token; MLP a two-layer head.
- `loss`: the run's training loss, read from its ClearML configuration (Eqs. (2.2)–(2.11)).
- `train recs`: the number of training recordings, from the run's recorded `data_split.json` (`docs/data/brain_age_run_splits.csv`). It is the same 2,145 for every phase-1 run (the check below the table); only the windows per recording differ (A: whole recordings; C: 30 s windows).
- `val` / `test`: MAE (Eq. (0.1), years) at the epoch with the lowest validation MAE. Phase 1 is **TUAB only**, in training and evaluation:
    - `val`: TUAB's validation split, 556 recordings (291 normal, 265 abnormal);
    - `test`: TUAB's official evaluation set, 274 recordings.
- `Norm` / `Ab`: the same TUAB test MAE restricted to its 149 normal and 125 abnormal recordings.
- `β`: the bias slope of Eq. (0.5) on the 274 TUAB test recordings. Every model regresses towards the mean age. The fine-tuned runs sit between −0.46 and −0.33, and the from-scratch runs S1–S3 at −0.49 to −0.52. The collapsed runs (C2, C2-head, C2-last, DX3, F2; test MAE above 13) reach −0.8 to −1.0, meaning they predict almost a constant age.
- `e*/E`: the selected epoch $e^\star$ and the number of epochs $E$.

**Same data split in every run.** Every run records the recordings of its train, val and test splits (the `data_split` artifact of its ClearML experiment), saved here as SHA-1 digests of the sorted recording lists (`docs/data/brain_age_run_splits.csv`, `rs.split_fingerprint`, regenerated by `rs.fetch_run_splits`). The check below the table compares them: all phase-1 runs train, validate and test on identical recordings, with no patient in two splits. Only the number of windows differs, where the run changes the windows: A scores whole recordings, and C uses 30 s windows. Phase 2's `Control` uses the same split.

**Why some `Norm` / `Ab` are missing ("–").** The per-cohort test metrics were added to the trainer on 2026-10-07 06:47 UTC (commit `62b159d`). 16 of the 21 runs without them started before that. The other five (E2, DX3, E1, E1c, E2c) are SageMaker jobs started 07:06–08:30 the same day whose job code did not include the change yet. All 21 logged only the overall val/test MAE. They saved no per-recording predictions, so the split by status cannot be recomputed from their logs; it needs a new inference pass of their checkpoints. Their `test` column is on the same 274 recordings as everyone else's. Table 2.4 fills these gaps: it re-scores every run's saved checkpoint, so `Norm` / `Ab` exist there for all of them.

The pre-study baseline (LaBraM fine-tuned with the original recipe adapted to regression) scored val 8.31 / test 9.04 (`docs/age_training_scenario_D.md`, Section 1).

| run | ClearML | family | change | loss | train recs | val | test | Norm | Ab | β | e*/E |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **A** | age_A_ep15_lr1e-4_fullrec_20261005_043322_643 | Evaluation protocol | B with whole-recording val/test | Huber δ 1 (2.3) | 2145 | 8.29 | 9.03 | – | – | -0.46 | 2/12 |
| **B** | age_B_eval5min_20261005_064356_435 | Evaluation protocol | CAR off, 15 ep, lr 1e-4, 5-min eval (pre-D baseline) | Huber δ 1 (2.3) | 2145 | 8.40 | 8.91 | – | – | -0.43 | 2/6 |
| **D** | age_D_eval5min_car_20261005_013250_973 | Baseline | B + CAR (finetune_tuab_age.json) | Huber δ 1 (2.3) | 2145 | 7.77 | 8.30 | – | – | -0.40 | 1/15 |
| **D-npy** | age_D_eval5min_car_npy_file_20261006_185443_758 | Baseline (repro) | D on the npy format | Huber δ 1 (2.3) | 2145 | 7.77 | 8.30 | – | – | -0.40 | 1/15 |
| **D-cohorts** | age_D_npy_cohorts_20261007_064710_242 | Baseline (repro) | D, local, cohort metrics | Huber δ 1 (2.3) | 2145 | 7.77 | 8.31 | 6.57 | 10.38 | -0.40 | 1/15 |
| **E2** | age_E2_npy_ep4_warm1_20261007_070635_244 | Schedule | D, 4 ep / 1 warmup | Huber δ 1 (2.3) | 2145 | 7.68 | 8.25 | – | – | -0.37 | 3/4 |
| **L1** | age_L1_ep12_mixup_ema_20261009_095540_331 | Schedule | M12 on 12 ep / 2 warmup | Huber δ 1 + mixup (2.3, 2.5) | 2145 | 7.51 | 8.47 | 6.65 | 10.63 | -0.42 | 3/12 |
| **C2** | age_C2_eval5min_regularized_20261005_012652_087 | Classic regularizers | B + ld 0.5, dp 0.2, drop 0.1, wd 0.1, clip 1 | Huber δ 1 (2.3) | 2145 | 12.47 | 13.52 | – | – | -0.81 | 14/15 |
| **DX3** | age_DX3_npy_regularized_20261007_070603_726 | Classic regularizers | D + C2 bundle | Huber δ 1 (2.3) | 2145 | 12.83 | 14.90 | – | – | -1.00 | 0/15 |
| **F1** | age_F1_short_ld05_20261008_215552_344 | Classic regularizers | E2 + layer decay 0.5 | Huber δ 1 (2.3) | 2145 | 7.74 | 8.36 | 6.42 | 10.68 | -0.36 | 3/4 |
| **F2** | age_F2_short_drop01_20261008_230105_724 | Classic regularizers | E2 + dropout 0.1 | Huber δ 1 (2.3) | 2145 | 13.25 | 15.17 | 13.37 | 17.32 | -0.97 | 0/4 |
| **F3** | age_F3_short_dp02_20261008_232500_075 | Classic regularizers | E2 + drop path 0.2 | Huber δ 1 (2.3) | 2145 | 7.70 | 8.41 | 6.31 | 10.91 | -0.35 | 3/4 |
| **F4** | age_F4_short_wd01_20261008_232504_307 | Classic regularizers | E2 + weight decay 0.1 | Huber δ 1 (2.3) | 2145 | 7.66 | 8.24 | 6.49 | 10.32 | -0.37 | 3/4 |
| **G1** | age_G1_short_init1_20261008_230259_560 | Head init | E2 + init_scale 1.0 | Huber δ 1 (2.3) | 2145 | 7.76 | 8.38 | 6.64 | 10.45 | -0.37 | 3/4 |
| **H1** | age_H1_short_ld075_20261009_003609_895 | Layer decay | E2 + layer decay 0.75 | Huber δ 1 (2.3) | 2145 | 7.71 | 8.67 | 6.64 | 11.09 | -0.40 | 1/4 |
| **H2** | age_H2_short_ld085_20261009_003617_662 | Layer decay | E2 + layer decay 0.85 | Huber δ 1 (2.3) | 2145 | 7.86 | 8.54 | 6.74 | 10.69 | -0.42 | 3/4 |
| **C2-head** | age_C2_eval5min_regularized_headonly_20261004_220639_025 | Partial fine-tuning | C2, head only | Huber δ 1 (2.3) | 2145 | 12.77 | 14.78 | – | – | -0.99 | 5/8 |
| **C2-last** | age_C2_eval5min_regularized_lastblock_head_20261005_000707_645 | Partial fine-tuning | C2, last block + head | Huber δ 1 (2.3) | 2145 | 12.50 | 13.64 | – | – | -0.87 | 6/15 |
| **M3** | age_M3_short_lora16_20261008_215551_826 | Partial fine-tuning | E2 + LoRA r16, lr 1e-3, ld 1.0 | Huber δ 1 (2.3) | 2145 | 7.68 | 9.24 | 7.41 | 11.42 | -0.44 | 3/4 |
| **M1** | age_M1_short_mixup04_20261007_231716_154 | Mixup / EMA | E2 + mixup 0.4 | Huber δ 1 + mixup (2.3, 2.5) | 2145 | 7.46 | 8.45 | 6.60 | 10.65 | -0.41 | 3/4 |
| **M1c** | age_M1c_short_cmixup04_s5_20261008_001050_123 | Mixup / EMA | E2 + C-Mixup 0.4, σ 5 y | Huber δ 1 + C-Mixup σ 5 y (2.3, 2.5, 2.6) | 2145 | 7.54 | 8.23 | 6.50 | 10.29 | -0.39 | 3/4 |
| **M2** | age_M2_short_ema_20261008_212509_991 | Mixup / EMA | E2 + EMA eval (0.9995) | Huber δ 1 (2.3) | 2145 | 7.58 | 8.17 | 6.35 | 10.35 | -0.36 | 2/4 |
| **M12** | age_M12_short_mixup_ema_20261008_204214_769 | Mixup / EMA | E2 + mixup 0.4 + EMA eval | Huber δ 1 + mixup (2.3, 2.5) | 2145 | 7.48 | 8.45 | 6.57 | 10.69 | -0.42 | 3/4 |
| **D2** | age_D2_eval5min_l1_20261005_000636_646 | Loss | B + L1 loss | L1 (2.4) | 2145 | 8.36 | 9.32 | – | – | -0.46 | 6/15 |
| **DX2** | age_DX2_huber_delta2_20261005_185121_984 | Loss | D + Huber δ 2 | Huber δ 2 (2.3) | 2145 | 7.79 | 8.41 | – | – | -0.38 | 3/15 |
| **M4** | age_M4_short_softlabel_20261008_215551_240 | Loss | E2 + soft labels σ 2.5 y | soft-label KL, σ 2.5 y (2.7) | 2145 | 7.67 | 8.59 | 6.92 | 10.57 | -0.40 | 3/4 |
| **B1** | age_B1_short_balance_sqrtinv_20261009_124653_371 | Loss | E2 + age-balanced, sqrt-inverse | Huber δ 1, age weights ρ^-1/2 (2.8) | 2145 | 7.83 | 8.48 | 6.66 | 10.63 | -0.37 | 2/4 |
| **B2** | age_B2_short_balance_inv_20261009_134321_186 | Loss | E2 + age-balanced, inverse | Huber δ 1, age weights ρ^-1 (2.8) | 2145 | 7.96 | 8.57 | 6.70 | 10.80 | -0.33 | 1/4 |
| **DX1** | age_DX1_amp_freq05_20261006_022243_393 | Codebook regularization | D + amplitude reconstruction (½ spectrum) | Huber δ 1 + amplitude reconstruction, w 1 (2.9) | 2145 | 7.72 | 8.54 | – | – | -0.39 | 4/15 |
| **DX4** | age_DX4_quant_20261005_160655_990 | Codebook regularization | D + quantization term, weight 1 | Huber δ 1 + quantization, w 1 (2.9) | 2145 | 7.67 | 8.44 | – | – | -0.39 | 3/15 |
| **DX4b** | age_DX4b_quant_w50_20261005_215817_973 | Codebook regularization | D + quantization term, weight 50 | Huber δ 1 + quantization, w 50 (2.9) | 2145 | 7.78 | 8.66 | – | – | -0.37 | 4/15 |
| **C** | age_C_eval5min_win30s_20261004_235942_658 | Augmentation / input | B + 30-s windows | Huber δ 1 (2.3) | 2145 | 8.24 | 9.35 | – | – | -0.42 | 8/15 |
| **D-clip1** | age_D_npy_clip1_20261007_003433_521 | Augmentation / input | D + clip 1.0 | Huber δ 1 (2.3) | 2145 | 7.70 | 8.36 | – | – | -0.41 | 1/15 |
| **E1** | age_E1_npy_random_crop_20261007_070629_845 | Augmentation / input | D + random crop | Huber δ 1 (2.3) | 2145 | 7.82 | 8.56 | – | – | -0.40 | 3/15 |

| run | ClearML | family | change | loss | train recs | val | test | Norm | Ab | β | e*/E |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **E1c** | age_E1c_npy_random_crop_clip1_20261007_070930_530 | Augmentation / input | D + random crop + clip 1.0 | Huber δ 1 (2.3) | 2145 | 7.72 | 8.13 | – | – | -0.38 | 1/15 |
| **E2c** | age_E2c_npy_ep4_warm1_clip1_20261007_083022_212 | Augmentation / input | E2 + clip 1.0 | Huber δ 1 (2.3) | 2145 | 7.63 | 8.24 | – | – | -0.38 | 3/4 |
| **D-cls** | age_D_npy_cls_head_20261007_003458_637 | Head | D, CLS token instead of mean pooling | Huber δ 1 (2.3) | 2145 | 7.90 | 8.24 | – | – | -0.38 | 2/15 |
| **D-clsmean** | age_D_npy_cls_mean_head_20261007_003450_736 | Head | D, [mean patch; CLS] concat | Huber δ 1 (2.3) | 2145 | 7.80 | 8.35 | – | – | -0.41 | 1/15 |
| **N2L** | age_N2L_ep12_mixup04_clip1_attnhead_20261009_232420_478 | Head | D + mixup 0.4 + clip 1 + attention head, 12 ep / 4 warmup | Huber δ 1 + mixup (2.3, 2.5) | 2145 | 7.63 | 8.49 | 6.68 | 10.64 | -0.43 | 4/12 |
| **MT1** | age_MT1_short_mixup04_multitask_20261010_234501_492 | Head (multi-task) | M1 + age/abnormal multi-task head (seed 0) | Huber δ 1 + mixup + 0.5 BCE abnormal (2.3, 2.5, 2.10) | 2145 | 7.54 | 8.38 | 6.73 | 10.35 | -0.40 | 3/4 |
| **MT1-seed1** | age_MT1_short_mixup04_multitask_seed1_local_20261011_003814_186 | Head (multi-task) | MT1, seed 1 | Huber δ 1 + mixup + 0.5 BCE abnormal (2.3, 2.5, 2.10) | 2145 | 7.62 | 8.54 | 6.75 | 10.69 | -0.41 | 3/4 |
| **S1** | age_S1_scratch_ep15_20261010_084032_716 | From scratch | D, no pretrained weights, ld 1.0 | Huber δ 1 (2.3) | 2145 | 8.79 | 9.42 | 7.75 | 11.42 | -0.49 | 14/15 |
| **S2** | age_S2_scratch_d6_mlp_ep15_20261010_092116_377 | From scratch | scratch, 6 blocks + MLP head, tuned (seed 0) | Huber δ 1 + mixup (2.3, 2.5) | 2145 | 8.53 | 9.30 | 7.48 | 11.48 | -0.50 | 12/15 |
| **S2-seed1** | age_S2_scratch_d6_mlp_ep15_seed1_local_20261010_142248_698 | From scratch | S2, seed 1 | Huber δ 1 + mixup (2.3, 2.5) | 2145 | 8.58 | 9.44 | 7.84 | 11.35 | -0.50 | 14/15 |
| **S3** | age_S3_scratch_d6_mlp_aug_ep15_local_20261010_215949_465 | From scratch | S2 + label-preserving augmentation + ±800 µV clip | Huber δ 1 + mixup (2.3, 2.5) | 2145 | 8.93 | 9.59 | 7.91 | 11.59 | -0.52 | 9/15 |

`Split check: all 45 phase-1 runs use the same recordings: train 2,145 (1,650 patients), val 556 (412), test 274 (251). Window counts differ only for A, C. Same split outside phase 1: Control.`

## Table 2.3. Phase 1 short list: one row per configuration, relative to D

Table 2.2 condensed to the 41 distinct configurations, each described and measured against D ( `rs.phase1_short_list` ).

- **Same results merged:** D-npy (the npy data format) and D-cohorts (a local rerun that also logs the cohort metrics) reproduce D exactly, so they are one row. D's `Norm` / `Ab` come from D-cohorts.
- **Seeds averaged:** MT1 + MT1-seed1 and S2 + S2-seed1 are the same configuration run twice; their row is the mean.
- **Every other configuration is kept**, grouped by family as in Table 2.2.

Columns:

- `change vs D` : what the configuration changes relative to D. `4/1 ep` is E2's schedule (4 epochs, 1 warmup) instead of D's 15 / 2; `no CAR` means without the common average reference.
- `val` , `test` , `Norm` , `Ab` , `β` , `e*/E` : as in Table 2.2.
- `train recs / patients` : training recordings / training patients, in thousands (K = 1,000): 2.15K / 1.65K, i.e. 2,145 recordings of 1,650 patients, for every phase-1 configuration (Table 2.2).
- `Δ val` , `Δ test` : Eq. (0.4) against D (val 7.77, test 8.30); negative is better.

- `vs D`: `better` / `worse` when $|\Delta_{\text{test}}|$ exceeds the single-run noise threshold $\tau = 0.38$ (Eq. (5.2)), otherwise `n.s.`. $\tau$ is conservative for the seed-averaged rows.
- `runs`: the runs merged into the row.

| family | config | change vs D | train recs / patients | val | Δ val | test | Δ test | vs D | Norm | Ab | β | e*/E | runs |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Evaluation protocol** | **A** | no CAR; scored on whole recordings | 2.15K / 1.65K | 8.29 | 0.52 | 9.03 | 0.73 | worse | – | – | -0.46 | 2/12 | |
| | **B** | no CAR | 2.15K / 1.65K | 8.40 | 0.63 | 8.91 | 0.61 | worse | – | – | -0.43 | 2/6 | |
| **Baseline** | **D** | — (reference) | 2.15K / 1.65K | 7.77 | 0.00 | 8.30 | 0.00 | reference | 6.57 | 10.38 | -0.40 | 1/15 | = D-npy, D-cohorts (same results) |
| **Schedule** | **E2** | 4/1 ep | 2.15K / 1.65K | 7.68 | -0.09 | 8.25 | -0.05 | n.s. | – | – | -0.37 | 3/4 | |
| | **L1** | mixup 0.4 + EMA eval, 12 ep / 2 warmup | 2.15K / 1.65K | 7.51 | -0.26 | 8.47 | 0.17 | n.s. | 6.65 | 10.63 | -0.42 | 3/12 | |
| **Classic regularizers** | **C2** | no CAR + layer decay 0.5, drop path 0.2, dropout 0.1, wd 0.1, clip 1 | 2.15K / 1.65K | 12.47 | 4.70 | 13.52 | 5.22 | worse | – | – | -0.81 | 14/15 | |
| | **DX3** | C2's regularizer bundle (with CAR) | 2.15K / 1.65K | 12.83 | 5.06 | 14.90 | 6.60 | worse | – | – | -1.00 | 0/15 | |
| | **F1** | 4/1 ep + layer decay 0.5 | 2.15K / 1.65K | 7.74 | -0.03 | 8.36 | 0.06 | n.s. | 6.42 | 10.68 | -0.36 | 3/4 | |
| | **F2** | 4/1 ep + dropout 0.1 | 2.15K / 1.65K | 13.25 | 5.48 | 15.17 | 6.87 | worse | 13.37 | 17.32 | -0.97 | 0/4 | |
| | **F3** | 4/1 ep + drop path 0.2 | 2.15K / 1.65K | 7.70 | -0.07 | 8.41 | 0.11 | n.s. | 6.31 | 10.91 | -0.35 | 3/4 | |
| | **F4** | 4/1 ep + weight decay 0.1 | 2.15K / 1.65K | 7.66 | -0.11 | 8.24 | -0.06 | n.s. | 6.49 | 10.32 | -0.37 | 3/4 | |
| **Head init** | **G1** | 4/1 ep + head init scale 1.0 | 2.15K / 1.65K | 7.76 | -0.01 | 8.38 | 0.08 | n.s. | 6.64 | 10.45 | -0.37 | 3/4 | |
| **Layer decay** | **H1** | 4/1 ep + layer decay 0.75 | 2.15K / 1.65K | 7.71 | -0.06 | 8.67 | 0.37 | n.s. | 6.64 | 11.09 | -0.40 | 1/4 | |
| | **H2** | 4/1 ep + layer decay 0.85 | 2.15K / 1.65K | 7.86 | 0.09 | 8.54 | 0.24 | n.s. | 6.74 | 10.69 | -0.42 | 3/4 | |
| **Partial fine-tuning** | **C2-head** | C2, only the head trained | 2.15K / 1.65K | 12.77 | 5.00 | 14.78 | 6.48 | worse | – | – | -0.99 | 5/8 | |
| | **C2-last** | C2, only the last block + head trained | 2.15K / 1.65K | 12.50 | 4.73 | 13.64 | 5.34 | worse | – | – | -0.87 | 6/15 | |
| | **M3** | 4/1 ep + LoRA r16 instead of full fine-tuning (lr 1e-3, ld 1.0) | 2.15K / 1.65K | 7.68 | -0.09 | 9.24 | 0.94 | worse | 7.41 | 11.42 | -0.44 | 3/4 | |
| **Mixup / EMA** | **M1** | 4/1 ep + mixup 0.4 | 2.15K / 1.65K | 7.46 | -0.31 | 8.45 | 0.15 | n.s. | 6.60 | 10.65 | -0.41 | 3/4 | |
| | **M1c** | 4/1 ep + C-Mixup (σ 5 y) | 2.15K / 1.65K | 7.54 | -0.23 | 8.23 | -0.07 | n.s. | 6.50 | 10.29 | -0.39 | 3/4 | |
| | **M2** | 4/1 ep + EMA eval | 2.15K / 1.65K | 7.58 | -0.19 | 8.17 | -0.13 | n.s. | 6.35 | 10.35 | -0.36 | 2/4 | |
| | **M12** | 4/1 ep + mixup 0.4 + EMA eval | 2.15K / 1.65K | 7.48 | -0.29 | 8.45 | 0.15 | n.s. | 6.57 | 10.69 | -0.42 | 3/4 | |
| **Loss** | **D2** | no CAR + L1 loss | 2.15K / 1.65K | 8.36 | 0.59 | 9.32 | 1.02 | worse | – | – | -0.46 | 6/15 | |
| | **DX2** | Huber δ 2 | 2.15K / 1.65K | 7.79 | 0.02 | 8.41 | 0.11 | n.s. | – | – | -0.38 | 3/15 | |
| | **M4** | 4/1 ep + soft labels (σ 2.5 y) | 2.15K / 1.65K | 7.67 | -0.10 | 8.59 | 0.29 | n.s. | 6.92 | 10.57 | -0.40 | 3/4 | |
| | **B1** | 4/1 ep + age-balanced loss (sqrt-inverse) | 2.15K / 1.65K | 7.83 | 0.06 | 8.48 | 0.18 | n.s. | 6.66 | 10.63 | -0.37 | 2/4 | |
| | **B2** | 4/1 ep + age-balanced loss (inverse) | 2.15K / 1.65K | 7.96 | 0.19 | 8.57 | 0.27 | n.s. | 6.70 | 10.80 | -0.33 | 1/4 | |
| **Codebook regularization** | **DX1** | codebook amplitude reconstruction (w 1) | 2.15K / 1.65K | 7.72 | -0.05 | 8.54 | 0.24 | n.s. | – | – | -0.39 | 4/15 | |
| | **DX4** | codebook quantization term (w 1) | 2.15K / 1.65K | 7.67 | -0.10 | 8.44 | 0.14 | n.s. | – | – | -0.39 | 3/15 | |
| | **DX4b** | codebook quantization term (w 50) | 2.15K / 1.65K | 7.78 | 0.01 | 8.66 | 0.36 | n.s. | – | – | -0.37 | 4/15 | |
| **Augmentation / input** | **C** | no CAR + 30 s windows | 2.15K / 1.65K | 8.24 | 0.47 | 9.35 | 1.05 | worse | – | – | -0.42 | 8/15 | |
| | **D-clip1** | gradient clipping 1.0 | 2.15K / 1.65K | 7.70 | -0.07 | 8.36 | 0.06 | n.s. | – | – | -0.41 | 1/15 | |
| | **E1** | random crop | 2.15K / 1.65K | 7.82 | 0.05 | 8.56 | 0.26 | n.s. | – | – | -0.40 | 3/15 | |
| | **E1c** | random crop + clipping 1.0 | 2.15K / 1.65K | 7.72 | -0.05 | 8.13 | -0.17 | n.s. | – | – | -0.38 | 1/15 | |
| | **E2c** | 4/1 ep + clipping 1.0 | 2.15K / 1.65K | 7.63 | -0.14 | 8.24 | -0.06 | n.s. | – | – | -0.38 | 3/4 | |
| **Head** | **D-cls** | class token instead of mean pooling | 2.15K / 1.65K | 7.90 | 0.13 | 8.24 | -0.06 | n.s. | – | – | -0.38 | 2/15 | |
| | **D-clsmean** | [mean; class token] concatenated | 2.15K / 1.65K | 7.80 | 0.03 | 8.35 | 0.05 | n.s. | – | – | -0.41 | 1/15 | |
| | **N2L** | mixup 0.4 + clip 1 + attention head, 12 ep / 4 warmup | 2.15K / 1.65K | 7.63 | -0.14 | 8.49 | 0.19 | n.s. | 6.68 | 10.64 | -0.43 | 4/12 | |
| **Head (multi-task)** | **MT1** | 4/1 ep + mixup 0.4 + age/abnormal multi-task head | 2.15K / 1.65K | 7.58 | -0.19 | 8.46 | 0.16 | n.s. | 6.74 | 10.52 | -0.41 | 3/4 | mean of 2 seeds (MT1, MT1-seed1) |
| **From scratch** | **S1** | no pretrained weights, layer decay 1.0 | 2.15K / 1.65K | 8.79 | 1.02 | 9.42 | 1.12 | worse | 7.75 | 11.42 | -0.49 | 14/15 | |
| | **S2** | no pretrained weights, 6 blocks + MLP head, mixup, tuned | 2.15K / 1.65K | 8.55 | 0.79 | 9.37 | 1.07 | worse | 7.66 | 11.41 | -0.50 | 13/15 | mean of 2 seeds (S2, S2-seed1) |

| family | config | change vs D | train recs / patients | val | Δ val | test | Δ test | vs D | Norm | Ab | β | e*/E | runs |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | S3 | S2 + label-preserving augmentation + ±800 µV clip | 2.15K / 1.65K | 8.93 | 1.16 | 9.59 | 1.29 | worse | 7.91 | 11.59 | -0.52 | 9/15 | |

**Reading Table 2.3.**

- **Nothing in phase 1 beats D on test beyond noise.** The lowest test MAEs (E1c 8.13, M2 8.17, M1c 8.23) are 0.07–0.17 years below D, well inside τ.
- **What clearly hurts** ( `worse` ):
  - dropping the common average reference: A, B, C and D2 are all without CAR;
  - the strong regularizer bundle and dropout, which collapse the model (C2, DX3, F2, and C2-head / C2-last);
  - LoRA instead of full fine-tuning (M3);
  - training from scratch (S1–S3).
- **Everything else is n.s.**: schedule, mixup / EMA, loss shape, codebook terms, heads, clipping and cropping all land within ±0.37 years of D on test.
- **Validation tells a slightly different story.** The mixup / EMA runs gain 0.2–0.3 years on val, the largest val gains of phase 1, which is why M1 was carried into phase 2. Their test gain is not significant. Section 5 tests each change against its own baseline.

## Table 2.4. Phase 1 re-evaluated in eval mode: train, val and test

Every phase-1 run's saved best checkpoint was scored again, **without any training**, on its train, val and test splits:

- **Code and jobs:** `scripts/eval_runs.py` , launched as SageMaker jobs by `scripts/submit_eval_runs.py` .
- **What a job does:** download the run's checkpoint from S3, rebuild the model and data from the run's own `run_config.yaml` on the TUAB v3.0.0 data it trained on, and predict every window in eval mode (no dropout, no mixup).
- **Results:** `docs/data/phase1_reeval_metrics.csv` and `docs/data/phase1_reeval_split_check.csv` . The per-recording predictions stay outside git under the TUH data-use agreement.

All 45 runs are re-scored. Their val and test MAE reproduce the registry (Table 2.2) to within 0.01 years. Rows are merged as in Table 2.3: same-result runs and seed repeats are averaged, and `runs` counts them. Metrics are recording-level (Section 0):

- **train**: scored on the run's own evaluation budget per recording (the first 5 minutes after the trim, like val and test), so the three splits compare directly.
- **train (all windows)**: MAE over every training window, as the model saw them in training.

The generalization gap is

$$\text{gap} = \text{MAE}_{\text{test}} - \text{MAE}_{\text{train}} \tag{2.12}$$

where a large gap is memorization. `split = recorded` is ✓ when every merged run's rebuilt train, val and test recordings are exactly those of its recorded `data_split.json`.

Columns: MAE (Eq. (0.1), years) and $R^2$ (Eq. (0.2)) per split; `test Norm` / `test Ab` over the 149 normal and 125 abnormal test recordings.

| family | config | train MAE | train R² | train MAE (all windows) | val MAE | val R² | test MAE | test R² | test Norm MAE | test Norm R² | test Ab MAE | test Ab R² | gap (test − train) | split = recorded | runs |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Evaluation protocol | A | 3.54 | 0.92 | 3.54 | 8.29 | 0.57 | 9.03 | 0.57 | 7.02 | 0.69 | 11.43 | 0.33 | 5.49 | ✓ | 1 |
| | B | 4.00 | 0.90 | 3.54 | 8.40 | 0.55 | 8.91 | 0.60 | 6.94 | 0.71 | 11.26 | 0.37 | 4.91 | ✓ | 1 |
| Baseline | D | 6.05 | 0.80 | 5.79 | 7.77 | 0.61 | 8.30 | 0.65 | 6.57 | 0.74 | 10.37 | 0.45 | 2.25 | ✓ | 3 |
| Schedule | E2 | 3.41 | 0.93 | 2.95 | 7.67 | 0.62 | 8.25 | 0.64 | 6.51 | 0.74 | 10.33 | 0.43 | 4.84 | ✓ | 1 |
| | L1 | 4.51 | 0.88 | 4.21 | 7.51 | 0.63 | 8.47 | 0.62 | 6.65 | 0.72 | 10.63 | 0.40 | 3.96 | ✓ | 1 |
| Classic regularizers | C2 | 13.29 | 0.15 | 13.36 | 12.47 | 0.06 | 13.52 | 0.14 | 12.58 | 0.13 | 14.65 | -0.09 | 0.24 | ✓ | 1 |
| | DX3 | 14.60 | 0.00 | 14.60 | 12.83 | 0.00 | 14.90 | -0.01 | 14.39 | -0.06 | 15.51 | -0.24 | 0.30 | ✓ | 1 |
| | F1 | 5.17 | 0.84 | 4.79 | 7.74 | 0.61 | 8.36 | 0.63 | 6.42 | 0.75 | 10.68 | 0.40 | 3.19 | ✓ | 1 |
| | F2 | 14.91 | -0.03 | 14.88 | 13.25 | -0.04 | 15.17 | -0.09 | 13.37 | 0.03 | 17.32 | -0.51 | 0.27 | ✓ | 1 |
| | F3 | 4.77 | 0.86 | 4.35 | 7.70 | 0.61 | 8.41 | 0.62 | 6.31 | 0.75 | 10.91 | 0.38 | 3.64 | ✓ | 1 |
| | F4 | 3.41 | 0.93 | 2.95 | 7.66 | 0.62 | 8.24 | 0.64 | 6.49 | 0.74 | 10.32 | 0.43 | 4.83 | ✓ | 1 |
| Head init | G1 | 3.36 | 0.93 | 2.91 | 7.76 | 0.60 | 8.38 | 0.63 | 6.64 | 0.73 | 10.45 | 0.43 | 5.02 | ✓ | 1 |
| Layer decay | H1 | 4.06 | 0.90 | 3.67 | 7.71 | 0.61 | 8.67 | 0.60 | 6.64 | 0.71 | 11.09 | 0.38 | 4.62 | ✓ | 1 |
| | H2 | 1.49 | 0.99 | 1.11 | 7.86 | 0.59 | 8.54 | 0.61 | 6.74 | 0.69 | 10.69 | 0.41 | 7.05 | ✓ | 1 |
| Partial fine-tuning | C2-head | 14.54 | 0.01 | 14.53 | 12.77 | 0.02 | 14.78 | 0.01 | 14.07 | -0.02 | 15.62 | -0.25 | 0.24 | ✓ | 1 |
| | C2-last | 13.86 | 0.08 | 13.79 | 12.50 | 0.07 | 13.64 | 0.18 | 13.24 | 0.10 | 14.12 | 0.03 | -0.22 | ✓ | 1 |
| | M3 | 1.61 | 0.99 | 1.23 | 7.68 | 0.60 | 9.24 | 0.57 | 7.41 | 0.65 | 11.42 | 0.35 | 7.63 | ✓ | 1 |
| Mixup / EMA | M1 | 4.50 | 0.88 | 4.18 | 7.46 | 0.63 | 8.45 | 0.61 | 6.60 | 0.73 | 10.65 | 0.39 | 3.95 | ✓ | 1 |
| | M1c | 4.38 | 0.88 | 4.02 | 7.54 | 0.62 | 8.23 | 0.63 | 6.50 | 0.72 | 10.29 | 0.43 | 3.85 | ✓ | 1 |
| | M2 | 4.11 | 0.90 | 3.70 | 7.58 | 0.63 | 8.17 | 0.65 | 6.35 | 0.75 | 10.35 | 0.44 | 4.07 | ✓ | 1 |
| | M12 | 4.67 | 0.87 | 4.36 | 7.48 | 0.63 | 8.45 | 0.61 | 6.57 | 0.72 | 10.69 | 0.39 | 3.78 | ✓ | 1 |
| Loss | D2 | 1.78 | 0.97 | 1.40 | 8.37 | 0.54 | 9.32 | 0.55 | 7.87 | 0.62 | 11.04 | 0.35 | 7.54 | ✓ | 1 |
| | DX2 | 3.19 | 0.94 | 2.72 | 7.79 | 0.60 | 8.41 | 0.62 | 6.49 | 0.72 | 10.70 | 0.41 | 5.22 | ✓ | 1 |
| | M4 | 2.25 | 0.96 | 1.93 | 7.67 | 0.61 | 8.59 | 0.61 | 6.92 | 0.68 | 10.57 | 0.42 | 6.34 | ✓ | 1 |
| | B1 | 4.10 | 0.91 | 3.66 | 7.83 | 0.60 | 8.48 | 0.61 | 6.66 | 0.72 | 10.63 | 0.39 | 4.38 | ✓ | 1 |
| | B2 | 5.56 | 0.84 | 5.10 | 7.96 | 0.58 | 8.57 | 0.60 | 6.70 | 0.72 | 10.80 | 0.36 | 3.01 | ✓ | 1 |
| Codebook regularization | DX1 | 3.48 | 0.93 | 3.01 | 7.72 | 0.60 | 8.54 | 0.63 | 7.06 | 0.70 | 10.30 | 0.45 | 5.06 | ✓ | 1 |
| | DX4 | 3.24 | 0.94 | 2.88 | 7.67 | 0.61 | 8.44 | 0.64 | 6.88 | 0.72 | 10.30 | 0.45 | 5.21 | ✓ | 1 |
| | DX4b | 3.15 | 0.94 | 2.64 | 7.78 | 0.60 | 8.66 | 0.61 | 6.89 | 0.72 | 10.77 | 0.39 | 5.51 | ✓ | 1 |
| Augmentation / input | C | 2.68 | 0.96 | 2.21 | 8.24 | 0.55 | 9.35 | 0.56 | 7.42 | 0.68 | 11.66 | 0.31 | 6.67 | ✓ | 1 |
| | D-clip1 | 6.12 | 0.79 | 5.86 | 7.70 | 0.61 | 8.36 | 0.65 | 6.74 | 0.73 | 10.29 | 0.47 | 2.24 | ✓ | 1 |
| | E1 | 3.49 | 0.93 | 3.04 | 7.82 | 0.60 | 8.56 | 0.61 | 6.65 | 0.71 | 10.85 | 0.40 | 5.08 | ✓ | 1 |
| | E1c | 6.25 | 0.78 | 5.93 | 7.72 | 0.61 | 8.13 | 0.64 | 6.31 | 0.73 | 10.29 | 0.45 | 1.87 | ✓ | 1 |
| | E2c | 3.56 | 0.92 | 3.10 | 7.63 | 0.62 | 8.24 | 0.64 | 6.49 | 0.73 | 10.33 | 0.43 | 4.68 | ✓ | 1 |
| Head | D-cls | 4.44 | 0.88 | 4.05 | 7.90 | 0.59 | 8.24 | 0.61 | 6.69 | 0.71 | 10.09 | 0.39 | 3.80 | ✓ | 1 |
| | D-clsmean | 6.03 | 0.80 | 5.78 | 7.80 | 0.61 | 8.35 | 0.64 | 6.62 | 0.73 | 10.41 | 0.45 | 2.32 | ✓ | 1 |
| | N2L | 4.47 | 0.88 | 4.22 | 7.63 | 0.62 | 8.49 | 0.61 | 6.68 | 0.71 | 10.64 | 0.40 | 4.01 | ✓ | 1 |
| Head (multi-task) | MT1 | 4.59 | 0.88 | 4.26 | 7.58 | 0.63 | 8.46 | 0.62 | 6.74 | 0.72 | 10.52 | 0.40 | 3.87 | ✓ | 2 |
| From scratch | S1 | 3.28 | 0.93 | 3.07 | 8.79 | 0.51 | 9.42 | 0.55 | 7.75 | 0.63 | 11.42 | 0.33 | 6.14 | ✓ | 1 |
| | S2 | 5.38 | 0.84 | 5.16 | 8.56 | 0.54 | 9.37 | 0.54 | 7.66 | 0.63 | 11.42 | 0.32 | 4.00 | ✓ | 2 |
| | S3 | 6.67 | 0.76 | 6.46 | 8.94 | 0.51 | 9.59 | 0.54 | 7.91 | 0.63 | 11.59 | 0.31 | 2.92 | ✓ | 1 |

### Reading Table 2.4.

- **The splits are verified.** All 45 runs rebuild exactly their recorded train (2,145), val (556) and test (274) recordings, so every phase-1 comparison is on the same data.
- **The stored results are reproduced.** Re-scoring the saved checkpoints gives the logged val / test MAE to within 0.01 years. The duplicates merged in Table 2.3 (D, D-npy, D-cohorts) also agree on train.
- **Memorization is common.** Most fine-tuned configurations reach a train MAE of 3–6 years against 8.1–8.7 on test, a gap of about 2–5.5 years (Eq. (2.12)).

    - The extreme cases fit the training recordings almost perfectly yet do not improve test: H2 (layer decay 0.85: train 1.5, test 8.5), M3 (LoRA: 1.6 vs 9.2), D2 (L1: 1.8 vs 9.3) and M4 (soft labels: 2.3 vs 8.6).
    - The smallest gaps among the working runs belong to the D-schedule runs with clipping or cropping (E1c 1.9, D-clip1 2.2), and to D itself (2.3). All three select an early epoch ( `e*/E` in Table 2.3).
- **The collapsed runs underfit, they do not overfit.** C2, DX3, F2, C2-head and C2-last have $R^2 \leq 0.15$ on train as well as test. They predict about the mean age for everyone.
- **Normal / abnormal for every run.** On test, $R^2$ is 0.62–0.75 on normal recordings and only 0.31–0.47 on abnormal ones for every configuration that did not collapse. Abnormal recordings stay the hard group throughout phase 1.

## What the age head reads: the feature space (patch tokens, class token, pathology labels)

Tables 2.5–2.9 have a `feature space` column. It says which vector the linear age head reads, so EEG-only runs and runs given clinical labels are not compared unawares. Let $t_{c,a} \in \mathbb{R}^{200}$ be the final-layer token of channel $c$ and 1-s patch $a$ ($C$ channels, $A$ patches per window), and $t_{\text{cls}}$ the class token, both after the final LayerNorm ( `labram/models/neural_transformer.py` ):

$$
\begin{aligned}
h_{\text{patch}} &= \frac{1}{CA}\sum_{c=1}^{C}\sum_{a=1}^{A} t_{c,a}, \qquad h_{\text{cls}} = t_{\text{cls}} \\
c &= \begin{bmatrix} s \\ p \end{bmatrix} \in \{0,1\}^{3+14} \\
\hat{z} &= w^{\top} h + b \qquad (\text{EEG only}) \\
\hat{z} &= w^{\top} \begin{bmatrix} h \\ W_c\, c \end{bmatrix} + b, \quad W_c \in \mathbb{R}^{32\times 17} \qquad (\text{with path label})
\end{aligned}
\tag{2.13}
$$

- **`patch_tokens`** : $h = h_{\text{patch}}$, the mean over all patch tokens. This is LaBraM's default ( `model.use_mean_pooling=true` ).
- **`cls_token`** : $h = h_{\text{cls}}$, the class token, which attends to every patch token in each layer ( `model.use_mean_pooling=false` ).
- **`+ path_label`** : the head also reads the recording's clinical **pathology-label vector** $c$, through a learned 32-d projection $W_c$ ( `model.session_features=head` ). How $c$ is built ( `dataset_maker/make_condition_features.py` , saved per cohort as `conditions_v1.npz` , loaded by `labram/data/session_features.py` ):

    1. **Status $s$, one-hot over 3:** normal / abnormal / not available. It is the TUAB folder label for a TUAB recording. For a TUEG session in the NEDC clinical sheet it is the

report finding: "abnormal due to" filled → abnormal, empty → normal. Otherwise it is not available (Eq. (1.2)).

2. **Pathology $p$, multi-hot over 14:** one entry per diagnosis group of the NEDC sheet (Table 1.9): clinical seizure, electrographic seizure, epilepsy, stroke, TBI, substance use, bipolar, anxiety, depression, schizophrenia, dementia (with Alzheimer's), multiple sclerosis, and other (Parkinson's, ADHD, under 50 recordings each). Entry 0 is "healthy or not available": it is set when no diagnosis is flagged or the session is not in the sheet. Several entries can be 1 at once.
3. **Concatenate** $c = [s; p]$, 17 numbers. They are categorical, so nothing is standardized. In training, mixup blends $c$ along with the EEG and the age (Eq. (3.3)).

The labels are inputs, not the target. They hold no age, so they leak no age. But a `path_label` model needs the recording's clinical status and diagnoses at prediction time, also on TUAB test. It is therefore not an EEG-only model. Its MAEs compare with the EEG-only rows only with that in mind.

- **`+ feat_tokens`** (Exp1t): the top-10 handcrafted EEG features enter as extra tokens, standardized on the cohort's training split. **`features only (no EEG)`** (Exp1f): an MLP on handcrafted features alone.

## Table 2.5. Phase 2: TUH cohort runs

One row per run of `docs/data/tuh_cohort_eval_v1.csv`, from `labram/eval/research_summary.py::PHASE2`:

- `exp`: the run, linked to its ClearML experiment (click the name);
- `cohort` (Table 1.4) and `train recs / patients`, its training recordings and patients, both in thousands with two decimals (2.15K / 1.65K = 2,145 recordings of 1,650 patients; the run's recorded split, `docs/data/brain_age_run_splits.csv`); `epochs` / `warmup`;
- `feature space`: what the age head reads (Eq. (2.13)): `patch_tokens`, `cls_token`, and `+ path_label` when it also receives the recording's status and pathology vector;
- `parent` and `change`: the run it is compared with in Section 5 and what differs;
- `loss`: the training loss (Eqs. (2.2)–(2.11)). Every phase-2 run is M1, Huber + mixup; the T1/T2 dropout runs add the consistency term.

Metrics:

- `test all` / `normal` / `abnormal`: MAE (Eq. (0.1)) on the **TUAB test set only**, all 274 recordings and its 149 normal / 125 abnormal ones. The phase-2 runs train on TUEG too, but none of them is tested on TUEG recordings: TUAB test is the one set every run shares (Section 0).
- `R² normal`: Eq. (0.2) on the normal TUAB test recordings;
- `TUEG val v2`: MAE on the TUEG part of the shared v2 validation split (2,325 recordings no v2/v3 model trained on), available only for runs scored on it. It is the only TUEG number

in the table. Each run's own validation split is not shown, because it differs from cohort to cohort.

- **The last three rows are phase 3**, for comparison on the same TUAB test set: each 5-fold CV experiment (Table 2.7) as the median of its 5 fold models ( `ens.` , Eq. (3.8)). Their `train recs / patients` is the mean over the fold models (3 of the 5 folds of the 29,035-recording pool), and their `R² normal` is the ensemble's.
- `cal. all` / `cal. normal` / `cal. abnormal` : the TUAB test MAE after a **leak-free age calibration**, and `cal. a / b` its two coefficients. Per experiment $r$ it takes three steps.
  1. **Fit set** $F_r$: normal recordings that $r$ never trained on and that are not TUAB test. For a single-split run (every row above the CV rows) these are the normal recordings of its **own validation split**. For a CV row they are its out-of-fold predictions of normal recordings (each recording predicted by the fold model whose in-pool test fold held it out, Table 3.5).
  2. **Fit** the least-squares line from predicted to true age over $F_r$, with $\hat{y}_j$ the run's recording-level prediction and $y_j$ the true age (years):

$$(a_r, b_r) = \arg\min_{a,b} \sum_{j \in F_r} \left(y_j - a\,\hat{y}_j - b\right)^2, \qquad a_r = \frac{\mathrm{Cov}_{F_r}(\hat{y}, y)}{\mathrm{Var}_{F_r}(\hat{y})}, \quad b_r = \bar{y}_{F_r} - a_r\,\bar{\hat{y}}_{F_r}$$

$$\hat{y}_i^{\,\mathrm{cal}} = a_r\,\hat{y}_i + b_r, \qquad \mathrm{MAE}^{\mathrm{cal}}(S) = \frac{1}{|S|} \sum_{i \in S} \left|\hat{y}_i^{\,\mathrm{cal}} - y_i\right|$$

  3. **Apply** it to each TUAB test prediction $\hat{y}_i$ and score the result with Eq. (0.1) on $S$ = all, normal or abnormal TUAB test recordings.

The TUAB test true age $y_i$ appears only in the final MAE, exactly as in every raw MAE. It is never used to fit $a_r$, $b_r$. The check below the table confirms that no fit set shares a recording or a patient with TUAB test. Eq. (2.14) is Eq. (3.11) of Section 3. Caveat for the single-split runs: their validation split also chose the best epoch, so their fit set was used once for model selection. It still holds no test data. The single-split numbers come from re-scoring each run's checkpoint on its val and test splits ( `scripts/eval_runs.py` ; `docs/data/phase2_calibration.csv` ). The re-score reproduces every run's raw test MAE within 0.01 except Exp3q (7.37 vs 7.34): it applies Exp3q's signal-quality filter to the test windows (8,210 of 8,220), which the table's number did not. Exp2's cohort has no test split, so it is scored on the same 274 TUAB test recordings through `normal_both_v1` , with its own training statistics. Table 3.5 gives the other corrections, including the gap correction that uses the test ages.

- No `β` : the cohort evaluation ( `docs/data/tuh_cohort_eval_v1.csv` ) stores MAE, RMSE and R² but not the bias slope, so $\beta$ is reported for phase 1 only.

**Same split or not.** The check below the table groups the runs by their recorded train/val/test recordings, as for Table 2.2:

- Runs on the same cohort share one split. Exp1f's `normal_both_v1_meta` holds the same recordings as `normal_both_v1`, and `all_both_v3_compact` the same as `all_both_v3`.
- **Exp3q is the exception** within `all_both_v1`. Its signal-quality filter drops windows beyond ±800 µV and windows with a flat channel, and recordings left without windows drop out. That leaves 19,815 instead of 20,368 train recordings and 2,836 instead of 2,905 val recordings, and its test loses 10 windows but keeps all 274 recordings.
- Splits differ across cohorts by design (Table 1.4): `v1`, `v2` and `v3` assign TUEG patients differently, so their val MAEs do not compare.
- **Exp4anymiss** trains on `all_both_v3_anymiss` (`dataset_maker/extend_cohort.py`). That is `all_both_v3` plus 2,003 training recordings of 545 training patients that lack A1/A2 or other channels: 1,115 without A1/A2, 876 without A1/A2/T1/T2, and 12 others. Its val and test recordings are identical to Exp4aug15's (same digests in `docs/data/brain_age_run_splits.csv`), so its own split differs only in training.
- **Exp4noear** uses Exp4anymiss's cohort and split (identical train, val and test digests) with A1, A2, T1 and T2 removed from every recording (`data.drop_channels`, 19 channels), no channel augmentation, and 8 / 2 epochs. It was trained locally.
- Every run with a test split tests on the same 274 TUAB recordings. Exp2 (`normal_TUEG_v1`) has no test split.

| exp | cohort | train recs / patients | feature space | test all | normal | abnormal | $R^2$ normal | TUEG val v2 | cal. all | cal. normal | cal. abnormal | cal. a / b | epochs | parent | change | los |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Control** | TUAB_v3.0.1_interp50-60 | 2.15K / 1.65K | patch_tokens | 8.23 | 6.42 | 10.39 | 0.75 | 9.29 | 8.12 | 6.30 | 10.29 | 1.09 / -3.8 | 4 / 1 | | M1 on TUAB only (the TUAB baseline of phase 2) | Hub δ 1 mixu (2. 2. |
| **Exp1** | normal_both_v1 | 5.45K / 3.00K | patch_tokens | 8.03 | 5.99 | 10.47 | 0.78 | 8.68 | 7.96 | 5.95 | 10.36 | 1.01 / 0.4 | 4 / 1 | Control | train on normal recordings only: TUAB normal + TUEG normal | Hub δ 1 mixu (2. 2. |
| **Exp2** | normal_TUEG_v1 | 3.76K / 2.15K | patch_tokens | 8.00 | 6.06 | 10.31 | 0.78 | 8.75 | 7.91 | 5.95 | 10.25 | 1.10 / -2.0 | 4 / 1 | Exp1 | TUEG normal only (no TUAB in training) | Hub δ 1 mixu (2. 2. |
| **Exp1b** | normal_both_v1 | 5.45K / 3.00K | patch_tokens | 8.11 | 6.09 | 10.52 | 0.77 | 8.66 | 8.15 | 6.13 | 10.54 | 0.96 / 2.0 | 10 / 3 | Exp1 | 10 epochs / 3 warmup instead of 4 / 1 | Hub δ 1 mixu (2. 2. |
| **Exp1t** | normal_both_v1 | 5.45K / 3.00K | patch_tokens + feat_tokens | 8.80 | 7.17 | 10.73 | 0.70 | – | 9.00 | 7.35 | 10.97 | 0.93 / 2.9 | 4 / 1 | Exp1 | classical EEG features as extra transformer tokens | Hub δ 1 mixu (2. 2. |
| **Exp1f** | normal_both_v1_meta | 5.45K / 3.00K | features only (no EEG) | 8.13 | 6.21 | 10.41 | 0.77 | – | 8.21 | 6.28 | 10.51 | 0.93 / 2.9 | 4 / 1 | Exp1 | classical EEG features only, MLP head, no EEG encoder | Hub δ 1 mixu (2. 2. |
| **Exp3** | all_both_v1 | 20.37K / 9.07K | patch_tokens | 7.26 | 5.57 | 9.28 | 0.80 | – | 7.23 | 5.53 | 9.26 | 1.03 / -1.6 | 4 / 1 | Control | add every TUEG recording (normal, abnormal, unlabelled): ~10x training data | Hub δ 1 mixu (2. 2. |
| **Exp3b** | all_both_v1 | 20.37K / 9.07K | patch_tokens | 7.70 | 6.17 | 9.51 | 0.75 | – | 7.48 | 5.96 | 9.29 | 1.06 / -2.3 | 10 / 3 | Exp3 | 10 epochs / 3 warmup instead of 4 / 1 | Hub δ 1 mixu (2. 2. |
| **Exp3q** | all_both_v1 | 19.82K / 8.97K | patch_tokens | 7.34 | 5.69 | 9.31 | 0.79 | – | 7.29 | 5.67 | 9.22 | 1.03 / -1.4 | 4 / 1 | Exp3 | drop windows beyond ±800 µV or with a flat channel | Hub δ 1 mixu (2. 2. |
| **Exp3c** | all_both_v1 | 20.37K / 9.07K | patch_tokens | 7.29 | 5.66 | 9.22 | 0.79 | – | 7.24 | 5.61 | 9.19 | 1.03 / -1.5 | 4 / 1 | Exp3 | clip the input to ±800 µV inside the model | Hub δ 1 mixu (2. 2. |
| **Exp3v2** | all_both_v2 | 20.47K / 9.09K | patch_tokens | 7.42 | 5.46 | 9.75 | 0.80 | 8.00 | 7.33 | 5.37 | 9.68 | 1.04 / -1.7 | 4 / 1 | Exp3 | cohort v2: one TUEG patient split shared by every cohort (data bookkeeping only) | Hub δ 1 mixu (2. 2. |
| **Exp3cond** | all_both_v2 | 20.47K / 9.09K | patch_tokens + path_label | 7.28 | 5.39 | 9.53 | 0.81 | 7.90 | 7.15 | 5.30 | 9.35 | 1.04 / -1.6 | 4 / 1 | Exp3v2 | status + pathology vectors concatenated to the pooled feature | Hub δ 1 mixu (2. 2. |
| **Exp3clscond** | all_both_v2 | 20.47K / 9.09K | cls_token + path_label | 7.25 | 5.36 | 9.51 | 0.81 | 7.98 | 7.18 | 5.27 | 9.45 | 1.04 / -1.3 | 4 / 1 | Exp3cond | head on the class token instead of the mean patch token | Hub δ 1 mixu (2. 2. |
| **Exp4drop** | all_both_v3 | 25.61K / 12.02K | patch_tokens + path_label | 7.28 | 5.26 | 9.68 | 0.82 | 7.76 | 7.10 | 5.04 | 9.55 | 1.07 / -3.0 | 4 / 1 | Exp3cond | cohort v3 (+ recordings missing only T1/T2) on 21 channels (T1/T2 dropped everywhere) | Hub δ 1 mixu (2. 2. |
| **Exp4base** | all_both_v3 | 25.61K / 12.02K | patch_tokens | 7.36 | 5.62 | 9.44 | 0.81 | 7.86 | 7.29 | 5.56 | 9.36 | 1.03 / -1.1 | 6 / 2 | Exp3v2 | cohort v3, 23 channels with missing T1/T2 left out per sample; | Hub δ 1 mixu (2. 2. |

| exp | cohort | train recs / patients | feature space | test all | normal | abnormal | R² normal | TUEG val v2 | cal. all | cal. normal | cal. abnormal | cal. a / b | epochs | parent | change | los |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | speed settings; 6 / 2 epochs | |
| **Exp4basec** | all_both_v3_compact | 25.61K / 12.02K | patch_tokens | 7.42 | 5.52 | 9.68 | 0.81 | 7.85 | 7.31 | 5.41 | 9.57 | 1.05 / -2.0 | 6 / 2 | Exp4base | compact copy of the cohort, same recorded split (reproduction check) | Hub δ 1 mixu (2. 2. |
| **Exp4ctrl** | all_both_v3 | 25.61K / 12.02K | cls_token + path_label | 7.30 | 5.35 | 9.63 | 0.81 | 7.92 | 7.16 | 5.18 | 9.51 | 1.06 / -2.3 | 10 / 3 | Exp4base | class token + condition vectors, 10 / 3 epochs with early stopping | Hub δ 1 mixu (2. 2. |
| **Exp4aug** | all_both_v3 | 25.61K / 12.02K | cls_token + path_label | 7.29 | 5.45 | 9.49 | 0.82 | 7.85 | 7.13 | 5.26 | 9.36 | 1.07 / -3.1 | 6 / 2 | Exp4ctrl | T1/T2 batch dropout (p 0.2) + consistency loss (λ 0.1); 6 / 2 epochs | Hub δ 1 mixu + 0 co si ten (2. 2. 2.1 |
| **Exp4aug15** | all_both_v3 | 25.61K / 12.02K | cls_token + path_label | 7.17 | 5.38 | 9.29 | 0.81 | 7.91 | 7.00 | 5.17 | 9.18 | 1.07 / -2.7 | 15 / 5 | Exp4aug | 15 / 5 epochs with early stopping (stopped at 11) | Hub δ 1 mixu + 0 co si ten (2. 2. 2.1 |
| **Exp4anymiss** | all_both_v3_anymiss | 27.62K / 12.19K | cls_token + path_label | 7.21 | 5.47 | 9.30 | 0.82 | 7.75 | 7.13 | 5.36 | 9.24 | 1.04 / -1.7 | 15 / 5 | Exp4aug15 | +2,003 training recordings missing A1/A2 or other channels; any 1-3 channels dropped (p 0.2) from epoch 4 (stopped at 12) | Hub δ 1 mixu + 0 co si ten (2. 2. 2.1 |
| **Exp4noear** | all_both_v3_anymiss | 27.62K / 12.19K | cls_token + path_label | 7.41 | 5.55 | 9.64 | 0.81 | 8.02 | 7.31 | 5.42 | 9.58 | 1.07 / -2.6 | 8 / 2 | Exp4anymiss | A1/A2/T1/T2 removed from every recording (19 channels), no channel augmentation; 8 / 2 epochs | Hub δ 1 mixu (2. 2. |
| **cvbase (CV ens.)** | all_both_v3_cvcompact | 17.42K / 8.13K | patch_tokens | 7.44 | 5.75 | 9.46 | 0.80 | – | 7.28 | 5.58 | 9.31 | 1.09 / -4.0 | 5 / 1 | Exp4base | 5-fold CV of Exp4base (5 / 1 epochs) | Hub δ 1 mixu (2. 2. |
| **cvctrl (CV ens.)** | all_both_v3_cvcompact | 17.42K / 8.13K | cls_token + path_label | 7.10 | 5.31 | 9.23 | 0.82 | – | 6.98 | 5.19 | 9.11 | 1.06 / -2.7 | 10 / 3 | cvbase | class token + condition vectors, 10 / 3 epochs (CV of Exp4ctrl) | Hub δ 1 mixu (2. 2. |
| **cvaug (CV ens.)** | all_both_v3_cvcompact | 17.42K / 8.13K | cls_token + path_label | 7.23 | 5.43 | 9.37 | 0.81 | – | 7.11 | 5.30 | 9.28 | 1.06 / -2.3 | 10 / 3 | cvctrl | T1/T2 batch dropout + consistency loss λ 0.1 (CV of Exp4aug) | Hub δ 1 mixu + 0 co si ten (2. 2. 2.1 |
| **cvaugval (CV ens.)** | all_both_v3_cvcompact | 23.23K / 10.84K | cls_token + path_label | 7.22 | 5.36 | 9.44 | 0.82 | – | 7.04 | 5.16 | 9.28 | 1.07 / -2.3 | 10 / 3 | cvaug | val-only CV: val = fold k, the other 4 folds train (~23.2K recordings), test = TUAB | Hub δ 1 mixu + 0 co si ten (2. 2. 2.1 |

```
Calibration fit sets (val, normal recordings) of the single-split runs: 291–1,388 recor
dings; shared with TUAB test: 0 recordings, 0 patients.
Re-scored raw TUAB test MAE vs the table: max difference 0.03 years (Exp3q +0.03).
same split: Exp1, Exp1b, Exp1t, Exp1f  (cohort normal_both_v1, normal_both_v1_meta)
same split: Exp3v2, Exp3cond, Exp3clscond  (cohort all_both_v2)
same split: Control  (cohort TUAB_v3.0.1_interp50-60)
same split: Exp4drop, Exp4base, Exp4basec, Exp4ctrl, Exp4aug, Exp4aug15  (cohort all_bo
th_v3, all_both_v3_compact)
same split: Exp4anymiss, Exp4noear  (cohort all_both_v3_anymiss)
same split: Exp3q  (cohort all_both_v1)
same split: Exp3, Exp3b, Exp3c  (cohort all_both_v1)
same split: Exp2  (cohort normal_TUEG_v1)
TUAB test recordings identical in every phase-2 run with a test split: True
```

## Table 2.6. Phase 2 and CV: the best runs (normal < 5.8, abnormal < 9.8)

The rows of Table 2.5 whose raw TUAB test MAE (Eq. (0.1)) is below **5.8 years on normal** and below **9.8 years on abnormal** recordings, sorted by `test all`. The columns are those of Table 2.5. `exp` links to ClearML, and `feature space` is defined by Eq. (2.13). The `cal.` columns are the leak-free calibration of Eq. (2.14).

| exp | cohort | train recs / patients | feature space | test all | nor-mal | abnor-mal | $R^2$ normal | cal. all | cal. normal | cal. abnormal |
|---|---|---|---|---|---|---|---|---|---|---|
| **cvctrl (CV ens.)** | all_both_v3_cvcompact | 17.42K / 8.13K | cls_token + path_label | 7.10 | 5.31 | 9.23 | 0.82 | 6.98 | 5.19 | 9.11 |
| **Exp4aug15** | all_both_v3 | 25.61K / 12.02K | cls_token + path_label | 7.17 | 5.38 | 9.29 | 0.81 | 7.00 | 5.17 | 9.18 |
| **Exp4anymiss** | all_both_v3_anymiss | 27.62K / 12.19K | cls_token + path_label | 7.21 | 5.47 | 9.30 | 0.82 | 7.13 | 5.36 | 9.24 |
| **cvaugval (CV ens.)** | all_both_v3_cvcompact | 23.23K / 10.84K | cls_token + path_label | 7.22 | 5.36 | 9.44 | 0.82 | 7.04 | 5.16 | 9.28 |
| **cvaug (CV ens.)** | all_both_v3_cvcompact | 17.42K / 8.13K | cls_token + path_label | 7.23 | 5.43 | 9.37 | 0.81 | 7.11 | 5.30 | 9.28 |
| **Exp3clscond** | all_both_v2 | 20.47K / 9.09K | cls_token + path_label | 7.25 | 5.36 | 9.51 | 0.81 | 7.18 | 5.27 | 9.45 |
| **Exp3** | all_both_v1 | 20.37K / 9.07K | patch_tokens | 7.26 | 5.57 | 9.28 | 0.80 | 7.23 | 5.53 | 9.26 |
| **Exp4drop** | all_both_v3 | 25.61K / 12.02K | patch_tokens + path_label | 7.28 | 5.26 | 9.68 | 0.82 | 7.10 | 5.04 | 9.55 |
| **Exp3cond** | all_both_v2 | 20.47K / 9.09K | patch_tokens + path_label | 7.28 | 5.39 | 9.53 | 0.81 | 7.15 | 5.30 | 9.35 |
| **Exp3c** | all_both_v1 | 20.37K / 9.07K | patch_tokens | 7.29 | 5.66 | 9.22 | 0.79 | 7.24 | 5.61 | 9.19 |
| **Exp4aug** | all_both_v3 | 25.61K / 12.02K | cls_token + path_label | 7.29 | 5.45 | 9.49 | 0.82 | 7.13 | 5.26 | 9.36 |
| **Exp4ctrl** | all_both_v3 | 25.61K / 12.02K | cls_token + path_label | 7.30 | 5.35 | 9.63 | 0.81 | 7.16 | 5.18 | 9.51 |
| **Exp3q** | all_both_v1 | 19.82K / 8.97K | patch_tokens | 7.34 | 5.69 | 9.31 | 0.79 | 7.29 | 5.67 | 9.22 |
| **Exp4base** | all_both_v3 | 25.61K / 12.02K | patch_tokens | 7.36 | 5.62 | 9.44 | 0.81 | 7.29 | 5.56 | 9.36 |
| **Exp4noear** | all_both_v3_anymiss | 27.62K / 12.19K | cls_token + path_label | 7.41 | 5.55 | 9.64 | 0.81 | 7.31 | 5.42 | 9.58 |
| **Exp3v2** | all_both_v2 | 20.47K / 9.09K | patch_tokens | 7.42 | 5.46 | 9.75 | 0.80 | 7.33 | 5.37 | 9.68 |
| **Exp4basec** | all_both_v3_compact | 25.61K / 12.02K | patch_tokens | 7.42 | 5.52 | 9.68 | 0.81 | 7.31 | 5.41 | 9.57 |
| **cvbase (CV ens.)** | all_both_v3_cvcompact | 17.42K / 8.13K | patch_tokens | 7.44 | 5.75 | 9.46 | 0.80 | 7.28 | 5.58 | 9.31 |

## Table 2.7. Phase 3: five-fold cross-validation

- **Folds:** the pool is `all_both_v3_cvcompact` train + val (29,035 recordings), split into 5 patient-disjoint folds, the same partition for every experiment. Two schemes:
  - `cvbase`, `cvctrl`, `cvaug`: for fold $k$, test = fold $k$, val = fold $k+1 \bmod 5$, and the other three folds train.
  - `cvaugval` (val-only, `cross_validation.test_set=held_out`): it is `cvaug` with val = fold $k$ and the other **four** folds training, with no in-pool test fold, so its `CV test` reads "–".

  The TUAB test set is in no fold, so each of the 5 fold models also scores it.

- **`exp`:** the CV experiment, linked to the ClearML project holding its five fold experiments (click the name).
- **`train recs / patients`:** training recordings / patients of a fold model, mean over the 5 folds, in thousands (each fold model trains on 3 of the 5 folds; `cvaugval` on 4).
- **`loss`:** the training loss (Eqs. (2.2)–(2.11)).
- **Per-fold columns:** mean ± sample sd over the 5 fold models of the MAE (Eq. (0.1)) on:
  - `CV val`: the fold's own val;
  - `CV test`: the fold's own test;
  - `TUAB test`, with `normal` and `abnormal`.
- **`ens.`:** the TUAB test MAE of the median of the 5 models' predictions (Eq. (3.8)). It needs the per-recording predictions.
- **`cal.`:** the same ensemble's TUAB test MAE after the leak-free bias correction of Eq. (3.11), all / normal / abnormal. The line age ≈ a · prediction + b is fitted on the experiment's own out-of-fold predictions of normal recordings (the CV test folds, never TUAB test), then applied to each test prediction alone, so no TUAB test true age is used (Table 3.6). Table 3.5 compares it with the other corrections.

Source: `docs/data/cv5_eval_v2.csv`.

| exp | train recs / patients | feature space | loss | CV val | CV test | TUAB test | nor-mal | ab-nor-mal | ens. TUAB test | ens. nor-mal | ens. abnor-mal | cal. TUAB test | cal. nor-mal | cal. abnor-mal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cvbase | 17.42K / 8.13K | patch_tokens | Huber δ 1 + mixup (2.3, 2.5) | 8.16 ± 0.11 | 8.18 ± 0.15 | 7.49 ± 0.09 | 5.81 ± 0.10 | 9.50 ± 0.11 | 7.44 | 5.75 | 9.46 | 7.28 | 5.58 | 9.31 |
| cvctrl | 17.42K / 8.13K | cls_token + path_label | Huber δ 1 + mixup (2.3, 2.5) | 8.09 ± 0.12 | 8.08 ± 0.13 | 7.28 ± 0.14 | 5.53 ± 0.08 | 9.37 ± 0.25 | 7.10 | 5.31 | 9.23 | 6.98 | 5.19 | 9.11 |
| cvaug | 17.42K / 8.13K | cls_token + path_label | Huber δ 1 + mixup + 0.1 consistency (2.3, 2.5, 2.11) | 8.06 ± 0.13 | 8.06 ± 0.15 | 7.33 ± 0.17 | 5.55 ± 0.17 | 9.44 ± 0.18 | 7.23 | 5.43 | 9.37 | 7.11 | 5.30 | 9.28 |
| cvaug-val | 23.23K / 10.84K | cls_token + path_label | Huber δ 1 + mixup + 0.1 consistency (2.3, 2.5, 2.11) | 7.90 ± 0.14 | – | 7.26 ± 0.10 | 5.41 ± 0.17 | 9.48 ± 0.09 | 7.22 | 5.36 | 9.44 | 7.04 | 5.16 | 9.28 |

**Reading Tables 2.5–2.7: bias correction.**

- **Raw, the CV ensembles are on par with the best single runs.** Their raw TUAB test MAE is 7.10–7.44, against 7.17–7.42 for the phase-2 runs on the same cohort (Exp4*), although each fold model trains on 17.4K (`cvaugval`: 23.2K) instead of 25.6K recordings.
- **The leak-free correction gains 0.1 years.** Calibrated, the ensembles reach 6.98–7.28 overall, 5.16–5.58 on normal and 9.11–9.31 on abnormal recordings: 0.09–0.20 years better than raw on every cohort. cvctrl stays the best (6.98).
- **No test label is used.** The `cal.` columns are fitted on out-of-fold predictions and applied to the prediction alone (Table 3.6). The much lower "corrected" numbers of the literature, e.g. Gemein et al.'s 6.60 / 12.85, use a correction applied with the test recordings' true ages. Table 3.5 lists ours of that kind (†) for comparison only.

- **Four folds instead of three (`cvaugval` vs `cvaug`, Table 2.7).** Training each fold model on 23.2K instead of 17.4K recordings improves the single model a little: the fold-mean TUAB test MAE goes from 7.33 to 7.26 and the fold-to-fold sd from 0.17 to 0.10. The ensemble barely moves, from 7.23 to 7.22 (7.18 with each fold's last-epoch checkpoint instead of the one val picked, `docs/data/cv5_eval_cvaugval_final.csv`). Calibrated, it reaches 7.04 overall and 5.16 on normal recordings, the lowest normal MAE of any CV run. All these differences are within the fold-to-fold sd. More data per model helps each model, but five models already average most of the gain away.
- **Missing-channel recordings and any-channel dropout (`Exp4anymiss` vs `Exp4aug15`, same val and test).** Exp4anymiss adds 2,003 training recordings with missing channels (mostly A1/A2) and, from epoch 4, drops any 1–3 channels from 20 % of the batches (`channel_drop.start_epoch`). Its validation MAE is slightly lower (7.58 vs 7.63), its TUEG val v2 MAE too (7.75 vs 7.91), and its TUAB test MAE about the same (raw 7.21 vs 7.17; normal 5.47 vs 5.38, abnormal 9.30 vs 9.29; calibrated 7.13 vs 7.00). All of these differences are within the epoch-to-epoch noise. The model now also covers recordings without A1/A2 and other channels, at no measurable cost on full-channel recordings.
- **Without the ear and anterior temporal channels (`Exp4noear` vs `Exp4anymiss`, same split).** Dropping A1, A2, T1 and T2 (19 channels, no channel augmentation, 8 / 2 epochs) costs about 0.2 years. Raw TUAB test goes from 7.21 to 7.41 (normal 5.47 → 5.55, abnormal 9.30 → 9.64), calibrated from 7.13 to 7.31, validation from 7.58 to 7.84, and TUEG val v2 from 7.75 to 8.02. Most of the loss is on abnormal recordings. The comparison is not clean: Exp4noear trained for 8 epochs with a schedule that decays over 8, against Exp4anymiss's 13 epochs (15 / 5, early-stopped). At the same epoch (7), the two are about level (test 7.41 vs 7.47, validation 7.84 vs 7.70). A 15-epoch run would separate the channels' effect from the schedule's.

## Table 2.8. From D to the best model: short list

The path from the base configuration D to the best models, one row per distinct configuration, all scored on the same 274 TUAB test recordings:

- **D:** the phase-1 reference. D-npy and D-cohorts reproduce it on another data format or scoring path with the same results (Table 2.3).
- **M1:** the phase-1 configuration carried into phase 2 (lowest validation MAE).
- **Every phase-2 run** (Table 2.5). One duplicate is merged: Exp4basec is Exp4base trained again on a compact copy of the same data with the same split, so its row is the mean of the two.
- **The four CV configurations** (Table 2.7), each as the mean over its 5 fold models and as the 5-model median ensemble.

Columns:

- `phase` and `config`.

- `cohort` (Table 1.4) and `train recs / patients`: the run's training recordings / training patients, in thousands (K = 1,000). For a CV row this is the mean over its 5 fold models, each of which trains on 3 of the 5 folds (17,239–17,607 recordings).
- `feature space` (Eq. (2.13)): what the head reads. A row is EEG-only when it is `patch_tokens` or `cls_token`, without `path_label`.
- `change`: what the run changes relative to its parent (Table 2.5).
- `test`, `Norm`, `Ab`: TUAB test MAE (Eq. (0.1)) over all, normal and abnormal recordings.
- `Δ test vs D`: Eq. (0.4) against D (8.30); negative is better.
- `note`: merged runs.

`◆` marks the lowest `test` among the EEG-only rows of each phase.

| phase | config | cohort | train recs / patients | feature space | change | test | Δ test vs D | Norm | Ab | note |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | **◆ D** | TUAB v3.0.0 | 2.15K / 1.65K | patch_tokens | base configuration (Table 2.1) | 8.30 | 0.00 | 6.57 | 10.38 | = D-npy, D-cohorts (same results) |
| | **M1** | TUAB v3.0.0 | 2.15K / 1.65K | patch_tokens | 4 / 1 epochs + mixup 0.4 (carried into phase 2) | 8.45 | 0.15 | 6.60 | 10.65 | |
| | **Control** | TUAB_v3.0.1_interp50-60 | 2.15K / 1.65K | patch_tokens | M1 on TUAB only (the TUAB baseline of phase 2) | 8.23 | -0.07 | 6.42 | 10.39 | |
| | **Exp1** | normal_both_v1 | 5.45K / 3.00K | patch_tokens | train on normal recordings only: TUAB normal + TUEG normal | 8.03 | -0.27 | 5.99 | 10.47 | |
| | **Exp2** | normal_TUEG_v1 | 3.76K / 2.15K | patch_tokens | TUEG normal only (no TUAB in training) | 8.00 | -0.30 | 6.06 | 10.31 | |
| | **Exp1b** | normal_both_v1 | 5.45K / 3.00K | patch_tokens | 10 epochs / 3 warmup instead of 4 / 1 | 8.11 | -0.19 | 6.09 | 10.52 | |
| | **Exp1t** | normal_both_v1 | 5.45K / 3.00K | patch_tokens + feat_tokens | classical EEG features as extra transformer tokens | 8.80 | 0.50 | 7.17 | 10.73 | |
| | **Exp1f** | normal_both_v1_meta | 5.45K / 3.00K | features only (no EEG) | classical EEG features only, MLP head, no EEG encoder | 8.13 | -0.17 | 6.21 | 10.41 | |
| | **◆ Exp3** | all_both_v1 | 20.37K / 9.07K | patch_tokens | add every TUEG recording (normal, abnormal, unlabelled): ~10x training data | 7.26 | -1.04 | 5.57 | 9.28 | |
| | **Exp3b** | all_both_v1 | 20.37K / 9.07K | patch_tokens | 10 epochs / 3 warmup instead of 4 / 1 | 7.70 | -0.60 | 6.17 | 9.51 | |
| | **Exp3q** | all_both_v1 | 19.82K / 8.97K | patch_tokens | drop windows beyond ±800 µV or with a flat channel | 7.34 | -0.96 | 5.69 | 9.31 | |
| | **Exp3c** | all_both_v1 | 20.37K / 9.07K | patch_tokens | clip the input to ±800 µV inside the model | 7.29 | -1.01 | 5.66 | 9.22 | |
| | **Exp3v2** | all_both_v2 | 20.47K / 9.09K | patch_tokens | cohort v2: one TUEG patient split shared by every cohort (data bookkeeping only) | 7.42 | -0.88 | 5.46 | 9.75 | |
| 2 | **Exp3cond** | all_both_v2 | 20.47K / 9.09K | patch_tokens + path_label | status + pathology vectors concatenated to the pooled feature | 7.28 | -1.02 | 5.39 | 9.53 | |
| | **Exp3clscond** | all_both_v2 | 20.47K / 9.09K | cls_token + path_label | head on the class token instead of the mean patch token | 7.25 | -1.05 | 5.36 | 9.51 | |
| | **Exp4drop** | all_both_v3 | 25.61K / 12.02K | patch_tokens + path_label | cohort v3 (+ recordings missing only T1/T2) on 21 channels (T1/T2 dropped everywhere) | 7.28 | -1.02 | 5.26 | 9.68 | |
| | **Exp4base** | all_both_v3 | 25.61K / 12.02K | patch_tokens | cohort v3, 23 channels with missing T1/T2 left out per sample; speed settings; 6 / 2 epochs | 7.39 | -0.91 | 5.57 | 9.56 | mean of Exp4base, Exp4basec (same data, compact copy) |
| | **Exp4ctrl** | all_both_v3 | 25.61K / 12.02K | cls_token + path_label | class token + condition vectors, 10 / 3 epochs with early stopping | 7.30 | -1.00 | 5.35 | 9.63 | |
| | **Exp4aug** | all_both_v3 | 25.61K / 12.02K | cls_token + path_label | T1/T2 batch dropout (p 0.2) + consistency loss (λ 0.1); 6 / 2 epochs | 7.29 | -1.01 | 5.45 | 9.49 | |
| | **Exp4aug15** | all_both_v3 | 25.61K / 12.02K | cls_token + path_label | 15 / 5 epochs with early stopping (stopped at 11) | 7.17 | -1.13 | 5.38 | 9.29 | |
| | **Exp4anymiss** | all_both_v3_anymiss | 27.62K / 12.19K | cls_token + path_label | +2,003 training recordings missing A1/A2 or other channels; any 1-3 channels dropped (p 0.2) from epoch 4 (stopped at 12) | 7.21 | -1.09 | 5.47 | 9.30 | |
| | **Exp4noear** | all_both_v3_anymiss | 27.62K / 12.19K | cls_token + path_label | A1/A2/T1/T2 removed from every recording (19 channels), no channel augmentation; 8 / 2 epochs | 7.41 | -0.89 | 5.55 | 9.64 | |
| | **cvbase (fold mean)** | all_both_v3_cvcompact | 17.42K / 8.13K | patch_tokens | 5-fold CV of Exp4base (5 / 1 epochs) | 7.49 | -0.81 | 5.81 | 9.50 | mean of 5 fold models |
| | **◆ cvbase (ensemble)** | all_both_v3_cvcompact | 17.42K / 8.13K | patch_tokens | 5-fold CV of Exp4base (5 / 1 epochs) | 7.44 | -0.86 | 5.75 | 9.46 | median of 5 fold models |
| | **cvctrl (fold mean)** | all_both_v3_cvcompact | 17.42K / 8.13K | cls_token + path_label | class token + condition vectors, 10 / 3 epochs (CV of Exp4ctrl) | 7.28 | -1.02 | 5.53 | 9.37 | mean of 5 fold models |
| | **cvctrl (ensemble)** | all_both_v3_cvcompact | 17.42K / 8.13K | cls_token + path_label | class token + condition vectors, 10 / 3 epochs (CV of Exp4ctrl) | 7.10 | -1.20 | 5.31 | 9.23 | median of 5 fold models |
| 3 | **cvaug (fold mean)** | all_both_v3_cvcompact | 17.42K / 8.13K | cls_token + path_label | T1/T2 batch dropout + consistency loss λ 0.1 (CV of Exp4aug) | 7.33 | -0.97 | 5.55 | 9.44 | mean of 5 fold models |
| | **cvaug (ensemble)** | all_both_v3_cvcompact | 17.42K / 8.13K | cls_token + path_label | T1/T2 batch dropout + consistency loss λ 0.1 (CV of Exp4aug) | 7.23 | -1.07 | 5.43 | 9.37 | median of 5 fold models |
| | **cvaugval (fold mean)** | all_both_v3_cvcompact | 23.23K / 10.84K | cls_token + path_label | val-only CV: val = fold k, the other 4 folds train (~23.2K recordings), test = TUAB | 7.26 | -1.04 | 5.41 | 9.48 | mean of 5 fold models |
| | **cvaugval (ensemble)** | all_both_v3_cvcompact | 23.23K / 10.84K | cls_token + path_label | val-only CV: val = fold k, the other 4 folds train (~23.2K recordings), test = TUAB | 7.22 | -1.08 | 5.36 | 9.44 | median of 5 fold models |

## Reading Table 2.8.

- **Phase 1 (TUAB, 2,145 training recordings) stays at about 8.3 test MAE.** No phase-1 change beats D beyond noise (Table 2.3).

- **More data is the big step.** Adding every TUEG recording (Exp3, about 20k training recordings) takes the test MAE to about 7.3, roughly 1 year better than D. The step is largest on abnormal recordings.
- **Everything after that moves the test MAE by a few tenths:** cohort versions, channel handling, condition vectors, epochs.
- **The lowest test MAE overall is the `cvctrl` 5-model ensemble (7.10)**, although each of its fold models trains on fewer recordings (about 17.4k) than the single-split v3 runs (25.6k).
- **EEG-only:** the single-split Exp3 (7.26) scores below the `cvbase` ensemble (7.44). But phase-2 single runs were compared on the test set itself (Section 0), so their numbers are optimistic, and `cvbase` is the unselected estimate.

## Figure 2.1. TUAB test MAE of every run

Every run's TUAB test MAE (Eq. (0.1), years), in the order the runs were made, left to right. Dotted vertical lines separate phase 1 (Table 2.2), phase 2 (Table 2.5) and the CV runs (Table 2.7).

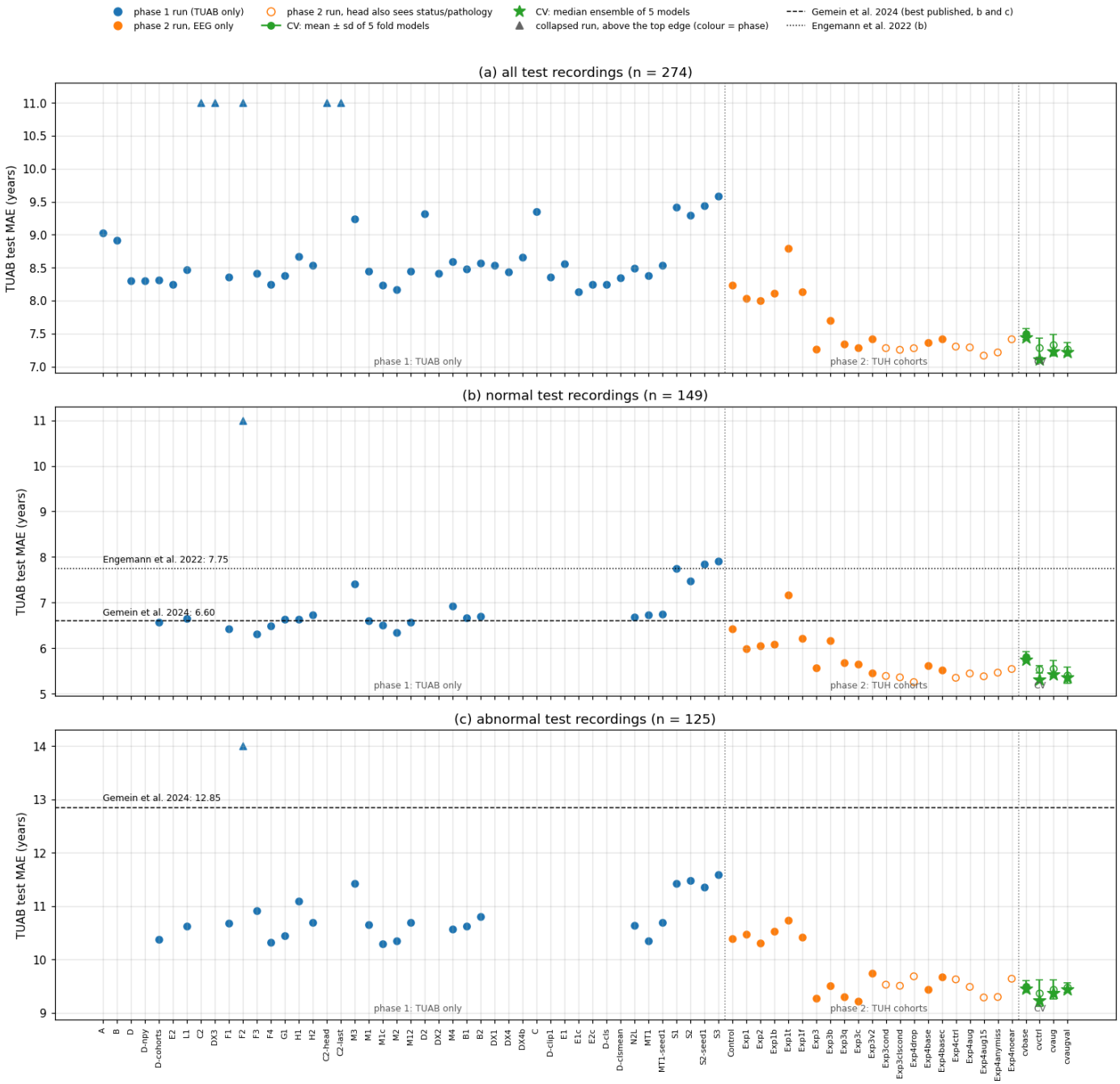
- (a) All 274 test recordings.
- (b) The 149 normal test recordings.
- (c) The 125 abnormal test recordings.

The 21 phase-1 runs without cohort metrics (Table 2.2) have no point in (b) or (c).

**Legend.**

- **Colour = phase:** blue phase 1 (TUAB only), orange phase 2 (TUH cohorts), green phase 3 (cross-validation).
- **Fill = what the head sees:**
  - filled marker: an EEG-only model;
  - hollow marker: the head also receives the recording's status and pathology vector (Eq. (3.6)), so the model is not EEG-only and its MAE is not comparable with published results.
- **Shape:**
  - circle: one model;
  - green circle with bars: the mean ± sample sd over the 5 CV fold models;
  - green star: the median ensemble of those 5 models (Eq. (3.8));
  - triangle: a collapsed run whose MAE is above the panel's top edge (11 years in (a) and (b), 14 in (c)). It is drawn at the edge so the axis stays readable; its true value is in Table 2.2.
- **Published results, black horizontal lines.** Both are the best published TUAB brain-age models. Their training data and splits differ from ours, so the lines are reference points, not a matched comparison.

- Dashed: Gemein et al. 2024, a TCN ensemble trained on TUEG normal recordings only. On TUAB's evaluation set it scores 6.60 on normal recordings (b) and 12.85 on abnormal ones (c), **after** their age-bias correction, which is applied with the test recordings' true ages; their raw normal MAE was 6.48. Our points are raw. Table 3.5 gives our numbers after the same kind of correction (marked †) and after a leak-free one. Their abnormal error comes from a model that never saw an abnormal recording; ours do.
- Dotted: Engemann et al. 2022, the best earlier result, 7.75 on TUAB normal recordings (b) (10-fold cross-validation).
- Panel (a) has no line: neither paper reports one MAE over all TUAB recordings.



## Table 2.9. The runs around the best result

The 12 runs with the lowest TUAB test MAE across all phases (Tables 2.2–2.7):

- `kind`:
  - `single split`: one run;
  - `CV mean`: mean over 5 fold models;
  - `CV ensemble`: median of 5 fold models.
- `run`: linked to its ClearML experiment (for CV: the project of its folds).
- `feature space` (Eq. (2.13)): what the head reads; EEG-only rows have no `path_label`.
- `test all` / `normal` / `abnormal`: MAE (Eq. (0.1)) on the **TUAB test set only**, all 274 recordings and its 149 normal / 125 abnormal ones. The same set is used in every kind of row: for the CV rows it is the held-out TUAB test set, which no fold trains on (Table 2.7). The status is TUAB's own normal/abnormal label.

TUEG columns (MAE, Eq. (0.1), years, on **TUEG recordings only**, those outside TUAB, that the run never trained on):

- `TUEG all` / `TUEG normal` / `TUEG abnormal`: all of them, and by status (Eq. (1.2));
- one column per NEDC diagnosis group of the pathology vector (Eq. (2.13), Table 1.9). A recording counts in every group it is flagged with, so groups overlap: `cl. seiz.` / `el. seiz.` clinical / electrographic seizure, `epil.` epilepsy, `stroke`, `TBI`, `subst.` substance use, `bipol.` bipolar, `anx.` anxiety, `depr.` depression, `schiz.` schizophrenia, `dement.` dementia (with Alzheimer's), `MS`, `other` (Parkinson's, ADHD).

Where the recordings come from:

- **Single-split run:** its own **validation** split, re-scored from its checkpoint (`scripts/eval_runs.py`). That is about 1.2–2.9K TUEG recordings, depending on the cohort.
- **CV row:** the out-of-fold predictions of the whole CV pool (26,685 TUEG recordings): the test folds, or for `cvaugval` its val folds, which also chose its epoch. The `CV mean` and `CV ensemble` rows therefore show the same TUEG values.
- **Phase-1 run:** trained and validated on TUAB only, so it has no TUEG numbers ("–").
- A group with fewer than 20 recordings for a run is "–". Many diagnosis groups are that small in a single validation split, while the CV pool has 36–2,064 per group.
- The single-split validation sets differ between cohorts, so their TUEG columns compare only within a cohort. The CV columns compare across the CV experiments.

The table is shown in two parts with the same rows: the TUAB test and TUEG status columns first, then the TUEG diagnosis-group columns.

Rows: the 12 lowest `test all`, listed single-split runs first, then the CV rows, each block ranked on `test all`. The best EEG-only run of each kind is marked `◆`. Source: `docs/data/tueg_pathology_mae.csv`.

| run | kind | phase | feature space | test all | normal | abnormal | TUEG all | TUEG normal | TUEG abnormal |
|---|---|---|---|---|---|---|---|---|---|
| Exp4aug15 | single split | 2 | cls_token + path_label | 7.17 | 5.38 | 9.29 | 7.84 | 7.26 | 8.24 |
| Exp4anymiss | single split | 2 | cls_token + path_label | 7.21 | 5.47 | 9.30 | 7.78 | 7.25 | 8.12 |
| Exp3clscond | single split | 2 | cls_token + path_label | 7.25 | 5.36 | 9.51 | 7.98 | 7.47 | 8.22 |
| ◆ Exp3 | single split | 2 | patch_tokens | 7.26 | 5.57 | 9.28 | 7.82 | 6.91 | 8.08 |
| Exp4drop | single split | 2 | patch_tokens + path_label | 7.28 | 5.26 | 9.68 | 7.72 | 6.97 | 8.27 |
| Exp3cond | single split | 2 | patch_tokens + path_label | 7.28 | 5.39 | 9.53 | 7.90 | 7.35 | 8.20 |
| Exp3c | single split | 2 | patch_tokens | 7.29 | 5.66 | 9.22 | 7.84 | 6.94 | 8.10 |
| cvaugval | CV mean | 3 | cls_token + path_label | 7.26 | 5.41 | 9.48 | 7.99 | 7.36 | 8.32 |
| cvctrl | CV mean | 3 | cls_token + path_label | 7.28 | 5.53 | 9.37 | 8.17 | 7.51 | 8.50 |
| cvctrl | CV ensemble | 3 | cls_token + path_label | 7.10 | 5.31 | 9.23 | 8.17 | 7.51 | 8.50 |
| cvaugval | CV ensemble | 3 | cls_token + path_label | 7.22 | 5.36 | 9.44 | 7.99 | 7.36 | 8.32 |
| cvaug | CV ensemble | 3 | cls_token + path_label | 7.23 | 5.43 | 9.37 | 8.14 | 7.48 | 8.50 |

| run | kind | cl. seiz. | el. seiz. | epil. | stroke | TBI | subst. | bipol. | anx. | depr. | schiz. | dement. | MS | other |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Exp4aug15 | single split | 7.07 | – | 8.46 | 7.21 | 8.53 | 8.42 | 7.48 | – | – | – | – | – | – |
| Exp4anymiss | single split | 7.02 | – | 8.35 | 7.08 | 8.84 | 7.60 | 7.46 | – | – | – | – | – | – |
| Exp3clscond | single split | 7.43 | – | 8.43 | 7.49 | 10.25 | 7.78 | – | – | – | – | – | – | – |
| ◆ Exp3 | single split | 6.70 | 8.58 | 5.93 | 9.79 | 10.49 | 9.20 | – | – | – | – | – | – | – |
| Exp4drop | single split | 7.03 | – | 9.08 | 7.76 | 11.09 | 8.32 | 6.77 | – | – | – | – | – | – |
| Exp3cond | single split | 7.13 | – | 8.49 | 7.89 | 10.52 | 7.33 | – | – | – | – | – | – | – |
| Exp3c | single split | 6.68 | 9.29 | 6.07 | 9.63 | 10.65 | 9.23 | – | – | – | – | – | – | – |
| cvaugval | CV mean | 7.31 | 8.36 | 7.85 | 8.69 | 10.03 | 7.38 | 5.98 | 6.17 | 7.16 | 8.39 | 7.88 | 10.26 | 7.14 |
| cvctrl | CV mean | 7.44 | 9.19 | 7.91 | 8.85 | 10.38 | 7.65 | 5.79 | 6.17 | 7.63 | 8.63 | 7.45 | 11.64 | 7.38 |
| cvctrl | CV ensemble | 7.44 | 9.19 | 7.91 | 8.85 | 10.38 | 7.65 | 5.79 | 6.17 | 7.63 | 8.63 | 7.45 | 11.64 | 7.38 |
| cvaugval | CV ensemble | 7.31 | 8.36 | 7.85 | 8.69 | 10.03 | 7.38 | 5.98 | 6.17 | 7.16 | 8.39 | 7.88 | 10.26 | 7.14 |
| cvaug | CV ensemble | 7.49 | 8.71 | 7.85 | 8.79 | 10.61 | 7.58 | 5.93 | 6.30 | 7.28 | 8.94 | 7.50 | 10.80 | 6.74 |

# 3. The best experiment

**Which run is best.** Single-split runs were compared on the TUAB test set itself (Section 0). Only the CV runs have an error bar, and their CV test / TUEG out-of-fold errors were never used for selection. So "best" is decided among the CV runs:

- **Best overall:** `cvctrl` (= Exp4ctrl cross-validated). It has the lowest TUAB test MAE, as a single fold model and as a 5-model ensemble (Table 2.7).
  - `cvaug` adds T1/T2 channel-dropout augmentation and ties with it: the differences are within noise in Table 5.2 (C2).
  - The simpler `cvctrl` is therefore the recommended configuration.
- **Best EEG-only:** `cvbase`**.** Its head sees only the EEG. `cvctrl` and `cvaug` also see the recording's status (normal / abnormal / no label) and 13 diagnosis flags from the clinical sheet. A deployed brain-age model would not have these, and published TUAB results do not use them. `cvbase` is therefore the number to compare with the literature (Table 3.3).

Both are LaBraM-base fine-tuned on `all_both_v3` with the M1 recipe. They differ in three settings (Table 3.2): class-token vs mean pooling, the condition vector, and 10 vs 5 epochs.

## What the model computes

**Input and target.** The raw EEG $u$ (µV) of a 10 s window is scaled to $x = u/100$ and cut into 1 s patches $x_{c,a,t}$:

- channel $c = 1..C$, with $C = 23$ minus the missing T1/T2 of that recording;

- patch $a = 1..10$, sample $t = 1..200$.

The common average reference removes what all channels share (`labram/data/preprocess.py::common_average_reference`):

$$\tilde{x}_{c,a,t} = x_{c,a,t} - \frac{1}{C}\sum_{c'=1}^{C} x_{c',a,t} \tag{3.1}$$

The age $y$ (years) is z-scored with the training windows' mean $\mu$ and standard deviation $\sigma$ (`labram/data/tuh_datasets.py::TUABAgeLoader._age_target`):

$$z = \frac{y - \mu}{\sigma} \tag{3.2}$$

**Mixup.** Each training batch is blended with a permuted copy of itself, using one $\lambda \sim \mathrm{Beta}(\alpha, \alpha)$ per batch with $\alpha = 0.4$ (`labram/train/mixup.py::mixup_batch`). The condition vector $c$ travels with the target, so it is blended too:

$$\bar{x} = \lambda x + (1-\lambda)\, x_\pi, \qquad \bar{z} = \lambda z + (1-\lambda)\, z_\pi, \qquad \bar{c} = \lambda c + (1-\lambda)\, c_\pi \tag{3.3}$$

**Loss.** Huber loss (Eq. (2.3)) with $\delta = 1$ on the z-scored residual $r = \hat{z} - \bar{z}$ (`labram/losses/regression.py::build_regression_criterion`):

$$\ell_\delta(r) = \begin{cases} \frac{1}{2} r^2 & |r| \le \delta \\ \delta\left(|r| - \frac{1}{2}\delta\right) & \text{otherwise} \end{cases} \tag{3.4}$$

**Encoder and head.** $f_\theta$ is the 12-block LaBraM-base transformer, initialized from the released `labram-base.pth`. Its output is $h \in \mathbb{R}^{200}$:

- the final LayerNorm's class token (`cvctrl`, `cvaug`);
- the mean over patch tokens (`cvbase`).

$$h = f_\theta(\tilde{x}) \tag{3.5}$$

The linear head reads $h$, concatenated with a 32-d projection of the 17-d condition vector $c \in \{0,1\}^{17}$ (3 one-hot status + 14 multi-hot pathology groups, `conditions_v1.npz`) when conditions are on (`labram/models/neural_transformer.py`, `model.session_features=head`):

$$\hat{z} = w^\top \begin{bmatrix} h \\ W_c\, c \end{bmatrix} + b, \qquad W_c \in \mathbb{R}^{32\times 17} \tag{3.6}$$

**Optimizer and schedule.** AdamW with weight decay 0.05. Layer-wise lr decay $\rho = 0.65$ gives block $l$ of $L = 12$ the rate $\mathrm{lr} \cdot \rho^{L+1-l}$ (`labram/optim_factory.py::LayerDecayValueAssigner`). The rate warms up linearly, then follows a cosine down to $10^{-6}$ (`labram/utils/training.py`).

**Scoring.** A recording's prediction is the de-normalized mean over its evaluation windows $W_i$ (the first 5 minutes after the trim, 30 windows; `evaluation.agg_windows=mean`):

$$\hat{y}_i = \mu + \sigma \, \frac{1}{|W_i|} \sum_{w \in W_i} \hat{z}_w \tag{3.7}$$

The fold ensemble takes the median over the $K = 5$ fold models (`labram/eval/cv_compare.py::ensemble`):

$$\hat{y}_i^{\text{ens}} = \text{median}_{k=1..K} \; \hat{y}_{i,k} \tag{3.8}$$

## Table 3.1. The best configuration (`cvctrl`), step by step

One row per stage of the pipeline:

- `setting`: the configuration field;
- `cvctrl`: its value, read from the configuration rebuilt by `rs.resolve_config` (`finetune_tuab_age.json` + the run's overrides, exactly as the trainer builds it);
- `cvbase`: the EEG-only variant's value, where it differs;
- `equation`: where the setting appears above.

| stage | what | setting | cvctrl | cvbase | equation |
|---|---|---|---|---|---|
| data | cohort | data.data_path | all_both_v3_cvcompact | | |
| | 5-fold CV, grouped by | cross_validation.split_by | subject | | |
| | CV pool | cross_validation.pool | train_val | | |
| windows | window length (s) | data.window_sec | 10 | | |
| | trim start / end (s) | data.trim_start_sec / data.trim_end_sec | 60 / 60 | | |
| | train minutes per recording per epoch | data.train_minutes_per_epoch | 5 | | |
| | eval minutes per recording | data.eval_minutes | 5 | | (3.7) |
| channels | channels left out where missing | channel_drop.channels | ["T1", "T2"] | | |
| | T1/T2 dropout probability (augmentation) | channel_drop.prob | 0 | | |
| input | common average reference | labram_plus.common_average_reference | True | | (3.1) |
| | per-patch z-scoring | labram_plus.z_score_patches | False | | |
| encoder | model | model.model | labram_base_patch200_200 | | (3.5) |
| | initial weights | finetune_checkpoint.finetune | labram-base.pth | | |
| | drop path | model.drop_path | 0.1 | | |
| | fused attention (speed only) | model.fused_attention | True | | |
| head | mean pooling (False = class token) | model.use_mean_pooling | False | True | (3.5) |
| | condition vector | model.session_features / data.session_features | head / conditions_v1.npz | off / | (3.6) |
| | condition projection size | model.session_head_dim | 32 | | (3.6) |
| | output init scale | model.init_scale | 0.001 | | |
| target | task | model.task | regression | | (3.2) |
| loss | regression loss / δ | loss.regression_loss / loss.huber_delta | huber / 1 | | (3.4) |
| augmentation | mixup α (on) | mixup.alpha / mixup.enabled | 0.4 / True | | (3.3) |
| optimizer | AdamW lr / batch | optimizer.lr / trainer.batch_size | 0.0002 / 128 | | |
| | weight decay / layer decay | optimizer.weight_decay / optimizer.layer_decay | 0.05 / 0.65 | | |
| schedule | epochs / warmup | trainer.epochs / optimizer.warmup_epochs | 10 / 3 | 5 / 1 | |
| | min lr | optimizer.min_lr | 1e-06 | | |
| selection | checkpoint | – | lowest val MAE (recording level) | | |
| scoring | window aggregation / case unit | evaluation.agg_windows / evaluation.agg_case_by | mean / recording | | (3.7) |

## Table 3.2. The three cross-validated configurations

Every configuration field on which `cvbase`, `cvctrl` and `cvaug` disagree (`labram/eval/research_summary.py::config_diff`, bookkeeping fields excluded).

| field | cvbase | cvctrl | cvaug |
|---|---|---|---|
| **channel_drop.consistency_weight** | 0 | 0 | 0.1 |
| **channel_drop.prob** | 0 | 0 | 0.2 |
| **data.session_features** | | conditions_v1.npz | conditions_v1.npz |
| **model.session_features** | off | head | head |
| **model.use_mean_pooling** | True | False | False |
| **optimizer.warmup_epochs** | 1 | 3 | 3 |
| **trainer.epochs** | 5 | 10 | 10 |

## Table 3.3. Results of the best experiment by evaluation set and cohort

`cvctrl` (best overall) and `cvbase` (best EEG-only), mean ± sample sd over the 5 fold models of the MAE (Eq. (0.1)) and the mean $R^2$ (Eq. (0.2)), per evaluation set and status cohort:

- `CV val`, `CV test`: the fold's own held-out folds, TUAB and TUEG recordings together;
- `TUEG test`: the TUEG recordings of each fold's test fold;
- `TUAB test`: the official test set, $n = 274$.

$R^2$ is relative to each cohort's own age spread, so it does not compare across cohorts. The last row gives the median ensemble (Eq. (3.8)) on the TUAB test set when the per-recording predictions are available. For reference, the best published TUAB normal-recording result is MAE 6.60, $R^2$ 0.73 (Gemein et al. 2024, trained on TUAB normal recordings only, on its own patient-wise re-split of TUAB).

| set | cohort | cvctrl MAE | cvctrl R² | cvbase MAE | cvbase R² |
|---|---|---|---|---|---|
| **CV val** | **all** | 8.09 ± 0.12 | 0.64 | 8.16 ± 0.11 | 0.64 |
| | **normal** | 7.33 ± 0.20 | 0.68 | 7.45 ± 0.17 | 0.68 |
| | **abnormal** | 8.49 ± 0.14 | 0.60 | 8.57 ± 0.13 | 0.59 |
| **CV test** | **all** | 8.08 ± 0.13 | 0.64 | 8.18 ± 0.15 | 0.63 |
| | **normal** | 7.39 ± 0.18 | 0.68 | 7.47 ± 0.14 | 0.67 |
| | **abnormal** | 8.46 ± 0.11 | 0.60 | 8.60 ± 0.15 | 0.59 |
| **TUEG test** | **all** | 8.17 ± 0.15 | 0.63 | 8.27 ± 0.15 | 0.62 |
| | **normal** | 7.52 ± 0.19 | 0.67 | 7.59 ± 0.16 | 0.67 |
| | **abnormal** | 8.50 ± 0.10 | 0.59 | 8.65 ± 0.13 | 0.58 |
| **TUAB test** | **all** | 7.28 ± 0.14 | 0.71 | 7.49 ± 0.09 | 0.70 |
| | **normal** | 5.53 ± 0.08 | 0.80 | 5.81 ± 0.10 | 0.79 |
| | **abnormal** | 9.37 ± 0.25 | 0.52 | 9.50 ± 0.11 | 0.51 |
| **TUAB test, ensemble** | **all** | 7.10 | 0.72 | 7.44 | 0.70 |
| | **normal** | 5.31 | 0.82 | 5.75 | 0.80 |
| | **abnormal** | 9.23 | 0.53 | 9.46 | 0.52 |

## Table 3.4. Predicted versus true age by subcohort and status

The three cross-validated models, scored per recording on three sets of recordings (`rs.prediction_sets`, from the per-recording CV predictions):

- `TUAB test, ensemble`: the 274 TUAB test recordings, predicted by the median of the 5 fold models (Eq. (3.8)).
- `TUAB out-of-fold`: the TUAB recordings of the CV pool (TUAB train + val), each predicted by the one fold model that never saw its patient.
- `TUEG out-of-fold`: the same for the pool's TUEG recordings that are not in TUAB.

`status` is the recording's normal / abnormal / no-label status (Eq. (1.2)), and `all` is every recording of the set. Per model, the columns are:

- `MAE` (Eq. (0.1), years);
- `bias` = mean(predicted − true) (Eq. (0.3), years): the signed drift. It is negative when the model predicts too young and positive when too old, so closer to 0 is better;
- $R^2$ (Eq. (0.2));
- the bias slope $\beta$ (Eq. (0.5)).

All come from `rs.prediction_metrics`. Needs the per-recording CV predictions and the TUH index.

| | | | cvbase | | | | cvctrl | | | | cvaug | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | n | MAE | bias | R² | β | MAE | bias | R² | β | MAE | bias | R² | β |
| set | status | | | | | | | | | | | | | |
| TUAB test, ensemble | all | 274 | 7.44 | -1.21 | 0.70 | -0.37 | 7.10 | -0.77 | 0.72 | -0.35 | 7.23 | -0.76 | 0.72 | -0.34 |
| | normal | 149 | 5.75 | -0.56 | 0.80 | -0.30 | 5.31 | -0.58 | 0.82 | -0.29 | 5.43 | -0.67 | 0.81 | -0.29 |
| | abnormal | 125 | 9.46 | -1.99 | 0.52 | -0.50 | 9.23 | -1.00 | 0.53 | -0.50 | 9.37 | -0.88 | 0.54 | -0.49 |
| TUAB out-of-fold | all | 2350 | 7.18 | -0.96 | 0.70 | -0.35 | 7.11 | -0.45 | 0.71 | -0.34 | 7.12 | -0.86 | 0.71 | -0.33 |
| | normal | 1258 | 6.55 | -0.85 | 0.70 | -0.38 | 6.37 | -0.61 | 0.71 | -0.34 | 6.41 | -1.15 | 0.71 | -0.35 |
| | abnormal | 1092 | 7.91 | -1.08 | 0.63 | -0.41 | 7.97 | -0.27 | 0.63 | -0.41 | 7.93 | -0.52 | 0.63 | -0.41 |
| TUEG out-of-fold | all | 26685 | 8.27 | -0.29 | 0.62 | -0.42 | 8.17 | 0.02 | 0.63 | -0.39 | 8.14 | -0.05 | 0.63 | -0.39 |
| | normal | 10197 | 7.59 | -0.24 | 0.67 | -0.38 | 7.51 | -0.30 | 0.68 | -0.36 | 7.48 | -0.45 | 0.68 | -0.36 |
| | abnormal | 14298 | 8.65 | -0.48 | 0.58 | -0.45 | 8.50 | 0.20 | 0.59 | -0.43 | 8.50 | 0.16 | 0.59 | -0.43 |
| | no label | 2190 | 8.99 | 0.78 | 0.51 | -0.50 | 9.07 | 0.32 | 0.51 | -0.50 | 8.94 | 0.38 | 0.51 | -0.49 |

## Figure 3.1. Predicted versus true age, per model and subcohort

Predicted age $\hat{y}_i$ against true age $y_i$ (years), one point per recording; the data of Table 3.4. One row per cross-validated model, and two subcohorts per row:

- left column: the 274 TUAB test recordings, predicted by the 5-model median ensemble;
- right column: the TUEG out-of-fold recordings ($n$ = 26,685), each predicted by the fold model that held it out.

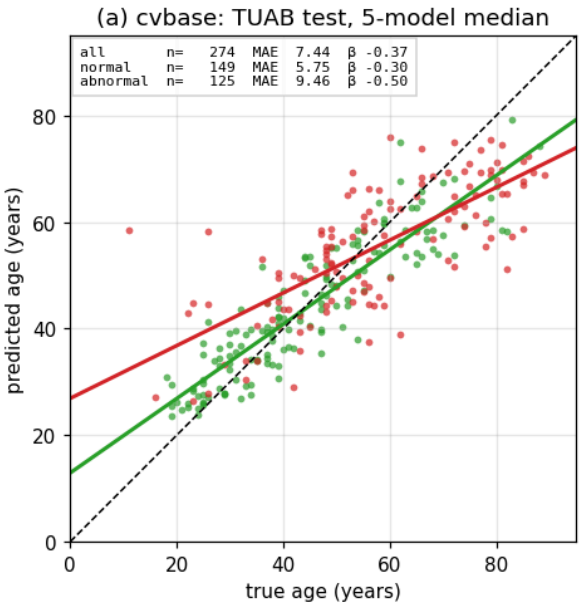
The panels are:

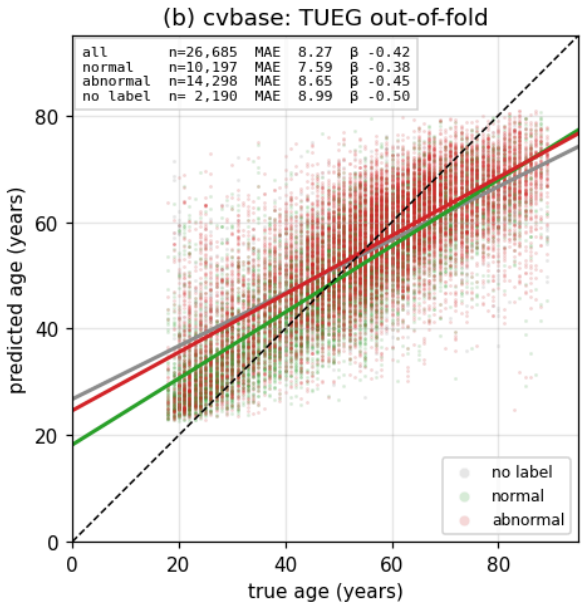
- (a), (b) `cvbase`, the best EEG-only model;
- (c), (d) `cvctrl`, the best overall;
- (e), (f) `cvaug`.

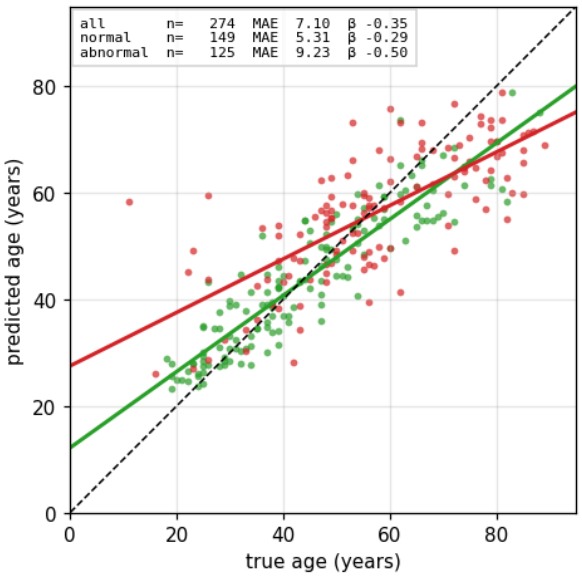
**Colour = status** (Eq. (1.2)): green normal, red abnormal, grey no label. The no-label group occurs only in TUEG.

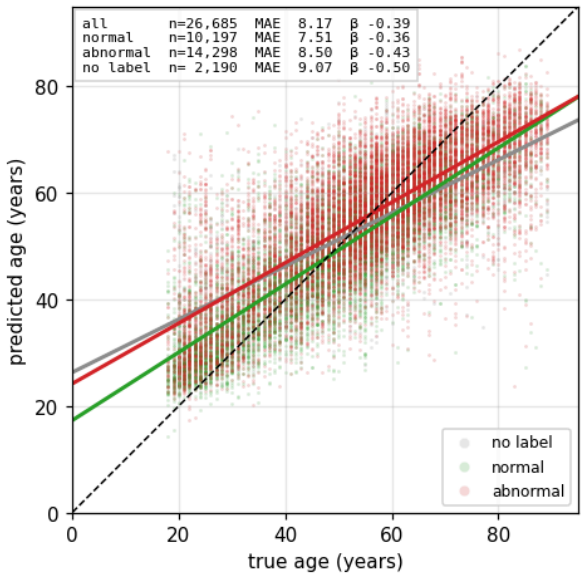
- **Dashed black diagonal:** a perfect prediction, $\hat{y} = y$.

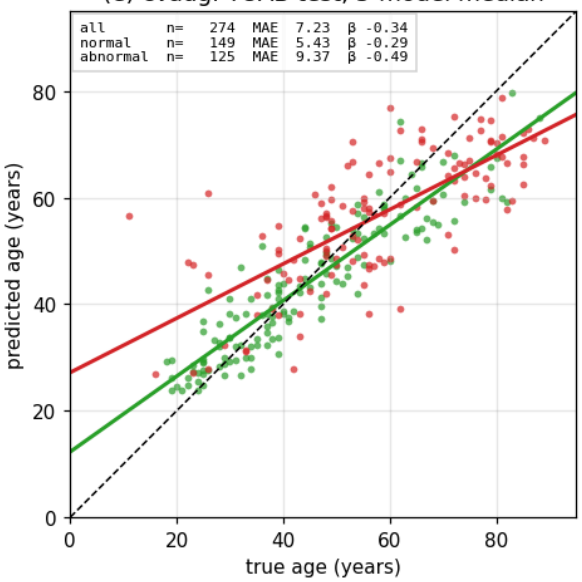
- **Coloured lines:** least-squares fits of $\hat{y}$ on $y$ per status. A fit flatter than the diagonal is the regression to the mean that $\beta$ measures: the fit's slope is $1 + \beta$ (Eq. (0.5)).
- **Box in each panel:** $n$, MAE and $\beta$ per status, and for all recordings together.

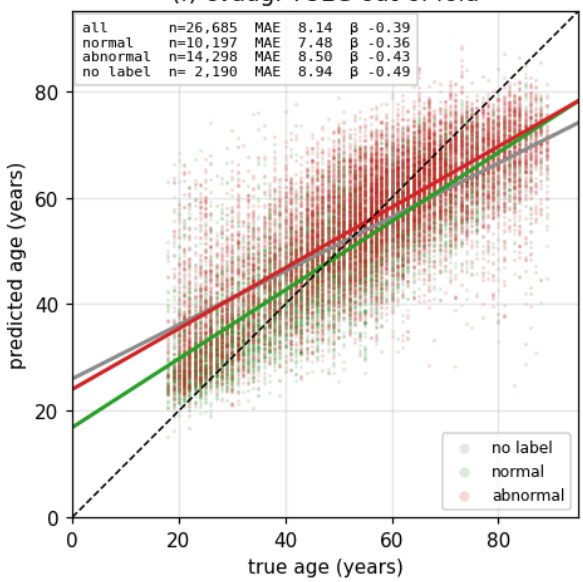

(a) cvbase: TUAB test, 5-model median
all n= 274 MAE 7.44 β -0.37
normal n= 149 MAE 5.75 β -0.30
abnormal n= 125 MAE 9.46 β -0.50
predicted age (years)
true age (years)


(b) cvbase: TUEG out-of-fold
all n=26,685 MAE 8.27 β -0.42
normal n=10,197 MAE 7.59 β -0.38
abnormal n=14,298 MAE 8.65 β -0.45
no label n= 2,190 MAE 8.99 β -0.50
no label
normal
abnormal
predicted age (years)
true age (years)


(c) cvctrl: TUAB test, 5-model median
all n= 274 MAE 7.10 β -0.35
normal n= 149 MAE 5.31 β -0.29
abnormal n= 125 MAE 9.23 β -0.50
predicted age (years)
true age (years)


(d) cvctrl: TUEG out-of-fold
all n=26,685 MAE 8.17 β -0.39
normal n=10,197 MAE 7.51 β -0.36
abnormal n=14,298 MAE 8.50 β -0.43
no label n= 2,190 MAE 9.07 β -0.50
no label
normal
abnormal
predicted age (years)
true age (years)


(e) cvaug: TUAB test, 5-model median
all n= 274 MAE 7.23 β -0.34
normal n= 149 MAE 5.43 β -0.29
abnormal n= 125 MAE 9.37 β -0.49
predicted age (years)
true age (years)


(f) cvaug: TUEG out-of-fold
all n=26,685 MAE 8.14 β -0.39
normal n=10,197 MAE 7.48 β -0.36
abnormal n=14,298 MAE 8.50 β -0.43
no label n= 2,190 MAE 8.94 β -0.49
no label
normal
abnormal
predicted age (years)
true age (years)

**Reading Table 3.4 and Figure 3.1.**

- **Abnormal recordings are harder in every set.**
  - On TUAB test, their MAE is 3.7–3.9 years above the normal recordings'.
  - In TUEG out-of-fold the gap is about 1 year (0.99–1.06). The no-label recordings are the hardest there (MAE 8.9–9.1).
- **Every model regresses to the mean** ($\beta \approx -0.3$ to $-0.5$). The fit lines in Figure 3.1 are visibly flatter than the diagonal: old patients are predicted too young and young ones too old. The effect is strongest for the abnormal and no-label recordings ($\beta \approx -0.43$ to $-0.50$ ).
- **Bias.** The overall drift is small: within ±1.2 years on TUAB test and ±0.3 on TUEG. But the EEG-only `cvbase` predicts abnormal TUAB test recordings 2.0 years too young on average, against 0.9–1.0 for `cvctrl` / `cvaug`.
- **TUEG is harder than TUAB.** TUEG out-of-fold MAE is 8.1–8.3, against 7.1–7.2 for the TUAB recordings of the same pool. Its normal recordings are worse too: 7.5–7.6 against 6.4–6.6. TUEG's status comes from report text rather than TUAB's curated folders. It also covers a broad clinical population: 38 % of its recordings in the clinical sheet list a sedating drug (Table 1.9).
- **cvctrl and cvaug beat cvbase on the TUAB test ensemble** by 0.2–0.35 years. Their gain is larger on normal recordings (0.3–0.4 years) than on abnormal ones (0.1–0.2).

## Table 3.5. TUAB test MAE before and after age-bias correction, and published / classical methods

**Every other number in this notebook is before bias correction.** That includes the MAEs of Sections 0–5 and Figure 2.1, except the `cal.` columns of Tables 2.5–2.7, which use Eq. (3.11) below. They score the model's raw prediction $\hat{y}_i$. Brain-age studies often also report a *bias-corrected* error. Gemein et al. 2024 does: its 6.60 / 12.85 are after correction, and its raw normal MAE was 6.48. This table puts our CV models on that footing, and lists the other methods evaluated on the TUAB test set.

**How a correction is made: three steps, and what each one sees.**

1. **Predict.** Each CV experiment's 5 fold models predict every TUAB test recording, and the ensemble is their median (Eq. (3.8)). The TUAB test set is in no fold. No fold model trained on it or chose its epoch on it (the epoch is chosen on the fold's val fold).
2. **Fit, on out-of-fold predictions only.** Each recording of the CV pool (TUAB train + val, TUEG train + val; 29,035 recordings of 13,550 patients) is predicted by the one fold model whose in-pool **test fold** held it out. These are the out-of-fold predictions. They are not the val folds that chose the epoch. The correction is fitted on them with *their* true ages, which is allowed: none of them is a TUAB test recording and none shares a patient with one (Table 3.6). `normal` in a column name means the fit uses only the normal out-of-fold recordings, as Gemein et al. do.
3. **Apply to the TUAB test predictions.** This is where the methods differ:

- **Recalibration** (Eq. (3.11)) and **Cole** (Eq. (3.10)) map each prediction $\hat{y}_i$ to a corrected one using $\hat{y}_i$ alone. They never read a TUAB test true age, so their MAE is a valid test error.
- The **gap correction** (Eq. (3.9)) subtracts the fitted age trend *evaluated at the test recording's true age* $y_i$. It **does use the TUAB test labels** (Table 3.6: permuting them changes the corrected predictions by more than 25 years). Its MAE is therefore not a prediction error. It is the size of the brain-age gap after its age trend is removed, the quantity Gemein et al. and much of the brain-age literature call "corrected MAE". It is shown only to compare with them, marked †, and is used nowhere else in the notebook.

The three corrections ( `rs.bias_corrected_test` , `rs.cole_corrected_test` , `rs.recalibrated_test` ):

$$e_i = \hat{y}_i - y_i \approx p(y_i), \qquad \hat{y}_i^{c} = \hat{y}_i - p(y_i) \tag{3.9}$$

where $p$ is a linear or quadratic polynomial fitted by least squares on the out-of-fold pairs $(y_j, e_j)$ and evaluated at the test recording's true age $y_i$;

$$\hat{y}_i' = (\hat{y}_i - c) \,/\, a, \qquad (a, c) = \arg\min \textstyle\sum_j (\hat{y}_j - a\, y_j - c)^2 \tag{3.10}$$

the inverse of the out-of-fold line $\hat{y} \approx a\, y + c$ (Cole et al.);

$$\hat{y}_i'' = a'\, \hat{y}_i + b', \qquad (a', b') = \arg\min \textstyle\sum_j (y_j - a'\, \hat{y}_j - b')^2 \tag{3.11}$$

the least-squares map from prediction to age over the out-of-fold recordings $j$. Eq. (3.11) gives the best linear estimate of the age from the prediction. It lowers the error, but it does not remove the age trend of the gap (it shrinks predictions further toward the mean). Eq. (3.10) removes that trend but amplifies the noise of the prediction by $1/a$.

**Which CV experiments** (each name links to its ClearML project; Table 2.7):

- **`cvbase`** , **`cvctrl`** , **`cvaug`** use the scheme with an in-pool test fold: for fold $k$, test = fold $k$, val = fold $k+1$, and three folds train. Their out-of-fold predictions, and so the fit set, are the test-fold predictions. These recordings never trained a model and never chose an epoch.
- **`cvaugval`** uses the val-only scheme: val = fold $k$, the other four folds train, and the test set is TUAB. Its out-of-fold predictions, and so its fit set, are the **val-fold** predictions ( `rs.CV_OOF_SPLIT` ). They are its best checkpoint's, so the same recordings also chose its epoch (epoch 7 in four folds, the last epoch, 9, in one). That makes the fit set slightly optimistic, but it still holds no TUAB test recording or patient (Table 3.6).

**`gap quad, normal †`** , **step by step** (with `cvctrl` 's numbers):

1. **Fit set.** The out-of-fold predictions of the **normal** recordings of the CV pool: 11,455 recordings of 6,387 patients. None is a TUAB test recording and none shares a patient with

one (Table 3.6).
2. **Fit the age trend of the error.** Least squares of the error $e_j = \hat{y}_j - y_j$ on a quadratic in the true age over that set: $p(y) = q_2 y^2 + q_1 y + q_0$. For `cvctrl` this gives $q_2 = -0.00257$, $q_1 = -0.102$, $q_0 = 11.36$. So $p(30) = +6.0$ and $p(75) = -10.8$: even on normal recordings, the model predicts 30-year-olds 6 years too old and 75-year-olds 11 years too young. That is regression to the mean, the trend the correction removes.
3. **Apply it with the test recording's true age**, Eq. (3.9): $\hat{y}_i^c = \hat{y}_i - p(y_i)$. Example: an abnormal TUAB test recording with true age 75 is predicted 70.4 (raw error −4.6). Then $\hat{y}^c = 70.4 - p(75) = 70.4 + 10.8 = 81.2$, a corrected gap of **+6.2 years**: this brain looks 6 years older than the model's prediction for a typical normal 75-year-old.
4. **Score:** MAE (Eq. (0.1)) of the corrected predictions, on all, normal or abnormal TUAB test recordings.

**Why this is not a data leak, and why it is still marked †.**

- **Nothing about TUAB test is fitted.** The fold models, their epoch choice and $p$ are all fitted without any TUAB test recording or patient (Table 3.6: 0 and 0).
- **The true age is used the way a brain-age study uses it.** In brain-age work the chronological age is always known, also when the model is applied to a new patient. The quantity of interest is the **brain-age gap** $\hat{y} - y$ (does this brain look older than it is?), not an estimate of the age. Step 3 removes from the gap the part that age alone explains. This is standard practice (Smith et al. 2019; de Lange & Cole 2020). **Gemein et al. 2024** do exactly this: a quadratic fitted on normal recordings, applied with the true ages. Their 6.60 (normal) / 12.85 (abnormal) are of this kind, which is why the `gap quad, normal †` column is the one to compare with them.
- **But it is no longer an age prediction.** $\hat{y}^c$ uses $y_i$, so its MAE measures the remaining gap. It compares only with other gap-corrected numbers (Gemein's), never with a raw MAE or with the leak-free recalibration. Table 3.6 shows this directly: permuting the test ages moves these predictions, and moves no other column.

**Columns.** TUAB test MAE (Eq. (0.1), years) per model and status:

- `raw`: no correction (as in Table 3.4);
- `recal, normal`: Eq. (3.11), fitted on the normal out-of-fold recordings. No test label;
- `cole, normal`: Eq. (3.10), fitted on the normal out-of-fold recordings. No test label;
- `gap lin, all †` / `gap quad, normal †`: Eq. (3.9), linear on all out-of-fold recordings or quadratic on the normal ones (Gemein et al.'s choice). † = **uses the TUAB test true age**;
- `bias, gap quad normal †`: the mean gap (Eq. (0.3)) after that correction. A positive value means the recordings look older than they are.

**Other methods**, with each source's own numbers; "–" means the source does not report it:

- **Gemein et al. 2024:** a TCN ensemble trained on TUEG normal recordings, scored on TUAB's evaluation set. Raw normal 6.48; after their quadratic gap correction (Eq. (3.9), so with the test ages), 6.60 normal and 12.85 abnormal.

- **Engemann et al. 2022:** the M/EEG brain-age benchmark. Best model 7.75 on TUAB normal recordings, raw, 10-fold CV on the TUAB normal training recordings (not the official evaluation set).
- **Handcrafted features + ridge** (this repo, `notebooks/eeg_feature_extraction_tuab.ipynb` §10). This is the classical approach of Engemann et al.: filterbank-Riemann, spectral, wavelet and other EEG features with a ridge regression. It is scored raw on **the same 274 TUAB test recordings**, and trained either on TUAB normal recordings only (the standard brain-age setup) or on all TUAB recordings.

| model | status | raw | recal, normal | cole, normal | gap lin, all † | gap quad, normal † | bias, gap quad normal † |
|---|---|---|---|---|---|---|---|
| cvbase | all | 7.44 | 7.28 | 9.09 | 6.06 | 5.61 | -0.05 |
| | normal | 5.75 | 5.58 | 7.77 | 5.65 | 4.77 | -1.63 |
| | abnormal | 9.46 | 9.31 | 10.67 | 6.56 | 6.60 | 1.84 |
| cvctrl | all | 7.10 | 6.98 | 8.86 | 6.03 | 5.59 | 0.39 |
| | normal | 5.31 | 5.19 | 7.35 | 5.53 | 4.62 | -1.55 |
| | abnormal | 9.23 | 9.11 | 10.67 | 6.61 | 6.74 | 2.70 |
| cvaug | all | 7.23 | 7.11 | 8.81 | 6.05 | 5.63 | 0.57 |
| | normal | 5.43 | 5.30 | 7.16 | 5.54 | 4.55 | -1.42 |
| | abnormal | 9.37 | 9.28 | 10.78 | 6.67 | 6.91 | 2.96 |
| cvaugval | all | 7.22 | 7.04 | 8.68 | 6.01 | 5.60 | 0.46 |
| | normal | 5.36 | 5.16 | 7.01 | 5.40 | 4.48 | -1.49 |
| | abnormal | 9.44 | 9.28 | 10.68 | 6.72 | 6.93 | 2.79 |
| Gemein et al. 2024 (TCN, TUEG normal) | normal | 6.48 | – | – | – | 6.60 | – |
| | abnormal | – | – | – | – | 12.85 | – |
| Engemann et al. 2022 (TUAB normal, 10-fold CV) | normal | 7.75 | – | – | – | – | – |
| features + ridge, trained on TUAB normal | all | 9.70 | – | – | – | – | – |
| | normal | 7.21 | – | – | – | – | – |
| | abnormal | 12.67 | – | – | – | – | – |
| features + ridge, trained on all TUAB | all | 8.53 | – | – | – | – | – |
| | normal | 6.95 | – | – | – | – | – |
| | abnormal | 10.41 | – | – | – | – | – |

## Table 3.6. Leakage check: does a correction see the TUAB test labels?

Two checks per correction of Table 3.5 ( `rs.correction_leak_check` ), over the four CV experiments:

- **Fit set. fit ∩ test recordings** / **fit ∩ test patients**: how many of the out-of-fold recordings the correction is fitted on are TUAB test recordings, or belong to a patient with a TUAB test recording (the patient is the first field of the recording name). Both must be 0. The `normal` fits use a subset of these recordings. The fit set is the whole CV pool, which is also everything the fold models trained on, so 0 shared patients also means no fold model ever saw a TUAB test patient.
- **Application.** The TUAB test true ages are randomly permuted within each experiment and the correction is run again on the same predictions. A correction that never reads the test ages returns identical corrected predictions. `max change (years)` is the largest

difference over all experiments and test recordings, and `uses TUAB test true age` is "yes" when it is above zero.

The `raw` row is the reference: it applies no correction, so it cannot change.

| method | fit ∩ test recordings | fit ∩ test patients | max change (years) | uses TUAB test true age |
|---|---|---|---|---|
| **raw** | 0 | 0 | 0.00 | no |
| **recal, normal** | 0 | 0 | 0.00 | no |
| **cole, normal** | 0 | 0 | 0.00 | no |
| **gap lin, all †** | 0 | 0 | 26.52 | yes |
| **gap quad, normal †** | 0 | 0 | 26.34 | yes |

**Reading Tables 3.5 and 3.6.**

- **No TUAB test recording or patient is in any fit set** (Table 3.6: 0 and 0). The fold models, their epoch choice and every correction fit are independent of the TUAB test set.
- **Only the gap correction reads the test labels**, when it is applied. Permuting the TUAB test ages leaves `recal` and `cole` unchanged (max change 0) but moves the gap-corrected predictions by more than 25 years. Its numbers (†) are brain-age-gap sizes, not test errors.
- **Leak-free, recalibration helps a little.** Eq. (3.11) lowers the MAE from 7.10–7.44 to 6.98–7.28 overall, 5.31–5.75 → 5.16–5.58 on normal and 9.23–9.46 → 9.11–9.31 on abnormal recordings. It is used for the `cal.` columns of Tables 2.5–2.7.
- **The Cole correction makes the error worse** ( `cole, normal` ): 7.10–7.44 raw becomes 8.68–9.09 overall. Removing the age trend this way spreads out the predictions, and their noise with them.
- **With the test ages (†), the gap shrinks a lot.** A quadratic fitted on normal recordings (Gemein's choice) gives 4.48–4.77 on normal and 6.60–6.93 on abnormal recordings, against Gemein's 6.60 / 12.85 with the same correction. Much of the raw abnormal error is regression to the mean ($\beta \approx -0.5$, Table 3.4), and Eq. (3.9) undoes it with the true age. These numbers compare only with other gap-corrected numbers, never with a raw or recalibrated MAE.
- **Abnormal recordings look older.** After the normal-fit gap correction they keep a positive gap of +1.8 to +3.0 years ( `bias, gap quad normal †` ), and normal recordings a negative one. This is the brain-age-gap effect Gemein et al. report: −4 years between normal and pathological subjects.
- **Against the other methods, raw on the same test set.** The EEG-only `cvbase` scores 7.44 all / 5.75 normal / 9.46 abnormal. The classical feature-ridge model scores 8.53 / 6.95 / 10.41 when trained on all of TUAB, and 9.70 / 7.21 / 12.67 trained on normal recordings only. Engemann et al.'s best is 7.75 on normal recordings (CV, not the evaluation set). The foundation model trained on TUAB + TUEG is better on every cohort. Against the feature model trained on all of TUAB it is 0.95–1.5 years ahead (smallest on abnormal recordings). Against the normal-only feature model it is 1.5–3.2 years ahead, most on abnormal recordings, where normal-only models fail (12.67 here, 12.85 for Gemein et al.).

## Table 3.7. Agreement between experiments on the TUAB test set

How similar are the models' predictions, recording by recording? The comparison covers two kinds of pair:

- every pair of experiment ensembles (cvbase / cvctrl / cvaug: the CV runs with an in-pool test fold; the val-only `cvaugval` is compared in Tables 2.7 and 3.5);
- as a reference, the 10 pairs of fold models *within* each experiment. These pairs share the configuration but differ in training data and seed.

For two predictors $a$ and $b$ on the same recordings, with errors $e_i^a = \hat{y}_i^a - y_i$ (`labram/eval/cv_compare.py::pairwise`):

$$\mathrm{agree}_\tau = \frac{1}{n}\sum_i 1\big[|\hat{y}_i^a - \hat{y}_i^b| \le \tau\big], \qquad \mathrm{sign} = \frac{1}{n}\sum_i 1\big[\operatorname{sgn} e_i^a = \operatorname{sgn} e_i^b\big] \tag{3.12}$$

Columns, per status (all 274, normal 149, abnormal 125):

- `r_pred`, `r_err`: the Pearson correlation of the two models' predictions and of their errors. `r_err` near 1 means they make the same mistakes on the same recordings.
- `agree_2y`, `agree_5y`: $\mathrm{agree}_\tau$ for $\tau$ = 2 and 5 years.
- `sign_agree`: Eq. (3.11, right). The share of recordings where both models err in the same direction (both predict too old, or both too young).
- `dMAE`: $\mathrm{MAE}_a - \mathrm{MAE}_b$ (Eq. (0.4)) for an ensemble pair; the mean $|\Delta\mathrm{MAE}|$ over the 10 pairs for a fold-model row.

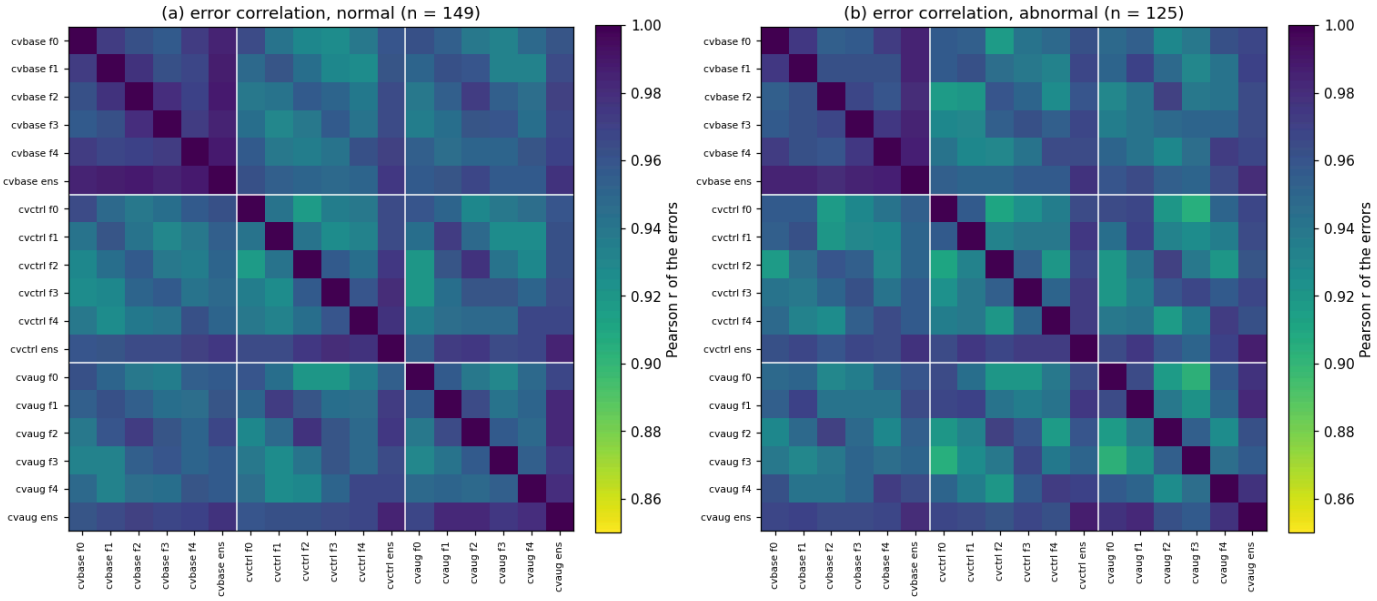
| status | pair | r_pred | r_err | agree_2y | agree_5y | sign_agree | dMAE |
|---|---|---|---|---|---|---|---|
| all | cvbase vs cvctrl (ensembles) | 0.987 | 0.974 | 0.693 | 0.960 | 0.912 | 0.340 |
| | cvbase vs cvaug (ensembles) | 0.989 | 0.977 | 0.701 | 0.978 | 0.912 | 0.214 |
| | cvctrl vs cvaug (ensembles) | 0.994 | 0.987 | 0.821 | 0.989 | 0.920 | -0.126 |
| | cvbase: fold model vs fold model (mean of 10 pairs) | 0.982 | 0.966 | 0.668 | 0.934 | 0.931 | 0.107 |
| | cvctrl: fold model vs fold model (mean of 10 pairs) | 0.969 | 0.936 | 0.532 | 0.876 | 0.889 | 0.177 |
| | cvaug: fold model vs fold model (mean of 10 pairs) | 0.972 | 0.939 | 0.520 | 0.886 | 0.872 | 0.208 |
| normal | cvbase vs cvctrl (ensembles) | 0.992 | 0.975 | 0.805 | 0.980 | 0.893 | 0.439 |
| | cvbase vs cvaug (ensembles) | 0.992 | 0.976 | 0.819 | 0.993 | 0.899 | 0.319 |
| | cvctrl vs cvaug (ensembles) | 0.995 | 0.984 | 0.899 | 1.000 | 0.886 | -0.120 |
| | cvbase: fold model vs fold model (mean of 10 pairs) | 0.989 | 0.969 | 0.759 | 0.982 | 0.914 | 0.108 |
| | cvctrl: fold model vs fold model (mean of 10 pairs) | 0.981 | 0.939 | 0.621 | 0.938 | 0.872 | 0.101 |
| | cvaug: fold model vs fold model (mean of 10 pairs) | 0.985 | 0.949 | 0.580 | 0.940 | 0.851 | 0.209 |
| abnormal | cvbase vs cvctrl (ensembles) | 0.976 | 0.976 | 0.560 | 0.936 | 0.936 | 0.222 |
| | cvbase vs cvaug (ensembles) | 0.980 | 0.980 | 0.560 | 0.960 | 0.928 | 0.088 |
| | cvctrl vs cvaug (ensembles) | 0.988 | 0.988 | 0.728 | 0.976 | 0.960 | -0.134 |
| | cvbase: fold model vs fold model (mean of 10 pairs) | 0.963 | 0.964 | 0.558 | 0.878 | 0.950 | 0.133 |
| | cvctrl: fold model vs fold model (mean of 10 pairs) | 0.936 | 0.936 | 0.427 | 0.802 | 0.909 | 0.311 |
| | cvaug: fold model vs fold model (mean of 10 pairs) | 0.941 | 0.937 | 0.450 | 0.822 | 0.898 | 0.214 |

## Figure 3.2. Error correlation between all TUAB-test predictors

The Pearson correlation of the errors $e_i = \hat{y}_i - y_i$ between every pair of the 18 predictors of the TUAB test set: each experiment's 5 fold models (`f0` – `f4`) and its median ensemble (`ens`).

- (a) The 149 normal recordings.
- (b) The 125 abnormal recordings.

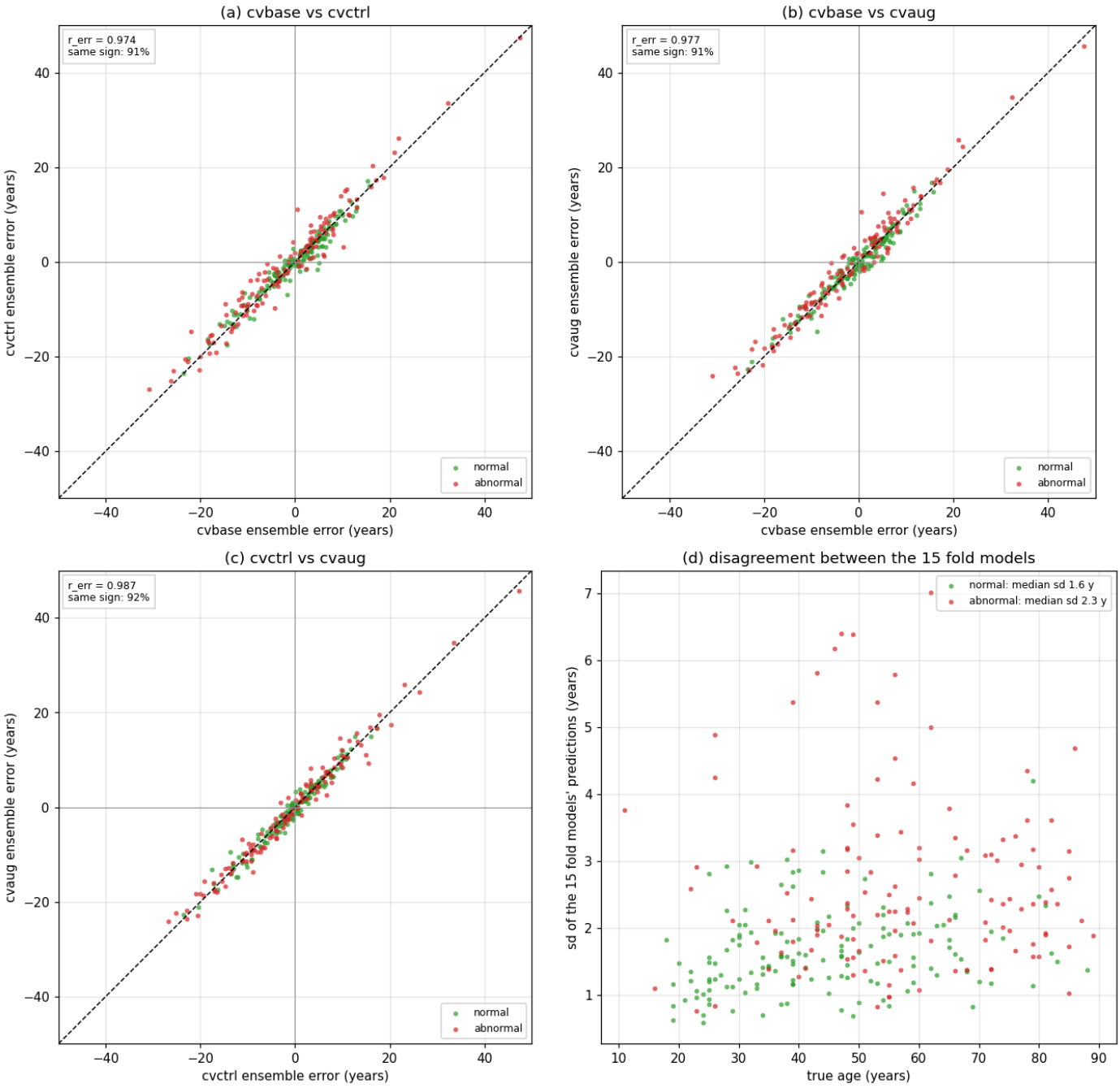
The colour scale runs from 0.85 (lighter) to 1 (darker). The thin white lines separate the three experiments.




## Figure 3.3. Per-recording agreement of the three experiments

TUAB test recordings, coloured by status: green normal, red abnormal. Errors $e = \hat{y} - y$ are in years.

- (a)–(c) The error of one experiment's ensemble against another's, for each pair of experiments. A point on the dashed diagonal means both models made the same error. A point in the upper-right or lower-left quadrant means both erred in the same direction.
- (d) **Disagreement per recording:** the sample standard deviation of the 15 fold models' predictions (3 experiments × 5 folds), against the true age.




**Reading Table 3.7 and Figures 3.2 and 3.3.**

- **The three experiments make almost the same errors.** Their ensembles' errors correlate at r = 0.974–0.987 (Figure 3.2, Figure 3.3(a)–(c)). On 91–92 % of the recordings both err in the same direction, and on 96–99 % they predict within 5 years of each other. The differences in MAE between them (Table 2.7) come from small shifts, not from different recordings going wrong.
- **The ensembles agree more than single fold models do.** The ensemble pairs agree more closely across experiments (r_err 0.97–0.99, within 2 years on 69–82 % of recordings) than two fold models of the same experiment (r_err 0.94–0.97, 52–67 %). Averaging five models removes much of the fold-to-fold noise, which is why the ensemble beats a single model (Table 5.2, C3).

- **Abnormal recordings: shared errors, more disagreement.** The models agree on the error direction more often on abnormal recordings: 93–96 % same sign, against 89–90 % on normal ones. But their predictions spread more there, with fewer abnormal recordings within 2 years (56–73 %, against 81–90 %). The fold models also disagree more on abnormal recordings (Figure 3.3(d)).
- **The largest errors are shared.** In Figure 3.3(a)–(c), the points far from the origin lie near the diagonal: the recordings that one model gets badly wrong, all models get badly wrong. Of the 20 largest `cvctrl` test errors, 16 are abnormal recordings (46 % of the test set is abnormal), and 17 are patients under 30 or over 70. These recordings are the main lever left for improving the MAE.

# 4. Changes versus the LaBraM recipe

The reference is **LaBraM as published for TUAB**, `labram/configs/defaults/finetune_tuab_paper.json`:

- LaBraM-base fine-tuned for normal/abnormal classification, the recipe of the paper and the README;
- lr $5 \cdot 10^{-4}$, batch 64, 50 epochs with 5 warmup, layer decay 0.65, drop path 0.1;
- no input re-referencing;
- a 50 Hz notch in preprocessing ( `dataset_maker/make_TUAB.py` ).

Table 4.1 lists every configuration field the best configuration changes. Table 4.2 lists the changes that live in the data or the code rather than the configuration. Section 5 then tests which of them moved the error.

## Table 4.1. Configuration changes: LaBraM TUAB recipe → best configuration

Every field on which `finetune_tuab_paper.json` and `cvctrl` (Table 3.1) differ, computed by `labram/eval/research_summary.py::config_diff`. Excluded:

- bookkeeping fields (paths, logging, infrastructure);
- fields a regression run never reads ( `INERT_FOR_REGRESSION` );
- fields of a section that is off in both.

Columns:

- `group` : the part of the pipeline;
- `meaning` : what the change does;
- `evidence` : the rows of Table 5.2 that test it, where one does.

| group | field | LaBraM recipe | best (cvctrl) | meaning | evidence |
|---|---|---|---|---|---|
| **augmentation** | **mixup.enabled** | False | True | mixup α 0.4 (Eq. (3.3)) | P3 |
| **channels** | **channel_drop.channels** | [] | ["T1", "T2"] | T1/T2 are the channels that may be missing | Q14 |
| | **channel_drop.consistency_weight** | 1 | 0 | 0: no consistency loss in cvctrl | Q17, C2 |
| | **channel_drop.enabled** | False | True | per-sample handling of missing T1/T2 (needed for cohort v3) | Q14 |
| | **channel_drop.max_drop** | 1 | 2 | up to both of T1/T2 | Q14 |
| | **channel_drop.prob** | 0.2 | 0 | 0: no channel-dropout augmentation in cvctrl | Q17, C2 |
| **data** | **data.data_format** | pickle | npy | storage only: one float32 npy per recording, windows bit-identical to the pickles | |
| **head** | **data.session_features** | | conditions_v1.npz | the condition vectors' file (conditions_v1.npz) | Q12, C1 |
| | **model.session_features** | off | head | status + pathology vector concatenated to the feature (Eq. (3.6)) | Q12, C1 |
| | **model.use_mean_pooling** | True | False | head on the class token instead of the mean patch token | P22, Q13, C1 |
| **input** | **labram_plus.enabled** | False | True | turns on the LaBraM++ input block (Eq. (3.1)) | P1 |
| | **labram_plus.z_score_patches** | True | False | LaBraM++ block with CAR only, no per-patch z-scoring | P1 |
| **optimizer** | **optimizer.lr** | 0.0005 | 0.0002 | 2e-4 at batch 128 = 1e-4 at batch 64, linearly scaled | Q14 |
| | **trainer.batch_size** | 64 | 128 | batch 128 (speed; lr scaled with it) | Q14 |
| **protocol** | **cross_validation.enabled** | False | True | 5-fold patient-grouped cross-validation | C3 |
| **schedule** | **optimizer.warmup_epochs** | 5 | 3 | 3 warmup epochs | P2, Q6, C1 |
| | **trainer.epochs** | 50 | 10 | 10 epochs instead of 50 (best epoch is early) | P2, Q6, C1 |
| **scoring** | **evaluation.agg_windows** | none | mean | one prediction per recording = mean of its windows (Eq. (3.7)) | |
| **speed** | **model.fused_attention** | False | True | fused attention kernel, numerically the same model | |
| **task** | **data.dataset** | TUAB | TUAB_AGE | age regression on TUH (TUAB_AGE loader) instead of normal/abnormal | |
| | **model.task** | classification | regression | scalar regression head (Huber on z-scored age, Eqs. (3.2), (3.4)) | |
| **windows** | **data.eval_minutes** | 0 | 5 | score each recording on its first 5 minutes after the trim | P26 |
| | **data.train_minutes_per_epoch** | 0 | 5 | 5 minutes per recording per epoch, redrawn each epoch (speed) | Q14 |
| | **data.trim_end_sec** | 0 | 60 | drop the last 60 s of every recording | |
| | **data.trim_start_sec** | 0 | 60 | drop the first 60 s of every recording | |

## Table 4.2. Changes outside the configuration

Changes in the training data, the preprocessing, the split and the code, with the Table 5.2 rows that test them.

| change | LaBraM recipe | this study | evidence |
|---|---|---|---|
| training data | TUAB train, 2,145 recordings (≈1.6k patients) | TUAB + TUEG, 25,613 recordings (≈12k patients, `all_both_v3`) | Q1, Q3 |
| mains interference | 50 Hz notch (TUH is recorded at 60 Hz, so the line stays in) | 50 + 60 Hz spectrum interpolation | L1 |
| channels | 23 fixed channels | 23; TUEG recordings missing only T1/T2 kept, with those channels left out per sample | Q14 |
| validation split | TUAB's train split, which leaks 16 patients between train and val | patient-disjoint, seeded; one TUEG patient split shared by all cohorts; TUAB-test patients removed from TUEG | Q11 |
| label | normal/abnormal | age from the EDF header (18–89 in the TUH cohorts) | – |
| unit of evaluation | 10 s window | recording (Eq. (3.7)), first 5 minutes after the trim | P26 |
| per-patient cap | none | at most 10 train/val recordings per patient | – |
| reported model | one run | median of 5 fold models (Eq. (3.8)) | C3 |
| reproducibility | ClearML silently reseeded every tracked run to 1337 | fixed: `distributed.seed` is honoured (commit `7b9a5fe`) | – |

# 5. Which changes mattered

**Noise floor.** Two runs of the same configuration with different seeds do not give the same MAE. The line-removal A/B study ran each of 3 preprocessing recipes $r$ with 3 seeds $s$, and each run is read at its best-validation epoch. The pooled within-recipe sample standard deviation of a metric $m$ ( `labram/eval/research_summary.py::seed_noise` ) is

$$\hat{\sigma}_m^2 = \frac{1}{N-R} \sum_r \sum_s \left(m_{r,s} - \bar{m}_r\right)^2 \tag{5.1}$$

with $N = 9$ runs and $R = 3$ recipes.

**Significance rule.** Each row of Table 5.2 compares a baseline with a variant that changes one thing (or a named bundle), using whichever evidence exists:

- **One run each** (phases 1 and 2). The difference of two single runs has standard deviation $\sqrt{2}\,\hat{\sigma}$, so a change counts only if ( `rs.pair_threshold` )

$$|\Delta\mathrm{MAE}| > \tau = 1.96\,\sqrt{2}\,\hat{\sigma}_{\mathrm{test}}. \tag{5.2}$$

- **Three seeds each** (line removal, L1/L2):

$$|\bar{\Delta}| > 1.96\,\hat{\sigma}_{\mathrm{test}}\,\sqrt{2/3}. \tag{5.3}$$

- **Five paired folds** (CV, C1/C2). Fold $k$ gives $\Delta_k$ on the same TUAB test recordings, and the 95 % interval is

$$\bar{\Delta} \pm t_{0.975,\,4}\,\frac{s_\Delta}{\sqrt{5}}, \qquad t_{0.975,\,4} = 2.776, \tag{5.4}$$

  where $\bar{\Delta}$ and $s_\Delta$ are the mean and sample sd of the $\Delta_k$. The change counts if the interval excludes 0.

- **Ensemble vs single model** (C3). A paired bootstrap over recordings (2,000 resamples, `cv_compare.py::bootstrap_delta_mae` ) of the ensemble's absolute error against the fold models' mean absolute error. The change counts if its 95 % interval excludes 0.

The verdict is `improves` / `hurts` / `n.s.` ( `rs.verdict` ). It is judged on the TUAB test MAE of all 274 recordings, the one set every run shares; the val and cohort columns are context.

**Limits.**

- $\hat{\sigma}$ is measured on one configuration, the E2 recipe on TUAB (Table 5.1), and assumed to hold for the others.
- A single-split row with $|\Delta|$ just under $\tau$ is not evidence of no effect, only of no detectable one.
- Several rows change more than one thing at once; they are marked *bundle*.

## Table 5.1. Seed-to-seed spread

**The model.** All 9 runs train the same model with the same recipe, **E2**: scenario D ( `finetune_tuab_age.json` ) on the short schedule. Read from each run's ClearML configuration:

- **Encoder:** LaBraM-base ( `labram_base_patch200_200` , 12 transformer blocks), initialized from the released `labram-base.pth` , all layers fine-tuned.
- **Input and head:** CAR input (Eq. (3.1)), mean patch-token pooling and a linear head.
- **Loss and optimizer:** Huber $\delta = 1$ on the z-scored age (Eq. (2.3)), no mixup. AdamW with lr $10^{-4}$, batch 64, weight decay 0.05, layer decay 0.65 and drop path 0.1.
- **Schedule and evaluation:** 4 epochs with 1 warmup, scored on the first 5 minutes of each recording after the 60 s trims.

**What varies.** The data is TUAB v3.0.1, on the same recordings and the same subject-disjoint split as phase 1 (Table 1.5). Only two things change:

- **Preprocessing:** the mains-interference removal, 3 variants. A is the 50 Hz notch (LaBraM as published), B a 60 Hz notch, C 50 + 60 Hz spectrum interpolation.
- **Seed:** 3 seeds per variant, which change the head initialization, the batch order, dropout and drop path. Seed 0 ran on a local g5.2xlarge; seeds 1 and 2 ran as SageMaker jobs on ml.g5.2xlarge ( `scripts/submit_line_removal_ab.sh` ).

Within a variant, the three runs differ only by the seed, which is what Eq. (5.1) pools. Each run is read at its best-validation epoch ( `docs/data/line_removal_ab.json` ). The table then gives the pooled standard deviation of Eq. (5.1) per metric and the single-pair threshold $\tau$ of Eq. (5.2).

Columns:

- `val_mae` : the TUAB validation split.
- `test_mae` , `test_normal_mae` , `test_abnormal_mae` : TUAB test, overall and per cohort.

| variant | seed | epoch | val_mae | test_mae | test_normal_mae | test_abnormal_mae |
|---|---|---|---|---|---|---|
| A_notch50 | 0 | 3 | 7.52 | 8.43 | 6.62 | 10.59 |
| | 1 | 3 | 7.66 | 8.40 | 6.60 | 10.55 |
| | 2 | 3 | 7.64 | 8.24 | 6.41 | 10.42 |
| B_notch60 | 0 | 2 | 7.46 | 8.01 | 6.39 | 9.96 |
| | 1 | 3 | 7.55 | 8.31 | 6.55 | 10.41 |
| | 2 | 3 | 7.54 | 8.07 | 6.38 | 10.08 |
| C_interp5060 | 0 | 2 | 7.46 | 8.03 | 6.39 | 9.99 |
| | 1 | 3 | 7.56 | 8.31 | 6.54 | 10.42 |
| | 2 | 3 | 7.56 | 8.10 | 6.39 | 10.13 |

```
pooled seed sd (years): val_mae 0.061, test_mae 0.138, test_normal_mae 0.100, test_abno
rmal_mae 0.194
single-pair threshold τ on TUAB test (Eq. 5.2): 0.38 years; 3-vs-3-seed threshold (Eq.
5.3): 0.22 years
```

## Table 5.2. Effect of each change

One row per tested change. Columns:

- `id` : the row's id, cited in Tables 4.1, 4.2 and 5.3.
- `phase` : 1 (registry, Table 2.2), 2 (Table 2.5), L (line-removal A/B, Table 5.1), C (CV, Table 2.7).

- `baseline → variant`: the two runs compared; `bundle` marks a change of several things at once.
- `Δ val`: Eq. (0.4) on the validation set the two runs share (phase 1: TUAB val; phase 2: TUEG val v2 where both runs were scored on it).
- `Δ test`, `Δ normal`, `Δ abnormal`: Eq. (0.4) on TUAB test, overall and per cohort.
- `noise`: $\pm\tau$ (Eqs. (5.2), (5.3)) or the 95 % interval (Eq. (5.4) / bootstrap).
- `verdict`: the rule above, on `Δ test`.

| id | phase | change | baseline → variant | Δ val | Δ test | Δ normal | Δ abnormal | noise | verdict |
|---|---|---|---|---|---|---|---|---|---|
| **P1** | 1 | common average reference on the input | B → D | -0.63 | -0.61 | – | – | ± 0.38 | improves |
| **P26** | 1 | score the first 5 min instead of whole recordings | A → B | 0.11 | -0.12 | – | – | ± 0.38 | n.s. |
| **P2** | 1 | 4 epochs / 1 warmup instead of 15 / 2 | D → E2 | -0.09 | -0.05 | – | – | ± 0.38 | n.s. |
| **P3** | 1 | mixup α 0.4 | E2 → M1 | -0.22 | 0.20 | – | – | ± 0.38 | n.s. |
| **P4** | 1 | C-Mixup (σ 5 y) | E2 → M1c | -0.14 | -0.02 | – | – | ± 0.38 | n.s. |
| **P5** | 1 | EMA weights for evaluation | E2 → M2 | -0.10 | -0.08 | – | – | ± 0.38 | n.s. |
| **P6** | 1 | layer decay 0.5 (stronger) | E2 → F1 | 0.06 | 0.11 | – | – | ± 0.38 | n.s. |
| **P7** | 1 | layer decay 0.75 (weaker) | E2 → H1 | 0.03 | 0.42 | – | – | ± 0.38 | hurts |
| **P8** | 1 | dropout 0.1 | E2 → F2 | 5.57 | 6.92 | – | – | ± 0.38 | hurts |
| **P9** | 1 | drop path 0.2 | E2 → F3 | 0.02 | 0.16 | – | – | ± 0.38 | n.s. |
| **P10** | 1 | weight decay 0.1 | E2 → F4 | -0.02 | -0.01 | – | – | ± 0.38 | n.s. |
| **P11** | 1 | head init scale 1.0 | E2 → G1 | 0.08 | 0.13 | – | – | ± 0.38 | n.s. |
| **P12** | 1 | LoRA r16 instead of full fine-tuning (bundle) | E2 → M3 | 0.00 | 0.99 | – | – | ± 0.38 | hurts |
| **P13** | 1 | L1 loss instead of Huber | B → D2 | -0.04 | 0.41 | – | – | ± 0.38 | hurts |
| **P14** | 1 | Huber δ 2 | D → DX2 | 0.02 | 0.11 | – | – | ± 0.38 | n.s. |
| **P15** | 1 | soft labels (σ 2.5 y) | E2 → M4 | -0.01 | 0.34 | – | – | ± 0.38 | n.s. |
| **P16** | 1 | age-balanced loss (sqrt-inverse) | E2 → B1 | 0.15 | 0.23 | – | – | ± 0.38 | n.s. |
| **P17** | 1 | codebook amplitude reconstruction | D → DX1 | -0.05 | 0.24 | – | – | ± 0.38 | n.s. |
| **P18** | 1 | codebook quantization term | D → DX4 | -0.10 | 0.14 | – | – | ± 0.38 | n.s. |
| **P19** | 1 | 30 s windows instead of 10 s | B → C | -0.16 | 0.44 | – | – | ± 0.38 | hurts |
| **P20** | 1 | random crop | D → E1 | 0.05 | 0.26 | – | – | ± 0.38 | n.s. |
| **P21** | 1 | gradient clipping 1.0 | D → D-clip1 | -0.07 | 0.06 | – | – | ± 0.38 | n.s. |
| **P22** | 1 | class token instead of mean pooling | D → D-cls | 0.13 | -0.06 | – | – | ± 0.38 | n.s. |
| **P23** | 1 | [mean ; class token] concatenated | D → D-clsmean | 0.03 | 0.05 | – | – | ± 0.38 | n.s. |
| **P24** | 1 | multi-task head (age + normal/abnormal) | M1 → MT1 | 0.08 | -0.07 | 0.13 | -0.30 | ± 0.38 | n.s. |
| **P25** | 1 | pretrained LaBraM weights instead of training from scratch | S1 → D | -1.02 | -1.12 | – | – | ± 0.38 | improves |
| **L1** | L | 50 + 60 Hz interpolation instead of 50 Hz notch | A_notch50 → C_interp5060 | -0.08 | -0.21 | -0.10 | -0.34 | ± 0.22 | n.s. |
| **L2** | L | 60 Hz notch instead of 50 Hz notch | A_notch50 → B_notch60 | -0.09 | -0.23 | -0.11 | -0.37 | ± 0.22 | improves |
| **Q1** | 2 | add every TUEG recording (~10x data) | Control → Exp3 | – | -0.97 | -0.85 | -1.11 | ± 0.38 | improves |
| **Q2** | 2 | train on normal recordings only (TUAB + TUEG) (bundle) | Control → Exp1 | -0.62 | -0.20 | -0.43 | 0.08 | ± 0.38 | n.s. |
| **Q3** | 2 | add abnormal + unlabelled recordings to normal-only | Exp1 → Exp3 | – | -0.77 | -0.41 | -1.19 | ± 0.38 | improves |
| **Q4** | 2 | TUEG normal only (no TUAB in training) | Exp1 → Exp2 | 0.07 | -0.03 | 0.07 | -0.16 | ± 0.38 | n.s. |
| **Q5** | 2 | 10 / 3 epochs on the normal cohort | Exp1 → Exp1b | -0.02 | 0.08 | 0.11 | 0.05 | ± 0.38 | n.s. |
| **Q6** | 2 | 10 / 3 epochs on the all cohort | Exp3 → Exp3b | – | 0.43 | 0.60 | 0.23 | ± 0.38 | hurts |
| **Q7** | 2 | window rejection (±800 µV, flat channel) | Exp3 → Exp3q | – | 0.07 | 0.12 | 0.03 | ± 0.38 | n.s. |
| **Q8** | 2 | input clip ±800 µV | Exp3 → Exp3c | – | 0.02 | 0.09 | -0.06 | ± 0.38 | n.s. |
| **Q9** | 2 | classical features as transformer tokens | Exp1 → Exp1t | – | 0.76 | 1.19 | 0.26 | ± 0.38 | hurts |
| **Q10** | 2 | classical features only, no EEG encoder | Exp1 → Exp1f | – | 0.10 | 0.22 | -0.06 | ± 0.38 | n.s. |
| **Q11** | 2 | shared TUEG patient split (v2) | Exp3 → Exp3v2 | – | 0.15 | -0.11 | 0.47 | ± 0.38 | n.s. |
| **Q12** | 2 | status + pathology condition vectors | Exp3v2 → Exp3cond | -0.10 | -0.14 | -0.07 | -0.22 | ± 0.38 | n.s. |
| **Q13** | 2 | class token (with conditions) | Exp3cond → Exp3clscond | 0.07 | -0.03 | -0.03 | -0.02 | ± 0.38 | n.s. |
| **Q14** | 2 | cohort v3 + missing-T1/T2 handling + speed settings (bundle) | Exp3v2 → Exp4base | -0.14 | -0.05 | 0.16 | -0.31 | ± 0.38 | n.s. |
| **Q15** | 2 | compact copy of the cohort (reproduction) | Exp4base → Exp4basec | -0.01 | 0.06 | -0.10 | 0.24 | ± 0.38 | n.s. |
| **Q16** | 2 | class token + conditions + 10 ep with early stopping (bundle) | Exp4base → Exp4ctrl | 0.05 | -0.06 | -0.27 | 0.19 | ± 0.38 | n.s. |
| **Q17** | 2 | T1/T2 dropout + consistency (6 vs 10 epochs) (bundle) | Exp4ctrl → Exp4aug | -0.07 | -0.01 | 0.09 | -0.14 | ± 0.38 | n.s. |
| **Q18** | 2 | T1/T2 dropped everywhere (21 channels) + cohort v3 (bundle) | Exp3cond → Exp4drop | -0.15 | -0.00 | -0.13 | 0.16 | ± 0.38 | n.s. |
| **Q19** | 2 | 15 / 5 epochs with early stopping (aug) | Exp4aug → Exp4aug15 | 0.06 | -0.12 | -0.06 | -0.20 | ± 0.38 | n.s. |
| **C1** | C | class token + conditions + 10 / 3 epochs (CV) (bundle) | cvbase → cvctrl | -0.07 | -0.21 | -0.28 | -0.13 | [-0.41, -0.01], better in 5/5 folds | improves |
| **C2** | C | T1/T2 dropout + consistency λ 0.1 (CV) | cvctrl → cvaug | -0.03 | 0.04 | 0.02 | 0.07 | [-0.16, +0.25], better in 2/5 folds | n.s. |
| **C3** | C | median of 5 fold models instead of one (cvctrl) | cvctrl single → cvctrl ensemble | – | -0.18 | -0.22 | -0.13 | [-0.27, -0.09] (bootstrap) | improves |
| **C3b** | C | median of 5 fold models instead of one (cvbase) | cvbase single → cvbase ensemble | – | -0.05 | -0.06 | -0.04 | [-0.11, +0.01] (bootstrap) | n.s. |

## Figure 5.1. Effect sizes

**What is plotted.** One row per tested change (Table 5.2). The point is how much that change moved the TUAB test MAE, Eq. (0.4):

$$\Delta\mathrm{MAE} = \mathrm{MAE}(\text{run with the change}) - \mathrm{MAE}(\text{run without it}),$$

in years, on all 274 test recordings. The two runs are named in brackets in the row label (baseline → variant).

This is a **difference of two errors, not a prediction error**. A point at −0.5 means the change made the model 0.5 years more accurate. It does not mean the model predicts ages 0.5 years too low.

- **Left of 0:** the change lowered the error, so it helped. Further left is better.
- **At 0:** no effect.
- **Right of 0:** the change raised the error, so it hurt.

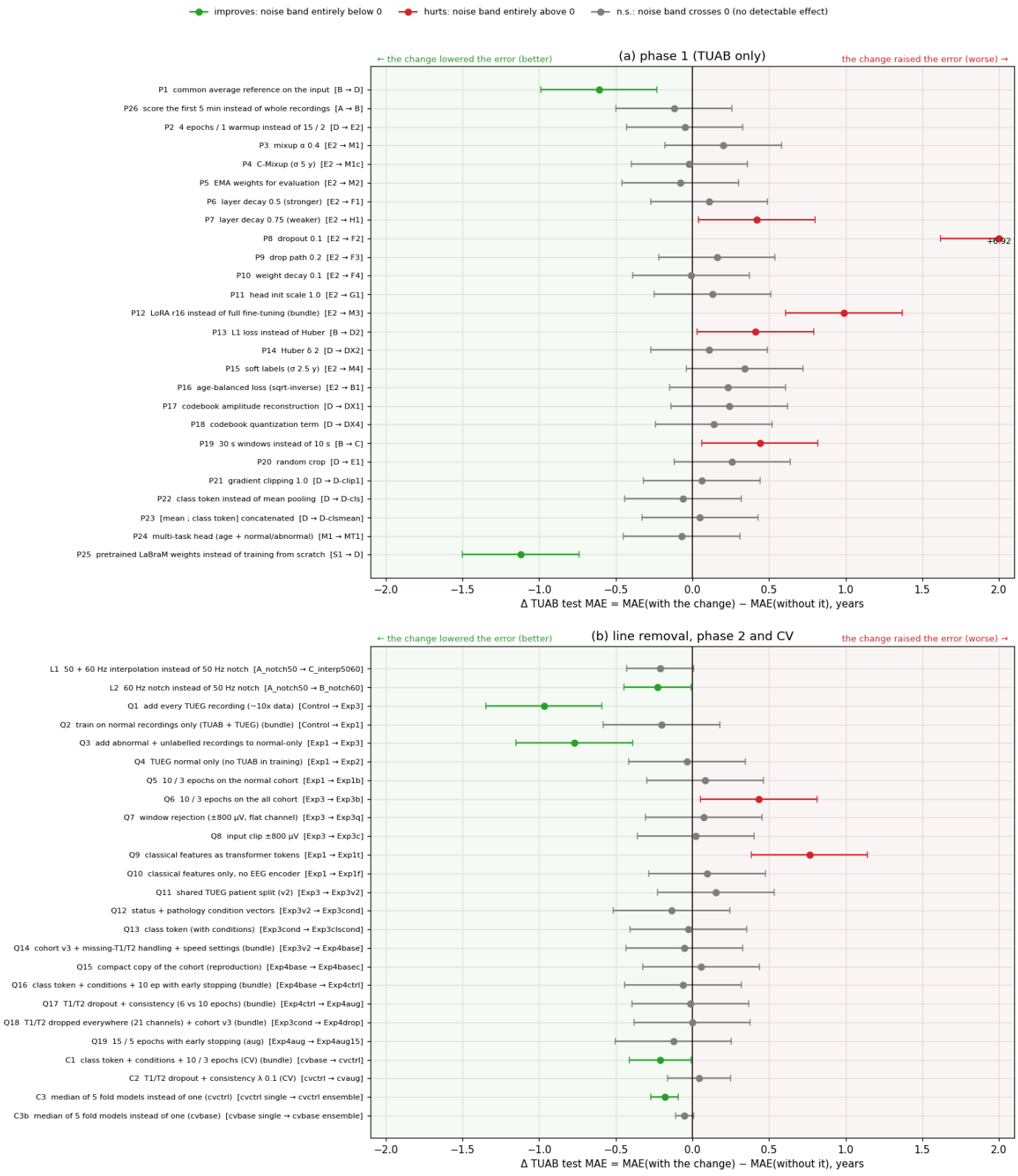
**Bars and colours.**

- The horizontal bar is the noise band: how far two runs differing only by chance would land apart. It is $\pm\tau$ (Eq. (5.2)) for single runs and 3-seed pairs, and the 95 % interval (Eq. (5.4) or the bootstrap) for CV and ensemble rows.
- **Green (`improves`):** the whole band lies left of 0, so the gain is larger than seed noise.
- **Red (`hurts`):** the whole band lies right of 0.
- **Grey (`n.s.`):** the band crosses 0. The effect cannot be told apart from noise.

Changes beyond ±2 years (collapsed runs) are drawn at the edge, with their value written next to them. Panels:

- (a) Phase 1, TUAB-only runs.
- (b) Line removal, phase 2 and CV.

The signed drift of the predictions themselves, mean(predicted − true), where closer to 0 is better, is the bias of Eq. (0.3). It is shown per model and status in Table 3.4 (`bias`) and in Figure 3.1.




# Table 5.3. Changes versus LaBraM: which mattered

The summary the study was after. One row per change from the LaBraM TUAB recipe to the best configuration (Tables 4.1 and 4.2), with:

- `estimate`: the Table 5.2 row that isolates the change best and its $\Delta$ test;

- `verdict` : that row's verdict;
- `in best config` : whether the change is part of `cvctrl` .

`not isolated` means no run changes it alone, so its effect is not known.

| change vs LaBraM | estimate | Δ test | verdict | in best config |
|---|---|---|---|---|
| **training data: TUAB → TUAB + TUEG (~10x recordings)** | Q1: Control → Exp3 | -0.97 | improves | yes |
| **abnormal and unlabelled recordings in training** | Q3: Exp1 → Exp3 | -0.77 | improves | yes |
| **pretrained LaBraM weights (vs from scratch)** | P25: S1 → D | -1.12 | improves | yes |
| **common average reference on the input** | P1: B → D | -0.61 | improves | yes |
| **mains: 50 + 60 Hz interpolation instead of 50 Hz notch** | L1: A_notch50 → C_interp5060 | -0.21 | n.s. | yes |
| **ensemble: median of the 5 fold models** | C3: cvctrl single → cvctrl ensemble | -0.18 | improves | yes |
| **class token + condition vectors + 10 epochs (bundle)** | C1: cvbase → cvctrl | -0.21 | improves | yes |
| **condition vectors alone (single split)** | Q12: Exp3v2 → Exp3cond | -0.14 | n.s. | yes |
| **class-token pooling alone (TUAB / TUH)** | P22: D → D-cls | -0.06 | n.s. | yes |
| **short schedule (4–10 epochs instead of 15–50)** | P2: D → E2 | -0.05 | n.s. | yes |
| **mixup α 0.4** | P3: E2 → M1 | 0.20 | n.s. | yes |
| **cohort v3 + missing-T1/T2 handling + speed settings (bundle)** | Q14: Exp3v2 → Exp4base | -0.05 | n.s. | yes |
| **5-minute evaluation window** | P26: A → B | -0.12 | n.s. | yes |
| **T1/T2 channel-dropout augmentation** | C2: cvctrl → cvaug | 0.04 | n.s. | no (cvaug) |
| **longer schedule on the full cohort (10 epochs)** | Q6: Exp3 → Exp3b | 0.43 | hurts | no |
| **dropout 0.1** | P8: E2 → F2 | 6.92 | hurts | no |
| **weaker layer decay 0.75** | P7: E2 → H1 | 0.42 | hurts | no |
| **LoRA instead of full fine-tuning** | P12: E2 → M3 | 0.99 | hurts | no |
| **30 s windows** | P19: B → C | 0.44 | hurts | no |
| **classical features as tokens** | Q9: Exp1 → Exp1t | 0.76 | hurts | no |
| **L1 loss** | P13: B → D2 | 0.41 | hurts | no |
| **weight decay, drop path, layer decay 0.5, head init** | P10: E2 → F4 | -0.01 | n.s. | no |
| **loss shape: Huber δ 2, soft labels, age-balanced** | P16: E2 → B1 | 0.23 | n.s. | no |
| **codebook regularization** | P18: D → DX4 | 0.14 | n.s. | no |
| **signal QC: window rejection / ±800 µV clip** | Q7: Exp3 → Exp3q | 0.07 | n.s. | no |
| **multi-task head** | P24: M1 → MT1 | -0.07 | n.s. | no |
| **trim 60 s at both ends; lr 1e-4 (vs 5e-4); batch 128** | – | – | not isolated | yes |

# 6. Conclusions

Generated from the tables above, so they stay in step with the data when the notebook is re-run.

- **Noise floor.** Seed-to-seed sd of the TUAB test MAE is 0.14 years (Table 5.1), so two single runs must differ by more than 0.38 years to count (Eq. (5.2)). Of 51 tested changes, 7 improve, 7 hurt and 37 are within noise (Table 5.2).
- **What improved the error (Table 5.2):** P25 pretrained LaBraM weights instead of training from scratch (-1.12); Q1 add every TUEG recording (~10x data) (-0.97); Q3 add abnormal + unlabelled recordings to normal-only (-0.77); P1 common average reference on the input (-0.61); L2 60 Hz notch instead of 50 Hz notch (-0.23); C1 class token + conditions + 10 / 3 epochs (CV) (bundle) (-0.21); C3 median of 5 fold models instead of one (cvctrl) (-0.18).
- **What hurt:** P13 L1 loss instead of Huber (+0.41); P7 layer decay 0.75 (weaker) (+0.42); Q6 10 / 3 epochs on the all cohort (+0.43); P19 30 s windows instead of 10 s (+0.44); Q9 classical features as transformer tokens (+0.76); P12 LoRA r16 instead of full fine-tuning (bundle) (+0.99); P8 dropout 0.1 (+6.92).
- **Best configuration: `cvctrl`.** TUAB test MAE 7.28 ± 0.14 per fold model (normal 5.53, abnormal 9.37; Table 3.3). Its head is given the status/pathology vector.
- **Best EEG-only: `cvbase`.** 7.49 ± 0.09 per fold model, normal recordings 5.81, against 6.60 for the best published TUAB normal result (Gemein et al. 2024, different split and training data).
- **Ensembles.** The median of the 5 fold models scores 7.10 ( `cvctrl` ) and 7.44 ( `cvbase` ) on the TUAB test, normal 5.75 ( `cvbase` ).
- **Most of the gain over LaBraM comes from data.** The largest effects are:
  - more and more varied training recordings (Q1, Q3);
  - pretrained weights (P25);
  - the common average reference (P1). Architecture, loss and regularization changes are within noise or harmful (Table 5.3).
- **The abnormal cohort carries the error.** Abnormal recordings stay about 4 years worse than normal ones in every configuration (Tables 2.5, 3.3).

**Caveats.**

- **Selection on test:** phases 1 and 2 chose among configurations on the TUAB test set, so their best single-split numbers are optimistic. The CV and TUEG out-of-fold numbers are not.
- **Bundles:** C1, Q14, Q16, Q17 and Q18 change several things at once, so their parts are not separately known.
- **Condition vectors:** `cvctrl` and `cvaug` use the status/pathology vector at test time. Only `cvbase` is comparable with EEG-only brain-age models.
- **Noise floor scope:** it comes from one configuration, the E2 recipe on TUAB (Table 5.1), and is assumed to transfer to the larger cohorts.